

# Categorical Homotopy Theory - Self Study

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## Abstract

This text contains my notes on Categorical Homotopy Theory, by Emily Riehl. I read this book independently during the summer of 2025, between my sophomore and junior years. My notes largely paraphrase the text. Going into the text, I lacked a precise understanding of model category theory, and had a weak intuition for simplicial sets/objects and methods. I also lacked some helpful algebraic topology knowledge. I continuously supplemented my readings with references to Hatcher's *Algebraic Topology*, Hovey's *Model Category Theory*, and on occasion May's *Simplicial Objects in Algebraic Topology*. I intended to, as a side effect of reading this text, improve my understanding of these subjects such that I might only need to skim texts on them in the future, rather than read them in their entirety.

**Notation.** Disjoint unions are denoted by  $\amalg$ . Composition of functors and morphisms is denoted by  $\circ$ , or juxtaposition when the notation can be well understood. Vertical composition of 2-cells is denoted by  $\circ$ , and horizontal composition is denoted by  $\cdot$ . Usually, juxtaposition of 2-cells denotes horizontal composition, but on occasion it may denote vertical composition if it can be well understood that there is no horizontal composition. For a category  $\mathcal{C}$ , I also write  $\mathcal{C}$  to denote the set of objects of  $\mathcal{C}$ . On occasion, I will instead write  $\text{Obj}(\mathcal{C})$  for emphasis. I write  $\text{Mor}(\mathcal{C})$  for the set of arrows of  $\mathcal{C}$ .







## Chapter 1

# All concepts are Kan extensions

I already know Kan extensions so I will breeze through this chapter.

## 1.1 Kan Extensions

**Definition 1.1** (Lan Extension). Given functors  $F : \mathcal{C} \rightarrow \mathcal{E}$  and  $K : \mathcal{C} \rightarrow \mathcal{D}$ , a *left Kan extension* of  $F$  along  $K$  is a functor  $\text{Lan}_K F : \mathcal{D} \rightarrow \mathcal{E}$  and a natural transformation  $\eta : F \Rightarrow \text{Lan}_K F \circ K$  (a pasting diagram) which is initial with respect to this property; for any other extension  $G : \mathcal{D} \rightarrow \mathcal{E}$  and  $\gamma : F \Rightarrow G \circ K$ , there is a unique natural transformation  $\nu : \text{Lan}_K F \Rightarrow G$  such that  $\gamma \cdot 1_K = (\nu \cdot 1_F) \circ \eta$ . Dually, a right Kan extension is terminal with respect to natural transformations  $\varepsilon : \text{Ran}_K F \circ K \Rightarrow F$ . ♥

**Remark.** These definitions generalize to any 2-category. We usually specialize in the 2-category  $\mathcal{Cat}$  for obvious reasons, so we reserve the term “Kan extension” for  $\mathcal{Cat}$ . When working in  $\mathcal{Cat}$ , I use “extension” to mean a pair  $(G, \gamma)$  satisfying the extension property, but not necessarily universally.

In unenriched category theory, universal properties (in this text) are representations of functors. A left Kan extension  $\text{Lan}_K F$  with notation as above is a representation for

$$\mathcal{E}^{\mathcal{C}}(F, - \circ K) : \mathcal{E}^{\mathcal{C}} \rightarrow \mathbf{Set}.$$

To justify this, recall here that universality gives for any  $G$ ,

$$\mathcal{E}^{\mathcal{C}}(F, G \circ K) \cong \mathcal{E}^{\mathcal{D}}(\text{Lan}_K F, G).$$

Hence,  $\text{Lan}_K F$  is a representation as the universal object. Alternatively, the Yoneda lemma promotes any extension  $\gamma : F \Rightarrow G \circ K$  to a natural transformation

$$\gamma : \mathcal{E}^{\mathcal{C}}(G \circ K, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, -).$$

Precomposing by  $1 \times K$  gives

$$\gamma \cdot 1 : \mathcal{E}^{\mathcal{C}}(G \circ K, - \circ K) = \mathcal{E}^{\mathcal{D}}(G, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, - \circ K).$$

Universality is then equivalent to that the natural transformation

$$\eta : \mathcal{E}^{\mathcal{D}}(\text{Lan}_K F, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, - \circ K)$$

be an isomorphism.

By universality,  $\text{Lan}_K -$  is functorial provided it always exists, so this gives an adjunction for precomposition  $K^*$ :

$$\mathcal{E}^{\mathcal{D}}(\text{Lan}_K F, G) \cong \mathcal{E}^{\mathcal{C}}(F, G \circ K) \quad \text{Lan}_K \dashv K^* \dashv \text{Ran}_K.$$

## 1.2 A formula

**Theorem 1.1.** *Let  $\mathcal{C}$  small,  $\mathcal{D}$  locally small, and  $\mathcal{E}$  cocomplete. Then any  $F : \mathcal{C} \rightarrow \mathcal{E}$  has an extension along any  $K : \mathcal{C} \rightarrow \mathcal{D}$  given pointwise by*

$$\mathrm{Lan}_K F(d) = \int^{c \in \mathcal{C}} \mathcal{D}(K(c), d) \cdot F(c)$$

for  $\cdot$  the copower/tensor and  $d \in D$ . Dually,

$$\mathrm{Ran}_K F(d) = \int_{c \in \mathcal{C}} F(c)^{\mathcal{D}(K(c), d)}.$$

Here, exponentiation is the power/cotensor.

| **Proof.** See my notes for Mac Lane. ■

**Theorem 1.2.** *With notation as in 1.1, we also have*

$$\mathrm{Lan}_K F(d) = \varinjlim \left( F \circ \Pi^d \right), \quad \Pi^d : (K \downarrow d) \rightarrow \mathcal{C},$$

for  $\Pi^d$  the comma category projection. We will prove these formulas again in the context of weighted limits later. Dually,

$$\mathrm{Ran}_K F(d) = \varprojlim \left( (d \downarrow K) \xrightarrow{\Pi_d} \mathcal{C} \xrightarrow{F} \mathcal{E} \right).$$

## 1.3 Pointwise Kan extensions

**Definition 1.2 (Preserves).** A functor  $L : \mathcal{E} \rightarrow \mathcal{F}$  preserves left Kan extensions if  $(L \mathrm{Lan}_K F, L\eta)$  is the left Kan extension  $\mathrm{Lan}_K LF$ . ♥

The following holds even when coends and colimits don't exist. Again, see my notes on Mac Lane for the (simple) proof.

**Lemma 1.1.** *Left adjoints preserve left Kan extensions. Right adjoints preserve right Kan extensions.*

It has generally been decided that Kan extensions in full generality are not nearly as important as the following special case, and the accompanying theorem which reconciles the terminology.

**Definition 1.3 (Pointwise Kan extensions).** If  $\mathcal{E}$  is locally small, a pointwise right Kan extension is one preserved by all representable functors  $\mathcal{E}(e, -)$ . By contravariance, a pointwise left Kan extension is one such that postcomposition by  $\mathcal{E}(-, e)$  with the opposite (right) Kan extension gives a right Kan extension. ♥

**Theorem 1.3.** *A right Kan extension is pointwise if and only if it is computed by limits as described in theorem 1.2. A left Kan extension is pointwise if and only if it is computed by colimits as in theorem 1.2.*

## 1.4 All concepts

The notion of Kan extensions subsumes all the other fundamental concepts of category theory.

**Theorem 1.4.** *Let  $F : \mathcal{C} \rightarrow \mathcal{E}$ . A limit of  $F$  is precisely a right Kan extension of  $F$  along the terminal functor  $! : \mathcal{C} \rightarrow \mathbf{1}$ . A left Kan extension along the terminal functor is precisely a colimit.*

**Theorem 1.5.** *Adjunctions  $(F, G, \eta, \varepsilon)$  are precisely left Kan extensions  $(G, \eta)$  of  $\mathbf{1}$  along  $F$  such that  $F$  preserves the Kan extension.*

Kan extensions also give the Yoneda lemma. For  $F : \mathcal{C} \rightarrow \mathbf{Set}$ ,

$$F(c) \cong \text{Ran}_{\mathbf{1}_e} F(c) \cong \int_{x \in \mathcal{C}} F(x)^{\mathcal{C}(c, x)} \cong \int_{x \in \mathcal{C}} \mathbf{Set}(\mathcal{C}(c, x), F(x)) \cong \mathbf{Set}^e(\mathcal{C}(c, -), F).$$

The first isomorphism is kind of just obvious, the second is the pointwise construction via ends, the third is definition of exponentials in  $\mathbf{Set}$ , and the fourth is an identity I proved in my notes on Mac Lane. This also gives us the coYoneda lemma:

$$F(c) \cong \int^{x \in \mathcal{C}} \mathcal{C}(x, c) \cdot F(x).$$

Finally, if  $\mathcal{E}$  is complete and  $K$  is fully faithful, then the counit is an isomorphism  $\text{Ran}_K F \circ K \cong F$ .

## 1.5 Adjunctions involving simplicial sets

We denote by  $\mathbf{sSet}$  the functor category  $\text{Fun}(\Delta^{\text{op}}, \mathbf{Set})$ , where  $\Delta$  is the simplex category, and  $\Delta_+$  is the augmented simplex category which includes  $[-1]$ .  $\mathbf{sSet}$  is the free colimit completion of  $\Delta$ , although this is not unique to  $\Delta$  and holds for any small category.

Let  $\mathcal{E}$  be cocomplete and locally small, and let  $\Delta^\bullet : \Delta \rightarrow \mathcal{E}$  be covariant. Write  $\Delta^n$  for the image  $\Delta^\bullet[n]$ ; similarly for  $X \in \mathbf{sSet}$ , write  $X_n$  for the set  $X[n]$  of  $n$ -simplices and  $X_\bullet : \Delta^{\text{op}} \rightarrow \mathbf{Set}$  when we wish to emphasize its functorial role, rather than its role as an assignment of sets. Define  $L = \text{Lan}_y \Delta^\bullet$  as the left Kan extension of  $\Delta^\bullet$  along the Yoneda embedding  $y$ :

$$\begin{array}{ccc} \Delta & \xrightarrow{\Delta^\bullet} & \mathcal{E} \\ & \searrow y & \downarrow \cong \\ & \mathbf{sSet} & \swarrow L \end{array}$$

Since the Yoneda embedding is fully faithful and  $\mathcal{E}$  is locally small and cocomplete, the unit is an isomorphism by the dual of the above statement. The functor  $L$  is then defined via theorem 1.1 as

$$L(X) := \int^{n \in \Delta} \mathbf{sSet}(y(n), X) \cdot \Delta^n \cong \int^{n \in \Delta} X_n \cdot \Delta^n.$$

The first isomorphism is once again theorem 1.1, and the second is the Yoneda lemma since  $y(n) = \Delta(-, n)$  and indeed  $\text{Nat}(\Delta(-, n), X) \cong X(n) = X_n$ . Since coends are colimits, which are characterized by coproducts and coequalizers, we obtain an explicit description as

$$L(X) = \text{coeq} \left( \coprod_{f \in \text{Mor}(\Delta)} X_{\text{cod } f} \cdot \Delta^{\text{dom } f} \rightrightarrows \coprod_{[n] \in \Delta} X_n \cdot \Delta^n \right).$$

The definition of  $L$  on the arrows is given by universality. We usually choose to denote by  $\Delta^n$  the simplicial set  $y[n]$ , and by cocompleteness we have

$$(L \circ y)[n] = \Delta^n,$$



somewhat justifying this different notation we have used here.

Since colimits commute,  $L$  preserves colimits, so the adjoint functor theorem gives an adjoint  $R : \mathcal{E} \rightarrow \mathbf{sSet}$ , which has for any  $e \in \mathcal{E}$ ,

$$R(e)_n \cong \mathbf{sSet}(y[n], R(e)) \cong \mathcal{E}(Ly[n], e) \cong \mathcal{E}(\Delta^n, e).$$

The first isomorphism is Yoneda, the second is the adjunction relation, and the third is our above discussion. Thus,  $n$ -simplices of  $R(e)$  can be defined as the set of maps from  $\Delta^n \rightarrow e$  in  $\mathcal{E}$ , making  $R = \mathcal{E}(\Delta^\bullet, -)$  with a mild abuse of notation. Face and degeneracy maps for the simplicial set  $R(e)$  are also thus given by precomposition by the appropriate maps to  $\Delta^n$  (called a cosimplicial object since it is covariant). Levelwise postcomposition makes  $R$  functorial.

The geometric realization of general simplicial sets is as follows. There is a natural functor  $\Delta \rightarrow \mathcal{T}\mathbf{op}$  given by  $[n] \mapsto \Delta^n$  for  $\Delta^n$  the standard  $n$ -simplex. It has both left- and right-adjoints. Since  $\mathcal{T}\mathbf{op}$  is cocomplete and locally small, plugging in  $\mathcal{T}\mathbf{op}$  for  $\mathcal{E}$  in the previous discussion gives an adjunction  $L \dashv R$  with  $L : \mathbf{sSet} \rightarrow \mathcal{T}\mathbf{op}$  the left Kan extension, and  $R : \mathcal{T}\mathbf{op} \rightarrow \mathbf{sSet}$  its right adjoint.  $R$  is called the *total singular complex functor*, denoted usually by  $S$ , and  $L$  is the *geometric realization*, denoted usually by  $|-|$ . It's computed as

$$|X| = \int^{n \in \Delta} X_n \times \Delta^n = \text{coeq} \left( \coprod_{f \in \text{Mor}(\Delta)} X_{\text{cod } f} \cdot \Delta^{\text{dom } f} \rightrightarrows \coprod_{[n] \in \Delta} X_n \cdot \Delta^n \right).$$

Here we note that copower is isomorphic to cartesian proeuct in  $\mathcal{T}\mathbf{op}$ . This gives  $|y(n) = \Delta^n| = \Delta^n$ , once again justifying the annoying notation.

Let  $\Delta^\bullet : \Delta \rightarrow \mathbf{Cat}$  now be the functor mapping  $[n]$  to the poset category  $n + 1$  and maps to functors. Indeed,  $\Delta^\bullet$  is a full embedding  $\Delta \hookrightarrow \mathbf{Cat}$ . Its right adjoint this time is the *nerve* functor  $N : \mathbf{Cat} \rightarrow \mathbf{sSet}$ . Recall that  $N(\mathcal{C})$  is a simplicial set with an  $n$ -simplex given by a string of  $n$  composable arrows in  $\mathcal{C}$ , or equivalently any functor  $[n] \rightarrow \mathcal{C}$ . The  $i$ th degeneracy maps insert the identity arrow, and the  $i$ th face map composes  $i$  and  $i + 1$ , unless  $i = 0$  or  $n$ , in which case the arrow is dropped.

The left adjoint  $h : \mathbf{sSet} \rightarrow \mathbf{Cat}$  maps a simplicial set to its homotopy category  $hX$ . The objects are 0-simplices (also called vertexes) of  $X$ , and its arrows are freely generated on 1-simplices with respect to relations governed by 2-simplices.

## Chapter 2

### Derived functors via deformation

I have some familiarity with the beginning material in this section, and will read those parts rather quickly compared to the other parts.

#### 2.1 Homotopical categories and derived functors

The ideas here were introduced around 20 years ago, but we depart slightly from the original terminology. I feel this is important to note to avoid confusions with the literature on the topic.

**Definition 2.1** (Homotopical Category). A *homotopical category* is a category  $\mathcal{M}$  together with a wide subcategory  $\mathcal{W}$  (wide meaning containing all objects) such that for any composable triple  $f, g, h$  of arrows, if  $hg$  and  $gf$  are in  $\mathcal{W}$ , then  $f, g, h, hgf \in \mathcal{W}$ . We call arrows in  $\mathcal{W}$  *weak equivalences*, and this property is called the 2-of-6 property. 

**Remark.** Weak equivalences in a model category satisfy the 2-of-6 property, so this construction is more general than model categories. However, the construction is not as general as the 2-of-3 property: if two of  $f, g, gf$  are in  $\mathcal{W}$ , then so is the third. This property required for a category to have weak equivalences, and some use this notion to be precisely the same as that of a homotopical category. We distinguish between them and choose to work in categories which satisfy the stronger 2-of-6 property.

Homotopical categories always include all isomorphisms. This is easily proven. Since  $\mathcal{W}$  is a subcategory including all objects, it equivalently includes all identities. Consider inverse morphisms  $f : a \rightarrow b$  and  $g : b \rightarrow a$ , and consider the composable triple  $(f, g, f)$ . Note  $fg = 1_b \in \mathcal{W}$  and  $gf = 1_a \in \mathcal{W}$ , so we must have  $f, g, f, fgf = f, g \in \mathcal{W}$ . Conversely, every category admits a minimal collection of weak equivalences whereby it may form a homotopical category, given by the set of isomorphisms in  $\mathcal{M}$ . In absence of another obvious notion of weak equivalence, we assume the minimal homotopical structure.

**Definition 2.2** (Homotopy Category). The homotopy category  $\mathrm{Ho} \mathcal{M}$  of a homotopical category  $(\mathcal{M}, \mathcal{W})$  is the formal localization of  $\mathcal{M}$  at  $\mathcal{W}$ . 

An explicit construction can be given by taking  $\mathrm{Obj}(\mathrm{Ho} \mathcal{M}) = \mathrm{Obj}(\mathcal{M})$ , and morphisms to be equivalence classes of zig-zags in  $\mathcal{M}$ <sup>1</sup> with only those in  $\mathcal{W}$  permitted to go backwards, quotiented by the relations

- adjacent arrows pointing in the same direction are equal to their composition;

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<sup>1</sup>Recall zig-zags are chains of morphisms not necessarily in the composable direction, meaning some are permitted to go backwards

- adjacent pairs  $\xleftarrow{w} \xrightarrow{w}$  and  $\xrightarrow{w} \xleftarrow{w}$  may be removed;
- identities may be removed.

This defines for each  $w \in \mathcal{W}$  a formal inverse  $w^{-1}$  which behaves in the expected way. For example, with  $\mathcal{W} \subseteq \mathcal{Top}$  the set of homotopy equivalences, we obtain that for  $fg \sim 1$ ,  $g \in [f^{-1}]$ .

There is a canonical localization functor  $\gamma : \mathcal{M} \rightarrow \mathcal{HoM}$  mapping  $x \mapsto x$  and  $f \mapsto [f]$ , and it is universal with respect to functors which invert weak equivalences. Hence, precomposition with  $\gamma$  is a bijection between functors  $\mathcal{M} \rightarrow \mathcal{N}$  which invert weak equivalences and functors  $\mathcal{HoM} \rightarrow \mathcal{N}$ .

The homotopy category of the minimal homotopical category is isomorphic to the original category. For another example, a *weak homotopy equivalence* in  $\mathcal{Top}$  is a continuous function  $f : X \rightarrow Y$  which is an isomorphism of sets of paths, meaning for  $U : \mathcal{Grpoid} \rightarrow \mathcal{Set}$  the forgetful functor  $\mathcal{G} \mapsto \mathcal{Obj}(\mathcal{G})$ , we have

$$(U\pi_0)(f) : (U\pi_0)(X) \cong (U\pi_0)(Y).$$

Taking weak homotopy equivalences to be weak equivalences, the homotopy category thereof is *equivalent* to the category of CW complexes and homotopy classes of maps.

**Remark.** In general, not all arrows inverted by localization are weak equivalences. A category for which those arrows inverted are precisely weak equivalences is called *saturated*. Quillen showed all model categories are saturated. We can say slightly more.

**Lemma 2.1.** *If  $\mathcal{M}$  is a category with a collection of arrows  $\mathcal{W}$  with a saturated localization, then  $(\mathcal{M}, \mathcal{W})$  is homotopical.*

**Proof.** Saturated means weak equivalences in  $\mathcal{M}$  are precisely isomorphisms in  $\mathcal{HoM}$ . Since isomorphisms satisfy the 2-of-6 property, so must the weak equivalences. ■

Annoyingly, homotopy categories need not be locally small. Quillen proved model categories induce locally small localizations, but in the more general setting it is preferable to avoid working directly in the homotopy category to avoid these issues.

**Definition 2.3 (Homotopical Functor).** A homotopical functor  $F : \mathcal{M} \rightarrow \mathcal{N}$  between homotopical categories is a functor which preserves weak equivalences. ♥

By univeraslity, homotopical functors induce unique functors

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{F} & \mathcal{N} \\ \gamma \downarrow & & \downarrow \delta \\ \mathcal{HoM} & \xrightarrow{F} & \mathcal{HoN} \end{array}$$

Since we often without loss of generality call both functors  $F$ , we say that  $F$  commutes with the localizations.

**Remark.** Let  $\mathcal{N}$  be a minimal homotopic category. Universality of localization  $\gamma : \mathcal{M} \rightarrow \mathcal{HoM}$  here is 2-categorical: natural transformations between *homotopical* functors  $\mathcal{M} \rightarrow \mathcal{N}$  are bijective with natural transformations  $\mathcal{HoM} \rightarrow \mathcal{N}$ . If  $\mathcal{N}$  is not minimal, we must distinguish between  $\mathcal{HoN}$  and  $\mathcal{N}$ . For  $\delta$  the localization functor, each natural transformation  $\alpha$  between functors  $\mathcal{M} \rightarrow \mathcal{N}$  lowers uniquely to a natural transformation  $\delta\alpha$  between functors  $\mathcal{HoM} \rightarrow \mathcal{HoN}$  (since  $\mathcal{HoN}$  inverts weak

equivalences and our functors are homotopical). Although this relation is unique, it is not necessarily bijective, and transformations between functors  $\text{Ho}\mathcal{M} \rightarrow \text{Ho}\mathcal{N}$  can always raise to one between functors  $\mathcal{M} \rightarrow \text{Ho}\mathcal{N}$ , but we may not be able to lift it along  $\delta$ .

**Example.** Let  $F : \mathcal{A} \rightarrow \mathcal{B}$  be a (not necessarily exact) additive functor between abelian categories. Denote by  $\text{Ch}_{\geq 0}(\mathcal{A})$  and  $\text{Ch}_{\geq 0}(\mathcal{B})$  the categories of chain complexes bounded below at 0. The induced functor  $F_{\bullet} : \text{Ch}_{\geq 0}(\mathcal{A}) \rightarrow \text{Ch}_{\geq 0}(\mathcal{B})$  is then homotopical if we take weak equivalences to be *chain complex homotopy equivalences*, but not necessarily if we take them to be quasi-isomorphisms. For example, consider a quasi-isomorphism  $f_{\bullet} : A_{\bullet} \rightarrow B_{\bullet}$  with  $\mathcal{A} = \mathcal{B} = \mathcal{A}b$ , given by

$$\begin{array}{ccccccc} \dots & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\sim} & \mathbb{Z}/4\mathbb{Z} & \xrightarrow{0} & \mathbb{Z}/2\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/4\mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0 \end{array}$$

This is shown to be a quasi-isomorphism by computing the cohomologies. The computation is not difficult, but takes enough space to type out to make it annoying. We can apply the hom-functor  $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$ , and we compute

$$\begin{aligned} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, 0) &= 0, & \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) &= \{0, 1\} = \mathbb{Z}/2\mathbb{Z}, \\ \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}) &= \{0, (0, 1) \mapsto (0, 2)\} = \mathbb{Z}/2\mathbb{Z}. \end{aligned}$$

Hence we obtain

$$\begin{array}{ccccccc} \dots & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\cong} & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{0} & \mathbb{Z}/2\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} 0 \\ & & \downarrow & & \downarrow 0 & & \downarrow 0 \\ \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0. \end{array}$$

The top row is exact, but note that the homology of the bottom complex at any  $\mathbb{Z}/2\mathbb{Z}$  is

$$\frac{\text{Ker}(\mathbb{Z}/2\mathbb{Z} \rightarrow 0)}{\text{Im}(0 \rightarrow \mathbb{Z}/2\mathbb{Z})} = \frac{\mathbb{Z}/2\mathbb{Z}}{0} = \mathbb{Z}/2\mathbb{Z} \neq 0.$$

Hence,  $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, f)_{\bullet}$  is not a quasi-isomorphism, and additive functors need not be homotopic for this class of weak equivalences.

Many other examples show there are interesting non-homotopical functors. One such class is that of *derived functors*, which are in some sense the closest approximation to homotopical.

**Definition 2.4 (Total Derived Functor).** A *total left derived functor*  $\mathbf{L}F$  of some functor  $F : \mathcal{M} \rightarrow \mathcal{N}$  for  $\mathcal{M}, \mathcal{N}$  homotopic is a *right* Kan extension  $\text{Ran}_{\gamma} \delta F$

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{F} & \mathcal{N} \\ \gamma \downarrow & \uparrow & \downarrow \delta \\ \text{Ho}\mathcal{M} & \xrightarrow[\mathbf{L}F]{} & \text{Ho}\mathcal{N} \end{array}$$

with  $\gamma, \delta$  localization functors. Dually, a *right* derived functor  $\mathbf{R}F$  is a *left* Kan extension  $\text{Lan}_{\gamma} \delta F$ . ♡

Universality tells us that total derived functors are equivalently homotopical functors  $\mathcal{M} \rightarrow \text{Ho } \mathcal{N}$ , which are often called “derived functors”. We reserve that terminology for a special case in which this correspondence is a lift along  $\delta$ . Sometimes these are called point-set derived functors.

**Definition 2.5 (Derived Functor).** A *left derived functor* of  $F : \mathcal{M} \rightarrow \mathcal{N}$  is a homotopical functor  $\mathbb{L}F : \mathcal{M} \rightarrow \mathcal{N}$  and a natural transformation  $\lambda : \mathbb{L}F \Rightarrow F$  such that  $\delta \cdot \lambda : \delta \circ \mathbb{L}F \rightarrow \delta \circ F$  is a total left derived functor of  $F$  (in the sense that it is equal to the Kan extension precomposed with  $\gamma$ ). Right derived functors are defined the same. ♥

## 2.2 Derived functors via deformations

**Definition 2.6 (Deformation).** A *left deformation* on a homotopical category  $\mathcal{M}$  consists of an endofunctor  $Q$  and a natural weak equivalence  $q : Q \Rightarrow 1_{\mathcal{M}}$ . Dually, a right deformation is a natural weak equivalence  $q : 1_{\mathcal{M}} \Rightarrow Q$ . ♥

Here, that  $q$  is a natural weak equivalence means that each component  $q_m$  for  $m \in \mathcal{M}$  is itself a weak equivalence. Deformations axiomatize a convenient setting in which derived functors exist, and are constructed rather easily. Necessarily  $Q$  is homotopical. This is easily shown. Let  $w : a \rightarrow b$  be a weak equivalence. We then have that  $q_b \circ Q(w) = w \circ q_a$ . Note that since  $w$  and  $q_a$  are weak equivalences, their composition must also be a weak equivalence by the 2-of-3 property, which is implied by the 2-of-6 property. Hence,  $q_b \circ Q(w)$  is a weak equivalence, and again by the 2-of-3 property we have that  $Q(w)$  is a weak equivalence. Now let  $\mathcal{M}_Q$  be any full subcategory containing the image of  $Q$ . The inclusion  $\mathcal{M}_Q \hookrightarrow \mathcal{M}$  and  $Q$  define an equivalence of categories between  $\text{Ho } \mathcal{M}$  and  $\text{Ho } \mathcal{M}_Q$ . The book calls  $\mathcal{M}_Q$  the subcategory of *cofibrant objects*, and specifies they use the term to mean something different than Quillen’s definition in model categories. I will try to abide by another convention and call it a *left deformation retract*.

**Definition 2.7 (Deformation for Functors).** A *left deformation* for a functor  $F : \mathcal{M} \rightarrow \mathcal{N}$  between homotopical categories is a left deformation  $Q$  for  $\mathcal{M}$  such that  $F$  is homotopical on a left deformation retract  $\mathcal{M}_Q \subseteq \mathcal{M}$ . We say  $F$  is *left deformable* when it admits a left deformation. ♥

In the context of model categories, Ken Brown’s lemma shows that a left Quillen functor (a cocontinuous<sup>2</sup> functor preserving cofibrations and trivial cofibrations) will preserve weak equivalences between cofibrant objects (objects where the unique morphism from the initial object are cofibrations). We prove this lemma later in the text.

As we continue, when we specify a left deformation for a category  $\mathcal{M}$ , we implicitly choose a left deformation retract.

**Theorem 2.1.** *If  $F : \mathcal{M} \rightarrow \mathcal{N}$  has a left deformation  $q : Q \Rightarrow 1_{\mathcal{M}}$ , then  $\mathbb{L}F = F \circ Q$  is a left derived functor of  $F$ .*

**Proof.** Let  $\delta : \mathcal{N} \rightarrow \text{Ho } \mathcal{N}$  the localization. By definition 2.5, we must show  $\delta FQ$  and  $\delta Fq : \delta FQ \Rightarrow \delta F$  form a left Kan extension of  $\delta F$  along  $\gamma$ . Equivalently, we show it satisfies the relevant universal property in  $\text{Fun}(\mathcal{M}, \text{Ho } \mathcal{N})$ . Let  $H : \mathcal{M} \rightarrow \text{Ho } \mathcal{N}$  be homotopical (hence meaning it inverts weak equivalences), and  $\alpha : G \Rightarrow \delta F$  the relevant natural transformation. Since  $G$  is homotopical and  $q$  is a natural weak equivalence,  $Gq : GQ \Rightarrow G$  is a natural isomorphism. Naturality of  $\alpha$

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<sup>2</sup>Conventions appear to vary, this book uses cocontinuous and says some texts require only that (trivial) cofibrations be preserved, but other sources I can find require that it be a left adjoint functor, strengthening the hypothesis.

gives that

$$\delta Fq \circ \alpha Q \circ (Gq)^{-1} = (\delta F \circ \alpha)(q \circ Q) \circ (Gq)^{-1} = \alpha q \circ (Gq)^{-1} = \alpha.$$

Hence,  $\alpha$  factors through  $\delta Fq$ . To prove uniqueness, first note that since  $Q$  is homotopical,  $qQ$  is a natural weak equivalence. Since  $Q$  is a deformation of  $F$ , we have that  $FqQ$  is a natural weak equivalence, and thus  $\delta FqQ$  is a natural isomorphism. Let  $\alpha'$  be another factorization of  $\alpha$  through  $\delta Fq$ . We then obtain

$$\begin{array}{ccc} GQ & \xrightarrow{\alpha'Q} & \delta FQ^2 \\ Gq \downarrow & & \downarrow \delta FQq \\ G & \xrightarrow{\alpha'} & \delta FQ \end{array}$$

This indeed commutes since  $Gq$  is still invertible, so flipping that arrow gives the same equality as above. This indeed also shows the required uniqueness.  $\blacksquare$

Hence for  $r : 1 \Rightarrow R$  a right deformation and  $\mathcal{M}_R$  a right deformation retract, we have also that  $\mathbb{R}F = FR$ .

Denote by  $\mathcal{L}\text{Def}$  the 2-category of homotopical categories equipped with specified left deformations and left deformation retracts, morphisms given by functors that restrict to homotopical functors on the deformation retracts (left deformable functors with respect to the specified deformation), and 2-morphisms as natural transformations. Note here that a *pseudofunctor* is a functor  $\mathcal{Z} \rightarrow \text{Cat}$  which only preserves identities and composition up to coherent natural isomorphism.

**Theorem 2.2.** *There is a pseudofunctor  $\mathbf{L} : \mathcal{L}\text{Def} \rightarrow \text{Cat}$  mapping homotopical categories to their homotopy category, left deformable functors to their total left derived functor, and natural transformations to the derived natural transformations.*

The proof of this theorem (which I omit :3) depends on our construction in theorem 2.1, and it does not hold in generality that the composite of total derived functors is total derived for the composite.

**Definition 2.8 (Deformable Adjunction).** An adjunction  $F \dashv G$  is a *deformable adjunction* if the left adjunct  $F$  is left deformable, and the right adjunct  $G$  is right deformable.  $\heartsuit$

**Theorem 2.3.** *If*

$$\mathcal{M} \begin{array}{c} \xrightarrow{F} \\ \leftarrow \perp \\ \xrightarrow{G} \end{array} \mathcal{N}$$

*is a deformable adjunction, then so is*

$$\mathbf{L}F : \quad \text{Ho } \mathcal{M} \begin{array}{c} \xrightarrow{\quad} \\ \leftarrow \perp \\ \xrightarrow{\quad} \end{array} \text{Ho } \mathcal{N} \quad : \mathbf{R}G.$$

*Furthermore, the total derived adjunction  $\mathbf{L}F \dashv \mathbf{R}G$  is the unique adjunction compatible with the adjunctions in the sense that*

$$\begin{array}{ccc} \mathcal{N}(F(m), n) & \cong & \mathcal{M}(m, G(n)) \\ \delta \downarrow & & \downarrow \gamma \\ \text{Ho } \mathcal{M}(F(m), n) & & \text{Ho } \mathcal{N}(m, G(n)) \\ Fq^* \downarrow & & \downarrow Gr_* \\ \text{Ho } \mathcal{N}(\mathbf{L}F(m), n) & \cong & \text{Ho } \mathcal{M}(m, \mathbf{R}G(n)) \end{array}$$

commutes for each  $m \in \mathcal{M}$  and  $n \in \mathcal{N}$ .

The proofs of this fact are very complex, and was omitted from the text. I am not going to go chase after a proof. Instead, I choose to make sense of the statement itself. The only point of confusion is the precomposition by  $Fq$  and postcomposition by  $Gr$ , but indeed by theorem 2.2 this gives the total derived functor.

**Proposition 2.1.** *The total left derived functor of a left deformable functor is an absolute Kan extension; in particular, a pointwise right Kan extension.*

**Proof.** Let  $(\delta FQ, \delta Fq)$  be the total left derived functor given by the deformation  $(Q, q)$  and theorem 2.2. Let  $H : \text{Ho } \mathcal{N} \rightarrow \mathcal{E}$ . We must show that  $H$  preserves the Kan extension  $\mathbf{L}F$ . To use our construction of  $\mathbf{L}F$ , we must transfer the universal property to  $\mathcal{E}^{\mathcal{M}}$ . Let  $G : \mathcal{M} \rightarrow \mathcal{E}$  homotopical and  $\alpha : G \Rightarrow H\delta F$ . It suffices to show it factors uniquely through  $H\delta Fq$ . Similarly to the last proof,  $\alpha$  factors as the composition

$$G \xrightarrow{(Gq)^{-1}} GQ \xrightarrow{\alpha Q} H\delta FQ \xrightarrow{H\delta Fq} H\delta F.$$

By the same logic as the above proof, this factorization is unique as well. ■

**Exercise.** Let  $F \dashv G$  and let  $\mathbf{L}F$  and  $\mathbf{R}G$  be absolute Kan extensions. Show that  $\mathbf{L}F \dashv \mathbf{R}G$ .

**Proof.** Let  $\eta : 1_{\mathcal{M}} \Rightarrow GF$  be the unit and  $\varepsilon : FG \Rightarrow 1_{\mathcal{N}}$  the counit of the adjunction  $F \dashv G$ . Let  $\gamma : \mathcal{M} \rightarrow \text{Ho } \mathcal{M}$  and  $\delta : \mathcal{N} \rightarrow \text{Ho } \mathcal{N}$  the localization functors. Let  $\beta : \delta G \Rightarrow \mathbf{R}G$  and  $\alpha : \mathbf{L}F \Rightarrow \delta F$  the derived natural transformations. ■

A double category  $\text{Der}$  with objects *deformable categories* (homotopical categories with both left and right deformations, such that either  $R$  preserves  $\mathcal{M}_Q$  or vice versa), horizontal 1-cells right deformable functors preserving fibrant objects, vertical 1-cells left deformable functors preserving cofibrant objects, and 2-cells appropriate natural transformations. There is then a double pseudofunctor  $\mathcal{D} \rightarrow \text{Cat}$  given by  $\mathcal{C} \mapsto \text{Ho } \mathcal{C}$  and  $F \mapsto \mathbf{L}F$ ; here  $\text{Cat}$  is the double category of categories, functors, and natural transformations. No way in hell I'm proving that.

### 2.3 Classical derived functors between abelian categories

Recall that  $\text{Mod}_R$  has enough projectives, meaning for any  $A$  there is a bounded below complex of projectives  $P_{\bullet}$  quasi-isomorphic to  $\iota(A)$  (a projective resolution of  $A$ ). Careful choice of projective resolutions is functorial  $Q : \text{Ch}_{\geq 0}(R) \rightarrow \text{Ch}_{\geq 0}(R)$  with a natural quasi-isomorphism  $q : Q \Rightarrow 1$ . Dually, modules admit injective resolutions which give a right deformation  $r : 1 \Rightarrow (R : \text{Ch}^{\geq 0}(R) \rightarrow \text{Ch}^{\geq 0}(R))$  on the category of bounded below cochain complexes.

Additive functors  $F : \text{Mod}_R \rightarrow \text{Mod}_S$  induce functors  $F_{\bullet} : \text{Ch}_{\geq 0}(R) \rightarrow \text{Ch}_{\geq 0}(S)$  and  $F^{\bullet} : \text{Ch}^{\leq 0}(R) \rightarrow \text{Ch}^{\leq 0}(S)$  preserving chain homotopies. When working with projective/injective resolutions, the quasi-isomorphisms become homotopy equivalences, and are thus preserved by  $F_{\bullet}$  and  $F^{\bullet}$ . Hence, with respect to quasi-isomorphisms,  $F^{\bullet}$  has a right derived functor and  $F_{\bullet}$  has a left derived functor, as described herein. The functor that we classically refer to as the left derived functor of  $F : \text{Mod}_R \rightarrow \text{Mod}_S$  in homological algebra is the composition

$$\text{Mod}_R \xrightarrow{\iota} \text{Ch}_{\geq 0}(R) \xrightarrow{\mathbb{L}F_{\bullet}} \text{Ch}_{\geq 0}(S) \xrightarrow{H_0} \text{Mod}_S.$$

One can also take the other derived functors, but the derivations we chose induce long exact sequences,

so we prefer those.

## 2.4 Preview of homotopy limits and colimits

Let  $\mathcal{M}$  a complete and cocomplete homotopical category, and  $\mathcal{D}$  a category indexing some small diagram. The colimit and limit functors  $\varinjlim, \varprojlim : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$  need not be homotopical with respect to pointwise weak equivalences in  $\mathcal{M}^{\mathcal{D}}$  (I assume meaning natural weak equivalences). However, since the functors are continuous, their respective Kan extensions exist, so we would want to define a left derived functor of  $\varinjlim$  and a right derived functor of  $\varprojlim$ , called hocolim and holim, respectively. Unfortunately, even if  $\mathcal{M}$  is a model category,  $\mathcal{M}^{\mathcal{D}}$  may not have a model structure such that  $\varinjlim$  and  $\varprojlim$  are Quillen. Hence the theory of deformations exists for construction derived functors without invoking model structures.



## Chapter 3

### Basic concepts of enriched category theory

There are many reasons to consider enrichments, but one notable one for this text is the desire for a “homotopy product”, which would be a product diagram that only commutes *up to homotopy*. In order to have some notion of homotopy for which this makes sense, we can assume hom-sets are equipped with topologies, and then homotopies between arrows would become paths in the underlying space. A homotopy product can then be equipped with a natural weak homotopy equivalence.

Since we have not defined a class of weak equivalences, the homotopy category  $h\mathcal{M}$  is different from the ones we worked with earlier. Denoting by  $\underline{\mathcal{M}}(a, b)$  the space assigned to  $\mathcal{M}(a, b)$ , we define  $h\mathcal{M}(a, b) := \pi_0 \underline{\mathcal{M}}(a, b)$ . Here note that  $\pi_0$  is the functor  $\mathcal{T}\text{op} \rightarrow \mathcal{S}\text{et}$  assigning to each space its path components; hence taking a space  $X$  and quotienting in  $\mathcal{S}\text{et}$  via the relation  $x \sim y$  iff there is a path from  $x$  to  $y$ .

#### 3.1 A first example

Let  $\text{Mod}_R$  be the category of left  $R$ -modules. We well know that the set  $\text{Mod}_R(A, B)$  admits an abelian group structure via pointwise addition, which we will denote by  $\underline{\text{Mod}}_R(A, B)$ . Moreover, composition

$$\circ : \text{Mod}_R(B, C) \times \text{Mod}_R(A, B) \rightarrow \text{Mod}_R(A, C)$$

distributes over addition, and hence is  $\mathbb{Z}$ -bilinear. Thus it factors through  $\otimes_{\mathbb{Z}}$ , inducing an abelian group homomorphism

$$\underline{\text{Mod}}_R(B, C) \otimes_{\mathbb{Z}} \underline{\text{Mod}}_R(A, B) \rightarrow \underline{\text{Mod}}_R(A, C)$$

in the category  $\mathcal{A}\text{b}$ .

Note also that since any abelian group  $a$  has  $\mathcal{A}\text{b}(\mathbb{Z}, a) \cong a$ , we can represent the identity morphism of  $A$  as an abelian group homomorphism

$$1_A : \mathbb{Z} \rightarrow \underline{\text{Mod}}_R(A, A).$$

Moreover,  $\mathbb{Z}$  is a unit for the monoidal product  $\otimes_{\mathbb{Z}} : \mathcal{A}\text{b} \times \mathcal{A}\text{b} \rightarrow \mathcal{A}\text{b}$  via left and right natural isomorphisms. This allows us to encode the usual composition axioms via diagrams in  $\mathcal{A}\text{b}$ . These facts show that  $\text{Mod}_R$  admits the structure of a category enriched over  $(\mathcal{A}\text{b}, \otimes_{\mathbb{Z}}, \mathbb{Z})$ .

#### 3.2 The base for enrichment

A symmetric monoidal category  $(\mathcal{V}, \times, *)$  is a category with a bifunctor  $-\times- : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$  called the monoidal product, and an element  $*$   $\in \mathcal{V}$  called the unit, together with specified natural isomorphisms

$$v \times w \cong w \times v, \quad u \times (v \times w) \cong (u \times v) \times w, \quad * \times v \cong v \cong v \times *$$

satisfying coherence criteria. Any category with finite products is symmetric monoidal with terminal object the unit; we call a category with such a monoidal structure *cartesian monoidal*. In most relevant examples,  $\mathcal{V}$  is complete and cocomplete, so we will generally assume this.

### 3.3 Enriched categories

**Definition 3.1** (Enriched Category). Let  $\mathcal{V}$  be a symmetric<sup>1</sup> monoidal category. A  $\mathcal{V}$ -category  $\underline{\mathcal{D}}$  is

- A collection of objects  $x, y, z \in \underline{\mathcal{D}}$ ;
- For each pair  $x, y \in \underline{\mathcal{D}}$ , a hom-object  $\underline{\mathcal{D}}(x, y) \in \mathcal{V}$ ;
- For each  $x \in \underline{\mathcal{D}}$ , a morphism  $\text{id}_x : * \rightarrow \underline{\mathcal{D}}(x, x)$  in  $\mathcal{V}$ ;
- For each triple  $x, y, z \in \underline{\mathcal{D}}$ , a morphism  $\circ : \underline{\mathcal{D}}(y, z) \otimes \underline{\mathcal{D}}(x, y) \rightarrow \underline{\mathcal{D}}(x, z)$  in  $\mathcal{V}$ ;

such that for all  $w, x, y, z \in \underline{\mathcal{D}}$ , the diagrams

$$\begin{array}{ccc}
 \underline{\mathcal{D}}(z, y) \otimes \underline{\mathcal{D}}(x, y) \otimes \underline{\mathcal{D}}(w, x) & \xrightarrow{1 \otimes \circ} & \underline{\mathcal{D}}(z, y) \otimes \underline{\mathcal{D}}(x, z) \\
 \downarrow \circ \otimes 1 & & \downarrow \circ \\
 \underline{\mathcal{D}}(y, w) \otimes \underline{\mathcal{D}}(w, x) & \xrightarrow{\circ} & \underline{\mathcal{D}}(w, z)
 \end{array}$$
  

$$\begin{array}{ccc}
 \underline{\mathcal{D}}(x, y) \otimes * & \xrightarrow{1 \otimes \text{id}_x} & \underline{\mathcal{D}}(x, y) \otimes \underline{\mathcal{D}}(x, x) \\
 \searrow \cong & & \downarrow \circ \\
 & & \underline{\mathcal{D}}(x, y)
 \end{array}
 \qquad
 \begin{array}{ccc}
 \underline{\mathcal{D}}(y, y) \otimes \underline{\mathcal{D}}(x, y) & \xleftarrow{\text{id}_y \otimes 1} & * \otimes \underline{\mathcal{D}}(x, y) \\
 \downarrow \circ & & \swarrow \cong \\
 \underline{\mathcal{D}}(x, y) & & 
 \end{array}$$

commute in  $\mathcal{V}$ . ♥

We omit a relevant associativity morphism in the first diagram because the text did it too and I only realized after copying it all down and I'm NOT editing it.

We already know of a plethora of examples. For example, a ring is an  $\mathcal{Ab}$ -category with a singleton for objects. A more interesting example is that topological spaces are *naturally* enriched over groupoids. Define  $\text{Top}(X, Y)$  to be the groupoid with objects as continuous maps  $X \rightarrow Y$  and morphisms as homotopy classes of homotopies between these maps.

If  $\mathcal{V}$  has copowers of the unit that are preserved by tensor in the following sense, then any unenriched  $\mathcal{C}$  has a free  $\mathcal{V}$ -category with objects the same as  $\mathcal{C}$ . Hom-objects from  $a$  to  $b$  are given by the copower  $\coprod_{\mathcal{C}(a, b)} *$ . Identity morphisms  $\text{id}_a$  are given by the inclusion of  $*$  into the copower  $\coprod_{\mathcal{C}(a, a)} *$  at the identity  $1_a$ . Since tensor preserves the relevant copower, we obtain

$$\left( \coprod_{\mathcal{C}(b, c)} * \right) \otimes \left( \coprod_{\mathcal{C}(a, b)} * \right) \cong \coprod_{\mathcal{C}(b, c)} \left( * \otimes \left( \coprod_{\mathcal{C}(a, b)} * \right) \right) \cong \coprod_{\mathcal{C}(b, c)} \left( \coprod_{\mathcal{C}(a, b)} (* \otimes *) \right) \cong \coprod_{\mathcal{C}(b, c)} \coprod_{\mathcal{C}(a, b)} * \cong \coprod_{\mathcal{C}(b, c) \times \mathcal{C}(a, b)} *.$$

Since there is a composition function  $\mathcal{C}(b, c) \times \mathcal{C}(a, b)$ , composition in the free  $\mathcal{V}$ -category simply reindexes along this arrow.

<sup>1</sup>We are using symmetric monoidal categories here, but the symmetric hypothesis can be dropped for general enrichments. We require it for some results, however. I will assume we only enrich over symmetric monoidal categories unless the text specifies otherwise.

**Definition 3.2** (Closed Monoidal Category). Let  $\mathcal{V}$  be a symmetric monoidal category. If each functor  $- \otimes v : \mathcal{V} \rightarrow \mathcal{V}$  has a right adjoint, denoted  $\underline{\mathcal{V}}(v, -)$ , the right adjoints assemble uniquely into a bifunctor

$$\underline{\mathcal{V}}(-, -) : \mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$$

such that for all  $u, v, w \in \mathcal{V}$ ,

$$\mathcal{V}(u \times v, w) \cong \mathcal{V}(u, \underline{\mathcal{V}}(v, w))$$

natural in  $u, v, w$ .

We call  $\underline{\mathcal{V}}(-, -)$  the *internal hom*, and  $\mathcal{V}$  is enriched over itself with composition

$$\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v) \rightarrow \underline{\mathcal{V}}(u, w)$$

given as follows. Note the adjunction  $- \otimes u \dashv \underline{\mathcal{V}}(u, -)$  gives the counit

$$\varepsilon^u : (- \otimes u) \circ \underline{\mathcal{V}}(u, -) = \underline{\mathcal{V}}(u, -) \otimes u \Rightarrow 1.$$

Thus the adjunctions give a composition

$$\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v) \otimes u \xrightarrow{1 \otimes \varepsilon_v^u} \underline{\mathcal{V}}(v, w) \otimes v \xrightarrow{\varepsilon_w^v} w.$$

Hence we have

$$\varepsilon_w^v \circ (\varepsilon_v^u \otimes 1) \in \mathcal{V}(\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v) \otimes u, w) \cong \mathcal{V}(\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v), \underline{\mathcal{V}}(u, w)).$$

The isomorphism follows by applying symmetry to the adjunction. We thus define composition as the adjunct of the relevant composition. The identity morphisms are defined as follows. The natural isomorphism  $\lambda_v : * \otimes v \cong v$  is adjunct under the same adjunction to the identity  $\text{id}_v$ :

$$\lambda \in \mathcal{V}(* \otimes v, v) \cong \mathcal{V}(*, \underline{\mathcal{V}}(v, v)).$$

Such a triple  $(\mathcal{V}, \otimes, *)$  is a *closed symmetric monoidal category*. ♥

When  $\mathcal{V}$  is cartesian monoidal, we call it *cartesian closed*. An example is  $\mathbf{sSet}$ . The product is the usual product of functors, and we desire an adjunction

$$\mathbf{sSet}(X \times Y, Z) \cong \mathbf{sSet}(X, Z^Y).$$

Here, we use exponential notation  $\mathbf{sSet}(X, Y) =: Y^X$ . If  $X$  is representable  $\Delta(-, [n])$ , which we write as  $\Delta^n$ , then Yoneda gives

$$(Z^Y)_n \cong \mathbf{sSet}(\Delta^n, Z^Y) \cong \mathbf{sSet}(\Delta^n \times Y, Z).$$

Hence we obtain a valid definition of  $Z^Y$  by setting the  $n$ -simplices as the set of maps  $\Delta^n \times Y \rightarrow Z$ . Hence  $- \times Y \dashv (-)^Y$  for each simplicial set  $Y$ , giving that  $\mathbf{sSet}$  is a closed cartesian monoidal category.

The category  $\mathbf{Cat}$  is cartesian closed, since functors and natural transformations between 2 fixed categories form another category. The precise definition of a 2-category is a category enriched over  $\mathbf{Cat}$ . Objects are 0-cells, the 1-cells are the objects of the hom-categories, and the 2-cells are the morphisms in the hom-categories. This interpretation gives us a notion of arrows in the enriched category, compatible with the notion of composition given in definition 3.2. Equivalently, we can take 1-cells to be the arrows of the *underlying category*, defined later.

The category  $\mathcal{T}\text{op}$  is *not* cartesian closed, but several large convenient subcategories are, such as  $\mathcal{C}\mathcal{G}\mathcal{H}\text{aus}$ . For now,  $\mathcal{T}\text{op}$  will denote one of these convenient categories, and we will make precise which categories we concern ourselves with later in the text.

We provide the following construction. Let  $(\mathcal{V}, \times, *)$  be a cartesian closed symmetric monoidal category, which we also assume to be complete and cocomplete. There is a general procedure for producing a closed symmetric monoidal category  $(\mathcal{V}_*, \wedge, S^0)$  of *pointed objects* or *based objects*, equivalently the slice category  $(* \downarrow \mathcal{V}) =: \mathcal{V}_*$ . We define  $S^0 := * \amalg *$  (the coproduct in  $\mathcal{V}$ ). Note that for each  $v, w$ , we can define a map  $v \rightarrow w$  by factoring through the terminal object  $v \rightarrow * \rightarrow w$  via the basepoint on  $w$ . This gives maps  $v, w \rightarrow v \times w$ , and universality of coproduct gives a canonical map  $f : v \amalg w \rightarrow v \times w$ . We then define the *smash product*  $\wedge$  to be the pushout in  $\mathcal{V}$ :

$$\begin{array}{ccc} v \amalg w & \xrightarrow{f} & v \times w \\ \downarrow & & \downarrow \\ * & \longrightarrow & v \wedge w \\ & \searrow & \downarrow \\ & & z. \end{array}$$

We then define the hom-objects  $\underline{\mathcal{V}}_*$  in  $\mathcal{V}_*$  to be pullbacks of internal homs in  $\mathcal{V}$ :

$$\begin{array}{ccc} & & z \\ & \searrow & \downarrow \\ & & \underline{\mathcal{V}}_*(v, w) \longrightarrow \underline{\mathcal{V}}(v, w) \\ & \downarrow & \downarrow \\ & & \underline{\mathcal{V}}(*, *) \longrightarrow \underline{\mathcal{V}}(*, w). \end{array}$$

The bottom map is postcomposition by the base map  $* \rightarrow w$ , and the downwards map is precomposition by the basepoint  $* \rightarrow v$ . This does not have a precise definition yet, but intuitively it makes sense, since hom-objects in cartesian monoidal categories are intuitively similar to hom-sets. There is then a forgetful functor  $U : \mathcal{V}_* \rightarrow \mathcal{V}$  forgetting the basepoint, which is left adjoint to the functor  $(-)_+ : \mathcal{V} \rightarrow \mathcal{V}_+$  given by computing the coproduct with the terminal object.

**Lemma 3.1.** *When  $\mathcal{V}$  is a cartesian closed symmetric monoidal category, the functor  $(-)_+ : \mathcal{V} \rightarrow \mathcal{V}_*$  is strong monoidal, meaning*

$$S^0 \cong (*)_+, \quad v_+ \wedge w_+ \cong (v \times w)_+$$

*with these isomorphisms natural, and appropriately commuting with associators and unitors.*

**Proof.** The unit isomorphism is by definition of  $S^0$ . By definition, the object  $v_+ \wedge w_+$  is given by the cofiber (the opposite arrow in the pushforward diagram) of

$$v_+ \amalg w_+ \rightarrow v_+ \times w_+ = (v \amalg *) \times (w \amalg *).$$

Since  $\mathcal{V}$  is cartesian closed and symmetric,  $v \times -$  and  $- \times v$  are left-adjoint and thus preserve

colimits, and thus preserve coproducts. Thus,  $\times$  distributes over  $\amalg$ , and we obtain

$$(v \amalg *) \times (w \amalg *) \cong (v \times (w \amalg *)) \amalg (* \times (w \amalg *)) \cong ((v \times w) \amalg (v \times *)) \amalg ((* \times w) \amalg (* \times *)).$$

Dropping the unnecessary parentheses due to associativity of  $\amalg$ , we also have

$$(v \times w) \amalg (v \times *) \amalg (* \times w) \amalg (* \times *) \cong (v \times w) \amalg v \amalg w \amalg *$$

since the terminal object  $*$  is a unit for  $\times$ . Universality of coproduct gives an arrow

$$v_+ \amalg w_+ \rightarrow v \amalg w \amalg *$$

which the canonical arrow  $v_+ \amalg w_+ \rightarrow v_+ \times w_+$  factors through by uniqueness. Hence we obtain a diagram

$$\begin{array}{ccccc} v_+ \amalg w_+ & \twoheadrightarrow & v \amalg w \amalg * & \longrightarrow & v \times w \amalg v \amalg w \amalg * \\ \downarrow & & & & \downarrow \\ * & \longrightarrow & * & \longrightarrow & v \times w \amalg * = (v \times w)_+ \end{array}$$

since the canonical map doesn't act on  $v \times w$ , which is a double pushout, since the top left arrow is epi. ■

A convenient category  $\mathcal{T}\text{op}$  of spaces gives a closed symmetric monoidal category  $(\mathcal{T}\text{op}_*, \wedge, S^0)$ . So does  $(\mathbf{sSet}_*, \wedge, \partial\Delta^1)$ .

### 3.4 Underlying categories of enriched categories

Here is some motivation which I'm copying down less for the motivation, and more because I think the new concepts are cool. Fix a group  $G$  to which the discrete topology has been associated, and take  $\mathcal{T}\text{op}$  some well-behaved category of topological spaces. The functor category  $\mathcal{T}\text{op}^G$  has as objects  $G$ -spaces and as morphisms  $G$ -equivariant maps. Unpacking this quickly, a functor is a group action of  $G$  on a topological space, meaning a choice of continuous automorphisms of a space that respect the group operation. Morphisms of  $G$ -spaces are natural transformations, but since groups are singletons, this amounts to a single continuous map that preserves the group action, called  $G$ -equivariant.

This category admits two natural enrichments. First, the set of  $G$ -equivariant maps can be viewed as a subspace  $\underline{\mathcal{T}\text{op}}^G(X, Y) \subseteq \underline{\mathcal{T}\text{op}}(X, Y)$ , where the latter has the *compact-open topology* generated by sets

$$V(K, U) := \{f \in \mathcal{T}\text{op}(X, Y) \mid f(K) \subseteq U\}, \quad K \subseteq X \text{ compact}, \quad U \subseteq Y \text{ open}.$$

This gives a topological enrichment.

Additionally,  $\mathcal{T}\text{op}^G$  is symmetric cartesian monoidal, with product given by the standard cartesian product together with the diagonal action  $(g, h)(x, y) = (gx, hy)$ . The singleton with the trivial action is a unit thereof. The space  $\underline{\mathcal{T}\text{op}}(X, Y)$  has a canonical  $G$ -action

$$g \cdot f(x) = gf(g^{-1}x).$$

This gives an enrichment over  $\mathcal{T}\text{op}^G$ , and we write  $\underline{\mathcal{T}\text{op}}_G(X, Y)$  for this space to distinguish from the other enrichment. With this definition  $\mathcal{T}\text{op}^G$  is cartesian closed.

We think of  $\mathcal{T}op^G$  as the underlying (unenriched) category for both categories, but this intuition is difficult to formalize in full generality. However, generality in this case helps extract the full definition.

To define an underlying category, we first need a way to construct underlying sets, and hence a functor  $\mathcal{V} \rightarrow \mathbf{Set}$ . For example, in  $\mathcal{A}b$ ,  $\mathbf{sSet}$ ,  $\mathcal{T}op$ ,  $\mathcal{C}at$ , and more, the underlying set functor is simply the representable functor  $\mathcal{V}(*, -) : \mathcal{V} \rightarrow \mathbf{Set}$ , since  $*$  is a singleton. Since this is the unit for the cartesian monoidal structures of these categories, we might generalize this to any symmetric monoidal category with  $*$  as the unit. This is on the right track, but we need a bit more.

**Definition 3.3 (Lax Monoidal).** A functor  $F : \mathcal{V} \rightarrow \mathcal{U}$  between symmetric monoidal categories  $(\mathcal{V}, \otimes, *)$  and  $(\mathcal{U}, \square, 1)$  is *lax monoidal* if there are natural transformations (NOT ISOMORPHISMS)

$$F(v) \otimes F(v') \rightarrow F(v \otimes v'), \quad 1 \rightarrow F(*).$$

♡

Clearly all strong monoidal functors are lax, with natural transformations as isomorphisms.

**Lemma 3.2 (Change of Base).** *Let  $F : \mathcal{V} \rightarrow \mathcal{U}$  be lax monoidal. Then any  $\mathcal{V}$ -category has an associated  $\mathcal{U}$ -category with the same objects.*

**Proof.** Let  $\underline{\mathcal{D}}_{\mathcal{V}}$  be a  $\mathcal{V}$ -category. Define a  $\mathcal{U}$ -category  $\underline{\mathcal{D}}_{\mathcal{U}}$  with the same objects, and with hom-objects  $\underline{\mathcal{D}}_{\mathcal{U}}(x, y) := F\underline{\mathcal{D}}_{\mathcal{V}}(x, y)$ . The lax monoidal structure of  $F$  gives composition and unit maps

$$\begin{aligned} F\underline{\mathcal{D}}_{\mathcal{V}}(y, z) \otimes F\underline{\mathcal{D}}_{\mathcal{V}}(x, y) &\rightarrow F(\underline{\mathcal{D}}_{\mathcal{V}}(y, z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x, y)) \xrightarrow{F(\circ)} F\underline{\mathcal{D}}_{\mathcal{V}}(x, z) = \underline{\mathcal{D}}_{\mathcal{U}}(x, z); \\ 1 &\rightarrow F(*) \xrightarrow{F(\text{id}_x)} F\underline{\mathcal{D}}_{\mathcal{V}}(x, x) = \underline{\mathcal{D}}_{\mathcal{U}}(x, x). \end{aligned}$$

■

The point is that  $\mathcal{V}(*, -)$  is lax monoidal. Let  $v, w \in \mathcal{V}$ . Then the bifunctor  $\otimes$  gives for any pair  $* \rightarrow v$  and  $* \rightarrow w$  a map  $* \otimes * \rightarrow v \otimes w$ . Precomposing with the  $\otimes$ -diagonal map, indeed an isomorphism here, gives rise to a function

$$\mathcal{V}(*, v) \times \mathcal{V}(*, w) \rightarrow \mathcal{V}(*, v \otimes w)$$

natural in  $v$  and  $w$ , and hence  $\mathcal{V}(*, -)$  is lax functorial. We then apply lemma 3.2 to change the base for the enrichment from  $\mathcal{V}$  to  $\mathbf{Set}$ .

**Definition 3.4 (Underlying Category).** Given a  $\mathcal{V}$ -category  $\underline{\mathcal{C}}$ , the underlying category  $\mathcal{C}_0$  has the same objects, and as hom-sets

$$\mathcal{C}_0(x, y) := \mathcal{V}(*, \underline{\mathcal{C}}(x, y)).$$

Definitions are as in lemma 3.2:  $1_x = \text{id}_x \in \mathcal{V}(*, \underline{\mathcal{C}}(x, x))$  and

$$\begin{array}{ccc} \mathcal{C}_0(y, z) \times \mathcal{C}_0(x, y) & \xrightarrow{\quad \circ \quad} & \mathcal{C}_0(x, z) \\ \downarrow & & \downarrow \\ \mathcal{V}(*, \underline{\mathcal{C}}(y, z)) \times \mathcal{V}(*, \underline{\mathcal{C}}(x, y)) & \xrightarrow[\otimes]{} \mathcal{V}(*, \underline{\mathcal{C}}(y, z) \otimes \underline{\mathcal{C}}(x, y)) \xrightarrow[\mathcal{V}(*, \circ)]{} & \mathcal{V}(*, \underline{\mathcal{C}}(x, z)). \end{array}$$

♡

Reconciling our previous example, first consider the  $\mathcal{T}\text{op}$ -enriched category  $\mathcal{T}\text{op}^G$ . The hom-sets of the underlying category are  $\mathcal{T}\text{op}(*, \mathcal{T}\text{op}^G(X, Y))$ , equivalently the set of points in  $\mathcal{T}\text{op}^G(X, Y)$ , equivalently all  $G$ -equivariant maps  $X \rightarrow Y$ . Hence the underlying category is  $\mathcal{T}\text{op}^G$ . Similarly, the hom-sets of the underlying category of the  $\mathcal{T}\text{op}^G$ -enriched category  $\mathcal{T}\text{op}_G$  are given by  $\mathcal{T}\text{op}^G(*, \mathcal{T}\text{op}_G(X, Y))$ . Since  $*$  is the trivial group action,  $G$ -equivariant maps  $* \rightarrow \mathcal{T}\text{op}_G(X, Y)$  are  $G$ -fixed points. Note that fixed points for the conjugation action are precisely  $G$ -equivariant maps:

$$g \cdot f(x) = gf(g^{-1}x) = gg^{-1}f(x) = f(x).$$

Hence, the underlying category is once again  $\mathcal{T}\text{op}^G$ .

These considerations with group actions generalize. For  $\mathcal{V}$  a complete and cocomplete closed symmetric monoidal category,  $\mathcal{V}^G$  inherits a canonical closed symmetrical monoidal structure such that the forgetful functor  $U : \mathcal{V}^G \rightarrow \mathcal{V}$  preserves the unit, tensor, and internal homs. The hom-object  $\underline{\mathcal{V}}^G(c, d)$  agrees with  $\underline{\mathcal{V}}(c, d)$ , and the underlying category is again the original  $\mathcal{V}^G$ . The discussion also generalizes in another way.

**Lemma 3.3.** *If  $(V, \otimes, *)$  is a closed symmetric monoidal category, the underlying category of the  $\mathcal{V}$ -category  $\underline{\mathcal{V}}$  is  $\mathcal{V}$ .*

**Proof.** Internal homs give the adjunction

$$\mathcal{V}(*, \underline{\mathcal{V}}(x, y)) \cong \mathcal{V}(* \otimes x, y) \cong \mathcal{V}(x, y),$$

with the second isomorphism the unitor. Hence, hom-sets of the underlying category agree with  $\mathcal{V}$ . We now need to show identities and composition agree. The first one is handled by functoriality of  $\mathcal{V}(*, -)$ , so now we need only show composition agrees.

Using this above isomorphism, we take  $f : x \rightarrow y$  in  $\mathcal{V}$  and the morphism  $f : * \rightarrow \underline{\mathcal{V}}(x, y)$  to be equivalent, hence we use the same label. Let  $\varepsilon_y^x : \underline{\mathcal{V}}(x, y) \otimes x \rightarrow y$  be the relevant counit of the relevant adjunction. Following the isomorphism gives a precise definition of  $f : x \rightarrow y$  in terms of  $f : * \rightarrow \underline{\mathcal{V}}(x, y)$  as the composition

$$x \cong * \otimes x \xrightarrow{f \otimes 1} \underline{\mathcal{V}}(x, y) \otimes x \xrightarrow{\varepsilon_y^x} y.$$

Now we follow composition as in definition 3.4. The composition in  $\underline{\mathcal{V}}$  is defined as

$$* \cong * \otimes * \xrightarrow{g \otimes f} \underline{\mathcal{V}}(y, z) \otimes \underline{\mathcal{V}}(x, y) \xrightarrow{\circ} \underline{\mathcal{V}}(x, z).$$

Recall composition was adjunct to

$$\underline{\mathcal{V}}(y, z) \otimes \underline{\mathcal{V}}(x, y) \otimes x \xrightarrow{1 \otimes \varepsilon_y^x} \underline{\mathcal{V}}(y, z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

Naturality and precomposition with a unitor gives that our definition of composition in its entirety is adjunct to

$$* \otimes x \cong * \otimes * \otimes x \xrightarrow{g \otimes f \otimes 1} \underline{\mathcal{V}}(y, z) \otimes \underline{\mathcal{V}}(x, y) \otimes x \xrightarrow{1 \otimes \varepsilon_y^x} \underline{\mathcal{V}}(y, z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

Functoriality and another unitor gives

$$x \cong * \otimes * \otimes x \xrightarrow{g \otimes 1 \otimes 1} \underline{\mathcal{V}}(y, z) \otimes * \otimes x \xrightarrow{1 \otimes f \otimes 1} \underline{\mathcal{V}}(y, z) \otimes \underline{\mathcal{V}}(x, y) \otimes x \xrightarrow{1 \otimes \varepsilon_y^x} \underline{\mathcal{V}}(y, z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

By functoriality of tensor, we can replace the part of the composition

$$* \otimes x \xrightarrow{f \otimes 1} \underline{\mathcal{V}}(x, y) \otimes x \xrightarrow{\varepsilon_y^x} y$$

with the definition of  $f : x \rightarrow y$  in  $\mathcal{V}$ :

$$x \cong * \otimes x \xrightarrow{g \otimes 1} \underline{\mathcal{V}}(y, z) \otimes x \xrightarrow{1 \otimes f} \underline{\mathcal{V}}(y, z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

Functoriality of  $\otimes$  gives that this composition is equivalent to

$$x \xrightarrow{f} y \cong * \otimes y \xrightarrow{g \otimes 1} \underline{\mathcal{V}}(y, z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

The tail of this composition is precisely the definition of  $g$  as an arrow in the underlying category, hence this is the composition  $g \circ f$ . ■

**Exercise.** Let  $\underline{\mathcal{D}}$  be a  $\mathcal{V}$ -category and  $g : * \rightarrow \underline{\mathcal{D}}(y, z)$  an arrow in  $\mathcal{D}_0$ . Define for any  $x \in \underline{\mathcal{D}}$  postcomposition:

$$g_* : \underline{\mathcal{D}}(x, y) \cong * \otimes \underline{\mathcal{D}}(x, y) \xrightarrow{g \otimes 1} \underline{\mathcal{D}}(y, z) \otimes \underline{\mathcal{D}}(x, y) \xrightarrow{\circ} \underline{\mathcal{D}}(x, z).$$

Precomposition is defined similarly. This definition gives an (unenriched) functor

$$\underline{\mathcal{D}}(x, -) : \mathcal{D}_0 \rightarrow \mathcal{V}, \quad y \mapsto \underline{\mathcal{D}}(x, y), \quad g \mapsto g_*.$$

Show that the composition with the underlying set functor  $\mathcal{V}(*, -) \circ \underline{\mathcal{D}}(x, -)$  is the representable functor  $\mathcal{D}_0(x, -) : \mathcal{D}_0 \rightarrow \mathbf{Set}$ .

**Proof.** By definition 3.4,  $\mathcal{V}(*, \underline{\mathcal{D}}(x, -)) =: \mathcal{D}_0(x, -)$ . Hence, the definitions agree on objects. We also have that  $\mathcal{D}_0(x, g) = g_*$  in the conventional sense of the notation, and by definition  $\underline{\mathcal{D}}(x, g) = g_*$  in the sense of the notation as above. We also have that for a morphism  $f$  in  $\mathcal{V}$ ,  $\mathcal{V}(x, f)$  is also precomposition in the conventional sense. Note that  $\mathcal{V}(*, \underline{\mathcal{D}}(x, g)) = \mathcal{V}(*, g_*)$ .

Let  $f : x \rightarrow y$  in  $\mathcal{D}_0$ , and equivalently  $f : * \rightarrow \underline{\mathcal{D}}(x, y)$  in  $\mathcal{V}$ . We must show that evaluating  $\mathcal{V}(*, g_*)(f)$  is precisely the same as  $\mathcal{D}_0(x, g)(f)$ . Since  $g_*$  is defined as an arrow in  $\mathcal{V}$ , the former is given as  $g_* \circ f$  with composition as defined for  $\mathcal{V}$ . The latter is given as  $g \circ f$ , with composition as defined in  $\mathcal{D}_0$ . Writing out the composition  $g_* \circ f$  as defined above gives the composition

$$* \xrightarrow{f} \underline{\mathcal{D}}(x, y) \cong * \otimes \underline{\mathcal{D}}(x, y) \xrightarrow{g \otimes 1} \underline{\mathcal{D}}(y, z) \otimes \underline{\mathcal{D}}(x, y) \xrightarrow{\circ} \underline{\mathcal{D}}(x, z)$$

with composition as in  $\underline{\mathcal{D}}$ . Functoriality simplifies this to

$$* \cong * \otimes * \xrightarrow{g \otimes f} \underline{\mathcal{D}}(y, z) \otimes \underline{\mathcal{D}}(x, y) \xrightarrow{\circ} \underline{\mathcal{D}}(x, z).$$



By definition 3.4, this is precisely the same as the definition of  $g \circ f$  in  $\mathcal{D}_0$ . Thus,  $\mathcal{V}(*, \underline{\mathcal{D}}(x, -)) = \mathcal{D}_0(x, -)$ . ■

Hence we have a good notion of precomposition and postcomposition. Importantly, pre- and postcomposition are continuous in topological categories, but we can indeed show more. These representable functors are  $\mathcal{V}$ -functors.

### 3.5 Enriched functors and enriched natural transformations

Small  $\mathcal{V}$ -categories can form a 2-category (if  $\mathcal{V}$  is allowed to be symmetric). To make precise this notion we require the notion of a  $\mathcal{V}$ -functor.

**Definition 3.5 ( $\mathcal{V}$ -Functor).** A  $\mathcal{V}$ -functor  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$  between  $\mathcal{V}$ -categories is an object map  $x \mapsto F(x)$  from  $\underline{\mathcal{C}}$  to  $\underline{\mathcal{D}}$ , together with morphisms

$$F_{x,y} : \underline{\mathcal{C}}(x, y) \rightarrow \underline{\mathcal{D}}(x, y)$$

in  $\mathcal{V}$  for each pair  $x, y \in \underline{\mathcal{C}}$ , such that the following diagrams commute for all  $x, y, z \in \underline{\mathcal{C}}$ :

$$\begin{array}{ccc} \underline{\mathcal{C}}(y, z) \otimes \underline{\mathcal{C}}(x, y) & \xrightarrow{\circ} & \underline{\mathcal{C}}(x, y) \\ \downarrow F_{y,z} \otimes F_{x,y} & & \downarrow F_{x,z} \\ \underline{\mathcal{D}}(F(y), F(z)) \otimes \underline{\mathcal{D}}(F(x), F(y)) & \xrightarrow{\circ} & \underline{\mathcal{D}}(F(x), F(z)) \end{array} \qquad \begin{array}{ccc} * & \xrightarrow{\text{id}_x} & \underline{\mathcal{C}}(x, x) \\ & \searrow \text{id}_{F(x)} & \downarrow F_{x,x} \\ & & \underline{\mathcal{D}}(F(x), F(x)). \end{array}$$

♡

For a regular category, hence a  $\mathbf{Set}$ -category, these diagrams demand that  $F(- \circ -) = F(-) \circ F(-)$  and  $F(1_{(-)}) = 1_{F(-)}$ . More importantly, a topological functor between  $\mathbf{Top}$ -categories consists of continuous maps between hom-spaces.  $\mathbf{Top}$ -functors are frequently called *continuous* functors in the literature, an unfortunate conflict of terminology from the uncommon, but still used, notion of a continuous functor being one which preserves small limits.

A simplicial functor between  $\mathbf{sSet}$ -categories consists of maps between the  $n$ -simplices of the appropriate hom-spaces. A 2-functor between  $\mathbf{Cat}$ -categories, equivalently 2-categories, is a map of objects, morphisms, and 2-cells that preserves domains, codomains, composition, and identities at all levels. Going back to a definition we explored earlier, a covariant  $\mathcal{A}\mathbf{b}$ -functor from a singleton  $\mathcal{A}\mathbf{b}$ -category  $R$  to  $\underline{\mathcal{A}\mathbf{b}}$  is a left  $R$ -module; a contravariant  $\mathcal{A}\mathbf{b}$ -functor is a right  $R$ -module.

**Exercise.** Extend definition 3.4 to define the underlying functor of  $\mathcal{V}$ -functors and show that your definition is functorial, i.e., defines a functor  $(-)_0 : \mathcal{V}\text{-Cat} \rightarrow \mathbf{Cat}$ .

**Proof.** Given a  $\mathcal{V}$ -functor  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$ , define  $F_0 : \mathcal{C}_0 \rightarrow \mathcal{D}_0$  to be the same as  $F$  on objects, and on morphisms is simply given by the relevant postcomposition in  $\mathcal{V}$ . More precisely, for  $f : x \rightarrow y$  in  $\mathcal{C}_0$ , we have  $f : * \rightarrow \underline{\mathcal{C}}(x, y)$  in  $\mathcal{V}$ , and we then define

$$F_0(f) := * \xrightarrow{f} \underline{\mathcal{C}}(x, y) \xrightarrow{F_{x,y}} \underline{\mathcal{D}}(x, y).$$

Hence  $F(f) \in \mathcal{V}(*, \underline{\mathcal{D}}(x, y)) = \mathcal{D}_0(x, y)$ . The fact that  $F$  commutes with  $\circ$  in  $\mathcal{V}$  and that it preserves identities makes it functorial. These are fairly easy to check, but since this is an exercise

I will spell it out in completeness. Let  $g : y \rightarrow z$  and  $f : x \rightarrow y$  in  $\mathcal{C}_0$ . We obtain the diagram

$$\begin{array}{ccccc}
 * \otimes * & \xrightarrow{g \otimes f} & \underline{\mathcal{C}}(y, z) \otimes \underline{\mathcal{C}}(x, y) & \xrightarrow{\circ} & \underline{\mathcal{C}}(x, y) \\
 & & \downarrow F_{y,z} \otimes F_{x,y} & & \downarrow F_{x,z} \\
 & & \underline{\mathcal{D}}(F(y), F(z)) \otimes \underline{\mathcal{D}}(F(x), F(y)) & \xrightarrow{\circ} & \underline{\mathcal{D}}(F(x), F(z)).
 \end{array}$$

This expresses the equality

$$\circ_{\mathcal{D}} \circ (F_{y,z} \otimes F_{x,y}) \circ (g \otimes f) = F_{x,z} \circ \circ_{\mathcal{C}} \circ (g \otimes f).$$

Functoriality of  $\otimes$  gives on the left

$$\circ_{\mathcal{D}} \circ (F_{y,z} \otimes F_{x,y}) \circ (g \otimes f) = \circ_{\mathcal{D}} \circ ((F_{y,z} \circ g) \otimes (F_{x,y} \circ f)).$$

We thus obtain by definition of  $F_0$  and of composition in  $\mathcal{C}_0$  that

$$F_0(g) \circ F_0(f) = F_0(g \circ f).$$

The identity is manifestly preserved. Thus,  $F_0$  is functorial.

Now for the functor  $(-)_0$ . We wish to show the assignment

$$\underline{\mathcal{C}} \mapsto \mathcal{C}_0, \quad F \mapsto F_0$$

is functorial. This is fairly simple. Let  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$  and  $G : \underline{\mathcal{D}} \rightarrow \underline{\mathcal{E}}$  be  $\mathcal{V}$ -functors. Then by definition of functor composition, we have

$$(G \circ F)_0(f) = (G \circ F)_{x,y} \circ f = G_{x,y} \circ F_{x,y} \circ f = (G_0 \circ F_0)(f).$$

Thus the assignment is indeed functorial. ■

When  $\underline{\mathcal{V}}$  is a closed  $\mathcal{V}$ -category, there is for any  $\underline{\mathcal{C}}$  a *representable functor*  $\underline{\mathcal{C}}(c, -) : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{V}}$  whose underlying functor is the representable functor  $\underline{\mathcal{C}}(c, -) : \mathcal{C}_0 \rightarrow \mathcal{V}$  as constructed earlier. The object map is merely  $x \mapsto \underline{\mathcal{C}}(c, x)$ ; the morphisms

$$\underline{\mathcal{C}}(c, -)_{x,y} : \underline{\mathcal{C}}(x, y) \rightarrow \underline{\mathcal{V}}(\underline{\mathcal{C}}(c, x), \underline{\mathcal{C}}(c, y))$$

are the adjoints of composition. More precisely, since  $\underline{\mathcal{V}}$  is closed, composition

$$\circ : \underline{\mathcal{C}}(x, y) \otimes \underline{\mathcal{C}}(c, x) \rightarrow \underline{\mathcal{C}}(c, y)$$

has an adjunct under definition 3.2 to

$$\underline{\mathcal{C}}(x, y) \rightarrow \underline{\mathcal{V}}(\underline{\mathcal{C}}(c, x), \underline{\mathcal{C}}(c, y)).$$

The definition of composition in the underlying category recovers  $g \mapsto g_*$  for the underlying functor.

Suppose still that  $\mathcal{V}$  is closed monoidal. Recall an unenriched category  $\mathcal{C}$  admits a free  $\mathcal{V}$ -categorical structure with objects the same as  $\mathcal{C}$ , and hom-objects given by

$$\underline{\mathcal{C}}(a, b) := \mathcal{C}(a, b) \cdot * = \coprod_{\mathcal{C}(a,b)} *.$$

A functor  $F$  from the free  $\mathcal{V}$ -category on  $\mathcal{C}$  to a  $\mathcal{V}$ -category  $\underline{\mathcal{D}}$  consists of an object function  $c \mapsto F(c)$ , together with morphisms

$$F_{x,y} : \coprod_{\mathcal{C}(x,y)} * \rightarrow \underline{\mathcal{D}}(F(x), F(y)).$$

Universality of coproduct gives that  $F_{x,y}$  is completely determined by some family of morphisms  $X := \{\varphi_f : * \rightarrow \underline{\mathcal{D}}(F(x), F(y))\}_{f \in \mathcal{C}(x,y)}$ . Since

$$X \subseteq \mathcal{V}(*, \underline{\mathcal{D}}(F(x), F(y))) =: \mathcal{D}_0(F(x), F(y)),$$

we have that  $F_{x,y}$  is equivalently an assignment  $\mathcal{C}(x,y) \rightarrow \mathcal{D}_0(F(x), F(y))$ . Assigning the notation  $F^V(\mathcal{C})$  to the free  $\mathcal{V}$ -category on  $\mathcal{C}$  (this notation will never be used again and is simply here to write down precisely my point), we have that functors  $F : F^V(\mathcal{C}) \rightarrow \underline{\mathcal{D}}$  are precisely functors  $F : \mathcal{C} \rightarrow \mathcal{D}_0$ . Hence, there is an isomorphism

$$\mathcal{V}\text{-Cat}(F^V(\mathcal{C}), \underline{\mathcal{D}}) \cong \text{Cat}(\mathcal{C}, \mathcal{D}_0)$$

natural in both entries by universality of coproduct. Thus there is an adjunction  $F^V \dashv (-)_0$ , and for this reason we will never again use the notation  $F^V$ , since functors  $F^V(\mathcal{C}) \rightarrow \underline{\mathcal{D}}$  are simply functors  $\mathcal{C} \rightarrow \mathcal{D}_0$ .

Recall that  $\text{sSet}$  admits a closed enrichment via  $\underline{\text{sSet}}(X, Y)_n = \text{sSet}(\Delta^n \times X, Y)$ . The terminal simplicial set  $\Delta^0 = \Delta(-, [0])$  in an unenriched sense also represents the functor  $(-) : \text{sSet} \rightarrow \text{Set}$ , which takes a simplicial set  $X$  to its set  $X_0$  of vertices. However viewing  $\text{sSet}$  as self-enriched, the represented functor  $\underline{\text{sSet}}(\Delta^0, -)$  becomes the identity on  $\text{sSet}$ :

$$\underline{\text{sSet}}(\Delta^0, X)_n = \text{sSet}(\Delta^n \times \Delta^0, X) \cong \text{sSet}(\Delta^n, X) \cong X_n.$$

The equality is definition, the first isomorphism is that  $\Delta^0$  is terminal, and the second is Yoneda.

The next natural definition is that of a  $\mathcal{V}$ -natural transformation. However, we need to recall that there are no morphisms to act on, so the natural way to reproduce the naturality square

$$\begin{array}{ccc} F(x) & \xrightarrow{F(f)} & F(y) \\ \alpha_x \downarrow & & \downarrow \alpha_y \\ G(x) & \xrightarrow{G(f)} & G(y) \end{array}$$

is via pre- and post-composition with components of  $\alpha$ .

**Definition 3.6 ( $\mathcal{V}$ -Natural Transformation).** A  $\mathcal{V}$ -natural transformation  $\alpha : F \Rightarrow G$  between  $\mathcal{V}$ -functors  $F, G : \mathcal{C} \rightarrow \underline{\mathcal{D}}$  is a morphism  $\alpha_x : * \rightarrow \underline{\mathcal{D}}(F(x), G(x))$  in  $\mathcal{V}$  for each  $x \in \mathcal{C}$ , such that for all  $x, y \in \mathcal{C}$ , the diagram

$$\begin{array}{ccc} \mathcal{C}(x, y) & \xrightarrow{F_{x,y}} & \underline{\mathcal{D}}(F(x), F(y)) \\ G_{x,y} \downarrow & & \downarrow (\alpha_y)_* \\ \underline{\mathcal{D}}(G(x), G(y)) & \xrightarrow{(\alpha_x)_*} & \underline{\mathcal{D}}(F(x), G(y)) \end{array}$$

with precomposition  $(\alpha_x)_*$  and postcomposition  $(\alpha_y)_*$  defined as earlier. ♥

The intuitive understanding of this in unenriched categories is that on the top right path, morphisms  $f : x \rightarrow y$  in  $\mathcal{C}$  should be carried to morphisms  $F(f) : F(x) \rightarrow F(y)$  in  $\mathcal{D}$ , and then postcomposed with  $\alpha_y$ . The bottom left path says that this should be equal to taking your morphism and applying  $G$  to obtain  $G(f) : G(x) \rightarrow G(y)$ , then precomposing with  $\alpha_x$ , giving an equality

$$\alpha_y \circ F(f) = G(f) \circ \alpha_x.$$

This is precisely the naturality condition.

**Example.** An  $\mathcal{A}\mathbf{b}$ -natural transformation between two left  $R$ -modules  $A, B : R \rightarrow \mathcal{A}\mathbf{b}$  is a single arrow  $\alpha_* : * \rightarrow \mathcal{A}\mathbf{b}(A, B)$  in  $\mathcal{A}\mathbf{b}$ , and hence a group homomorphism, that must commute with scalar multiplication by our above discussion. Hence  $\mathcal{A}\mathbf{b}$ -natural transformations here are  $R$ -module homomorphisms, and thus  $\mathcal{A}\mathbf{b}\text{-Fun}(R, \mathcal{A}\mathbf{b}) \cong \text{Mod}_R$ .

The addition of  $\mathcal{V}$ -natural transformations assembles  $\mathcal{V}\text{-Cat}$  into a 2-category. We've shown already that the underlying set functor  $(-)_0$  extends to  $\mathcal{V}$ -functors, making it a functor  $\mathcal{V}\text{-Cat} \rightarrow \mathcal{C}\mathbf{at}$ . We now claim we can use it to define underlying  $\mathcal{V}$ -natural transformations.

**Proposition 3.1.** *Underlying  $(-)_0$  is a 2-functor  $\mathcal{V}\text{-Cat} \rightarrow \mathcal{C}\mathbf{at}$ . Additionally, lax monoidal functors  $F : \mathcal{V} \rightarrow \mathcal{U}$  induce 2-functors  $\mathcal{V}\text{-Cat} \rightarrow \mathcal{U}\text{-Cat}$ .*

**Proof.** The statement is obvious, yet the proof is messy. Let  $(\mathcal{V}, \otimes, *)$  and  $(\mathcal{U}, \otimes, 1)$  be monoidal, and let  $F : \mathcal{V} \rightarrow \mathcal{U}$  be lax monoidal. Let  $\lambda : F(x) \otimes F(x) \rightarrow F(x \otimes x)$  be the natural transformation, and let  $\varphi : 1 \rightarrow F(*)$  be the other arrow.

We prove the second statement first. By lemma 3.2, we already have an object function by mapping some  $\mathcal{V}$ -category  $\underline{\mathcal{D}}_{\mathcal{V}}$  to the  $\mathcal{U}$ -category  $\underline{\mathcal{D}}_{\mathcal{U}}$  with the same objects, and hom-objects

$$\underline{\mathcal{D}}_{\mathcal{U}}(x, y) := F(\underline{\mathcal{D}}_{\mathcal{V}}(x, y)).$$

Let  $G : \underline{\mathcal{C}}_{\mathcal{V}} \rightarrow \underline{\mathcal{D}}_{\mathcal{V}}$  be a  $\mathcal{V}$ -functor. Define a new functor  $F(G)$  by  $F(G) = G$  on objects, and

$$(F(G))_{x,y} = F(G_{x,y}).$$

Since  $G$  is  $\mathcal{V}$ -functorial, the diagram

$$\begin{array}{ccc} \underline{\mathcal{C}}_{\mathcal{V}}(y, z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x, y) & \xrightarrow{\circ} & \underline{\mathcal{C}}_{\mathcal{V}}(x, z) \\ G_{y,z} \otimes G_{x,y} \downarrow & & \downarrow G_{x,z} \\ \underline{\mathcal{D}}_{\mathcal{V}}(y, z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x, y) & \xrightarrow{\circ} & \underline{\mathcal{D}}_{\mathcal{V}}(x, z) \end{array}$$

commutes in  $\mathcal{V}$ . Hence by functoriality of  $F$ , the diagram

$$\begin{array}{ccc} F(\underline{\mathcal{C}}_{\mathcal{V}}(y, z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x, y)) & \xrightarrow{F(\circ)} & \underline{\mathcal{C}}_{\mathcal{U}}(x, z) \\ F(G_{y,z} \otimes G_{x,y}) \downarrow & & \downarrow F(G_{x,z}) \\ F(\underline{\mathcal{D}}_{\mathcal{V}}(y, z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x, y)) & \xrightarrow{F(\circ)} & \underline{\mathcal{D}}_{\mathcal{U}}(x, z) \end{array}$$



To show  $F(G)$  preserves identities, recall by definition 3.5 that since  $G$  is  $\mathcal{V}$ -functorial, for any  $x$  the diagram

$$\begin{array}{ccc} * & \xrightarrow{\text{id}_x} & \underline{\mathcal{C}}_{\mathcal{V}}(x, x) \\ & \searrow \text{id}_{G(x)} & \downarrow G_{x,x} \\ & & \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(x)) \end{array}$$

commutes. Functoriality of  $F$  gives that the diagram

$$\begin{array}{ccc} F(*) & \xrightarrow{F(\text{id}_x)} & \underline{\mathcal{C}}_{\mathcal{U}}(x, x) \\ & \searrow F(\text{id}_{G(x)}) & \downarrow F(G_{x,x}) \\ & & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(x)) \end{array}$$

commutes. To avoid confusion, denote by  $1$  the unit in  $\mathcal{U}$ . Since  $F$  is lax monoidal, there is a natural transformation  $1 \rightarrow F(*)$ , giving that

$$\begin{array}{ccccc} 1 & \longrightarrow & F(*) & \xrightarrow{F(\text{id}_x)} & \underline{\mathcal{C}}_{\mathcal{U}}(x, x) \\ & & & \searrow F(\text{id}_{G(x)}) & \downarrow F(G_{x,x}) \\ & & & & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(x)) \end{array}$$

commutes. Recall also that lemma 3.2 defines the identity as the composition

$$\text{id}_x : 1 \longrightarrow F(*) \xrightarrow{F(\text{id}_x)} \underline{\mathcal{D}}_{\mathcal{U}}(x, x).$$

Note this is the top composition on the above diagram, and also note that we have by the same result that

$$\text{id}_{G(x)} : 1 \longrightarrow F(*) \xrightarrow{F(\text{id}_{G(x)})} \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(x)).$$

Note that this is the other side of the diagram. Thus, the diagram is equivalent to

$$\begin{array}{ccc} 1 & \xrightarrow{\text{id}_x} & \underline{\mathcal{C}}_{\mathcal{U}}(x, x) \\ & \searrow \text{id}_{G(x)} & \downarrow F(G_{x,x}) \\ & & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(x)). \end{array}$$

Thus,  $F(G)$  is a  $\mathcal{U}$ -functor, and  $F$  is a functor.

Finally, we must show  $F$  is a 2-functor by showing it preserves domain, codomain, identity, and composition of natural transformations both horizontally and vertically. Let  $G, H : \underline{\mathcal{C}}_{\mathcal{V}} \rightarrow \underline{\mathcal{D}}_{\mathcal{V}}$  be  $\mathcal{V}$ -functors, and let  $\alpha : G \Rightarrow H$  be a  $\mathcal{V}$ -natural transformation. Let  $\varphi : 1 \rightarrow F(*)$  be the relevant

lax monoidal arrow. Since the components  $\alpha_x$  are morphisms in  $\mathcal{V}$ , define  $F(\alpha)$  componentwise as  $F(\alpha)_x = F(\alpha_x) \circ \varphi$ . This is indeed an arrow  $1 \rightarrow \underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(x))$  by definition of  $\underline{\mathcal{D}}_{\mathcal{U}}$ . By definition 3.6, the square

$$\begin{array}{ccc} \underline{\mathcal{C}}_{\mathcal{V}}(x, y) & \xrightarrow{G_{x,y}} & \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y)) \\ \downarrow H_{x,y} & & \downarrow (\alpha_y)_* \\ \underline{\mathcal{D}}_{\mathcal{V}}(H(x), H(y)) & \xrightarrow{(\alpha_x)^*} & \underline{\mathcal{D}}_{\mathcal{V}}(G(x), H(y)) \end{array}$$

commutes. Functoriality of  $F$  gives that the square

$$\begin{array}{ccc} \underline{\mathcal{C}}_{\mathcal{U}}(x, y) & \xrightarrow{F(G_{x,y})} & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ \downarrow F(H_{x,y}) & & \downarrow F((\alpha_y)_*) \\ \underline{\mathcal{D}}_{\mathcal{U}}(H(x), H(y)) & \xrightarrow{F((\alpha_x)^*)} & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(y)) \end{array}$$

commutes. Recall that postcomposition is given by the composition

$$(\alpha_y)_* : * \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y)) \xrightarrow{\alpha_y \otimes 1} \underline{\mathcal{D}}_{\mathcal{V}}(G(x), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y)) \xrightarrow{\circ} \underline{\mathcal{D}}_{\mathcal{V}}(G(x), H(y)).$$

Applying  $F$  and naturality of the lax functorial arrow  $\lambda$  gives the diagram

$$\begin{array}{ccc} F(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))) & \xrightarrow{F(\alpha_y \otimes 1)} & F(\underline{\mathcal{D}}_{\mathcal{V}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))) \xrightarrow{F(\circ)} \underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(y)) \\ \uparrow \lambda & & \uparrow \lambda \\ F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) & \xrightarrow{F(\alpha_y) \otimes 1} & \underline{\mathcal{D}}_{\mathcal{U}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \end{array}$$

commuting. Applying this to the side of the above diagram gives

$$\begin{array}{ccccc} \underline{\mathcal{C}}_{\mathcal{U}}(x, y) & \xrightarrow{F(G_{x,y})} & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) & & \\ \downarrow F(H_{x,y}) & & \downarrow F(\ell_{\mathcal{V}}^{-1}) & & \\ & & F(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))) & \xleftarrow{\lambda} & F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ & & \downarrow F(\alpha_y \otimes 1) & & \downarrow F(\alpha_y) \otimes 1 \\ & & F(\underline{\mathcal{D}}_{\mathcal{V}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))) & \xleftarrow{\lambda} & \underline{\mathcal{D}}_{\mathcal{U}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ & & \downarrow F(\circ) & & \\ \underline{\mathcal{D}}_{\mathcal{U}}(H(x), H(y)) & \xrightarrow{F((\alpha_x)^*)} & \underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(y)) & & \end{array}$$

which commutes for  $\ell_{\mathcal{V}}$  the left unitor in  $\mathcal{V}$ . One of the coherence conditions for lax monoidal functors states that, for  $\varphi : 1 \rightarrow F(*)$  the unit arrow for the lax monoidal functor and for  $\ell_{\mathcal{U}}$  the left unitor in  $\mathcal{U}$ , for any  $x \in \mathcal{V}$ , the diagram

$$\begin{array}{ccc} 1 \otimes F(x) & \xrightarrow{\varphi \otimes 1} & F(*) \otimes F(x) \\ \ell_{\mathcal{U}} \downarrow & & \downarrow \lambda \\ F(x) & \xleftarrow{F(\ell_{\mathcal{V}})} & F(* \otimes x) \end{array}$$

commutes. Postcomposing with  $F(\ell_{\mathcal{V}}^{-1}) = F(\ell_{\mathcal{V}})^{-1}$  and precomposing with  $\ell_{\mathcal{U}}^{-1}$  give the equalities

$$F(\ell_{\mathcal{V}}^{-1}) \circ \ell_{\mathcal{U}} = \lambda \circ (\varphi \otimes 1) \implies F(\ell_{\mathcal{V}}^{-1}) = \lambda \circ (\varphi \otimes 1) \circ \ell_{\mathcal{U}}^{-1}.$$

Plugging in  $x = \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))$  and applying this equality gives the commutative square

$$\begin{array}{ccc} 1 \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) & \xrightarrow{\varphi \otimes 1} & F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ \uparrow \ell_{\mathcal{U}}^{-1} & & \downarrow \lambda \\ \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) & \xrightarrow{F(\ell_{\mathcal{V}}^{-1})} & F(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))). \end{array}$$

Rotating 90 degrees clockwise allows us to plug this into the top right corner of the above diagram, giving the commutative diagram (left side omitted for simplicity)

$$\begin{array}{ccc} \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) & \xrightarrow{\ell_{\mathcal{U}}^{-1}} & 1 \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ \downarrow F(\ell_{\mathcal{V}}^{-1}) & & \downarrow \varphi \otimes 1 \\ F(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))) & \xleftarrow{\lambda} & F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ \downarrow F(\alpha_y \otimes 1) & & \downarrow F(\alpha_y) \otimes 1 \\ F(\underline{\mathcal{D}}_{\mathcal{V}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))) & \xleftarrow{\lambda} & \underline{\mathcal{D}}_{\mathcal{U}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \\ \downarrow F(\circ) & & \\ \underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(y)). & & \end{array}$$

This finally gives an expression for  $F((\alpha_y)_*)$  as the composition (here recognizing that  $\circ := F(\circ) \circ \lambda$  in  $\underline{\mathcal{D}}_{\mathcal{U}}$ )

$$F((\alpha_y)_*) = F(\circ) \circ \lambda \circ F(\alpha_y) \otimes 1 \circ (\varphi \otimes 1) \circ \ell_{\mathcal{U}}^{-1}.$$

This is precisely the definition of  $(F(\alpha_y) \circ \varphi)_* = (F(\alpha_y))_*$ . Verification of precomposition is the exact same, since  $\varphi$  moves with the morphism to the other side of the monoidal product. Therefore,  $F(\alpha) : F(G) \rightarrow F(H)$  is  $\mathcal{U}$ -natural. In particular, domain and codomain are preserved.



Note that the identity  $\mathcal{V}$ -natural transformation  $1_G : G \Rightarrow G$  is obviously given by  $(1_G)_x = \text{id}_{G(x)}^\mathcal{V}$ . By definition,  $F(1_G)_x = F(\text{id}_{G(x)}^\mathcal{V}) \circ \varphi$ . Recall that we have by definition

$$F(\text{id}_{G(x)}^\mathcal{V}) \circ \varphi = \text{id}_{G(x)}^\mathcal{U}.$$

Moreover,  $F(G(x)) = G(x)$ , and thus  $(1_{F(G)})_x = \text{id}_{G(x)}$ . Therefore,  $F(1_G) = 1_{F(G)}$  preserves identities.

It thus suffices to show  $F$  preserves horizontal and vertical composition. First, we show vertical composition. Let  $G, H, I : \underline{\mathcal{C}}_\mathcal{V} \rightarrow \underline{\mathcal{D}}_\mathcal{V}$  be  $\mathcal{V}$ -functors, and let  $\alpha : G \Rightarrow H$  and  $\beta : H \Rightarrow I$  be  $\mathcal{V}$ -natural transformations. Define the horizontal composite  $(\beta \circ \alpha)$  componentwise by composing in the underlying category as

$$(\beta \circ \alpha)_x : * \xrightarrow{\ell_\mathcal{V}^{-1}} * \otimes * \xrightarrow{\beta_x \otimes \alpha_x} \underline{\mathcal{D}}_\mathcal{V}(I(x), H(x)) \otimes \underline{\mathcal{D}}_\mathcal{V}(H(x), G(x)) \xrightarrow{\circ} \underline{\mathcal{D}}_\mathcal{V}(I(x), G(x)).$$

It suffices to show  $F(\beta \circ \alpha)_x = F(\beta)_x \circ F(\alpha)_x$ . We defined

$$F(\beta \circ \alpha)_x = F((\beta \circ \alpha)_x) \circ \varphi.$$

This is viewed by functoriality of  $F$  on the above composition as

$$1 \xrightarrow{\varphi} F(*) \xrightarrow{F(\ell_\mathcal{V}^{-1})} F(* \otimes *) \xrightarrow{F(\beta_x \otimes \alpha_x)} F(\underline{\mathcal{D}}_\mathcal{V}(I(x), H(x)) \otimes \underline{\mathcal{D}}_\mathcal{V}(H(x), G(x))) \xrightarrow{F(\circ)} \underline{\mathcal{D}}_\mathcal{U}(I(x), G(x)).$$

By naturality of  $\lambda$ , it follows that

$$\begin{array}{ccc} F(*) \otimes F(*) & \xrightarrow{F(\beta_x) \otimes F(\alpha_x)} & \underline{\mathcal{D}}_\mathcal{U}(I(x), H(x)) \otimes \underline{\mathcal{D}}_\mathcal{U}(H(x), G(x)) \\ \lambda \downarrow & & \downarrow \lambda \\ 1 \xrightarrow{\varphi} F(*) & \xrightarrow{F(\ell_\mathcal{V}^{-1})} F(* \otimes *) & \xrightarrow{F(\beta_x \otimes \alpha_x)} F(\underline{\mathcal{D}}_\mathcal{V}(I(x), H(x)) \otimes \underline{\mathcal{D}}_\mathcal{V}(H(x), G(x))) \xrightarrow{F(\circ)} \underline{\mathcal{D}}_\mathcal{U}(I(x), G(x)) \end{array}$$

commutes. Applying the coherence condition for the left unitor as we did before, we obtain that the diagram

$$\begin{array}{ccccc} 1 \otimes F(*) & \xrightarrow{\varphi \otimes 1} & F(*) \otimes F(*) & \xrightarrow{F(\beta_x) \otimes F(\alpha_x)} & \underline{\mathcal{D}}_\mathcal{U}(I(x), H(x)) \otimes \underline{\mathcal{D}}_\mathcal{U}(H(x), G(x)) \\ \ell_\mathcal{U}^{-1} \uparrow & & \lambda \downarrow & & \downarrow \lambda \\ 1 \xrightarrow{\varphi} F(*) & \xrightarrow{F(\ell_\mathcal{V}^{-1})} & F(* \otimes *) & \xrightarrow{F(\beta_x \otimes \alpha_x)} & F(\underline{\mathcal{D}}_\mathcal{V}(I(x), H(x)) \otimes \underline{\mathcal{D}}_\mathcal{V}(H(x), G(x))) \xrightarrow{F(\circ)} \underline{\mathcal{D}}_\mathcal{U}(I(x), G(x)) \end{array}$$

also commutes. Note that since  $\ell_{\mathcal{U}} : * \otimes - \Rightarrow 1_{\mathcal{U}}$ , is natural, we also have that  $\ell_{\mathcal{U}}^{-1} : 1 \Rightarrow * \otimes -$  is natural. Hence by naturality of  $\ell_{\mathcal{U}}^{-1}$ ,

$$\begin{array}{ccc} 1 & \xrightarrow{\varphi} & F(*) \\ \ell_{\mathcal{U}}^{-1} \downarrow & & \downarrow \ell_{\mathcal{U}}^{-1} \\ 1 \otimes 1 & \xrightarrow{1 \otimes \varphi} & 1 \otimes F(*) \end{array}$$

commutes. Gluing this into the top left of our above diagram gives the commutative diagram gives us that the diagram

$$\begin{array}{ccccccc} 1 \otimes 1 & \xrightarrow{1 \otimes \varphi} & 1 \otimes F(*) & \xrightarrow{\varphi \otimes 1} & F(*) \otimes F(*) & \xrightarrow{F(\beta_x) \otimes F(\alpha_x)} & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \\ \uparrow \ell_{\mathcal{U}}^{-1} & & \uparrow \ell_{\mathcal{U}}^{-1} & & \downarrow \lambda & & \downarrow \lambda \\ 1 & \xrightarrow{\varphi} & F(*) & \xrightarrow{F(\ell_{\mathcal{V}}^{-1})} & F(*) \otimes F(*) & \xrightarrow{F(\beta_x \otimes \alpha_x)} & F(\underline{\mathcal{D}}_{\mathcal{V}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(H(x), G(x))) \xrightarrow{F(\circ)} \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x)) \end{array}$$

commutes. Note once again that as defined in lemma 3.2,  $\lambda \circ F(\circ)$  is simply composition in  $\underline{\mathcal{D}}_{\mathcal{U}}$ , and since  $(\varphi \otimes 1) \circ (1 \otimes \varphi) = \varphi \otimes \varphi$  by functoriality, this gives us the equality

$$\begin{array}{c} \xrightarrow{\quad F(\beta \circ \alpha)_x \quad} \\ 1 \xrightarrow{\ell_{\mathcal{U}}^{-1}} F(*) \otimes F(*) \xrightarrow{\varphi \otimes \varphi} F(*) \otimes F(*) \xrightarrow{F(\beta_x) \otimes F(\alpha_x)} \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \xrightarrow{\circ} \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x)), \end{array}$$

where we have recalled that the bottom row above is the expanded version of  $F(\beta \circ \alpha)_x = F(\beta_x \circ \alpha_x) \circ \varphi$ . Functoriality of  $\otimes$  once again gives that

$$(F(\beta_x) \otimes F(\alpha_x)) \circ (\varphi \otimes \varphi) = (F(\beta_x) \circ \varphi) \otimes (F(\alpha_x) \circ \varphi) = F(\beta)_x \otimes F(\alpha)_x.$$

Applying this to the above diagram gives

$$\begin{array}{c} \xrightarrow{\quad F(\beta \circ \alpha)_x \quad} \\ 1 \xrightarrow{\ell_{\mathcal{U}}^{-1}} F(*) \otimes F(*) \xrightarrow{F(\beta)_x \otimes F(\alpha)_x} \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \xrightarrow{\circ} \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x)). \end{array}$$

Note that this is precisely the definition of composition  $F(\beta)_x \circ F(\alpha)_x$  in  $\underline{\mathcal{D}}_{\mathcal{U}}$ . Thus, the diagram expresses the equality

$$F(\beta \circ \alpha)_x = F(\beta)_x \circ F(\alpha)_x.$$

Therefore,  $F$  preserves vertical composition of  $\mathcal{V}$ -natural transformations.

Finally, let

$$\begin{array}{ccccc} \mathcal{C}_{\mathcal{V}} & \xrightarrow{G^1} & \mathcal{D}_{\mathcal{V}} & \xrightarrow{G^2} & \mathcal{E}_{\mathcal{V}} \\ & \downarrow \alpha & & \downarrow \beta & \\ & H^1 & & H^2 & \end{array}$$

be a diagram of  $\mathcal{V}$ -natural transformations between  $\mathcal{V}$ -functors. Recall the horizontal composite  $\beta \cdot \alpha$  is defined componentwise as

$$(\beta \cdot \alpha)_x = (\beta_{H^1(x)}) \circ \left( G_{G^1(x), H^1(x)}^2 \circ \alpha_x \right)$$

where the center composition  $\circ$  is componentwise vertical composition of  $\mathcal{V}$ -natural transformations, and the right composition is composition in  $\mathcal{V}$ . We need to show  $F(\beta \cdot \alpha) = F(\beta) \cdot F(\alpha)$ . By definition,

$$F(\beta \cdot \alpha)_x = F \left( \beta_{H^1(x)} \circ \left( G_{G^1(x), H^1(x)}^2 \circ \alpha_x \right) \right) \circ \varphi.$$

Since  $F$  preserves vertical composition, we have

$$F \left( \beta_{H^1(x)} \circ \left( G_{G^1(x), H^1(x)}^2 \circ \alpha_x \right) \right) \circ \varphi = F(\beta_{H^1(x)}) \circ F \left( G_{G^1(x), H^1(x)}^2 \circ \alpha_x \right) \circ \varphi.$$

Expanding out  $F(\beta)_x \cdot F(\alpha)_x$  similarly gives

$$\begin{aligned} F(\beta)_x \circ F(\alpha)_x &= F(\beta)_{F(H^1)(x)} \circ \left( F(G^2)_{G^1(x), H^1(x)} \circ F(\alpha)_x \right) \\ &= F(\beta)_{H^1(x)} \circ \left( F \left( G_{G^1(x), H^1(x)}^2 \right) \circ F(\alpha)_x \right) \\ &= (F(\beta_{H^1(x)}) \circ \varphi) \circ \left( F \left( G_{G^1(x), H^1(x)}^2 \right) \circ F(\alpha)_x \circ \varphi \right). \end{aligned}$$

At this point the diagrams become too large to fit on the screen, so we trust the reader to use the definition of the horizontal composite (and the equalities induced when  $F$  preserves it) and to reference the large diagram used in the proof of the vertical composite being preserved, in order to see that these two equations are indeed equal. Therefore,  $F : \mathcal{V}\text{-Cat} \rightarrow \mathcal{U}\text{-Cat}$  is a 2-functor.

Note that  $(-)_0$  is simply the functor  $\mathcal{V}(*, -) : \mathcal{V} \rightarrow \mathbf{Set}$ . Since  $\mathbf{Set}\text{-Cat} = \mathbf{Cat}$  by definition, it suffices to show  $(-)_0$  is lax monoidal with  $\mathbf{Set}$  endowed with the cartesian monoidal structure. First, let  $\lambda : \mathcal{V}(*, -) \times \mathcal{V}(*, -) \Rightarrow \mathcal{V}(*, - \otimes -)$  be given by tensoring and precomposing with  $\ell_{\mathcal{V}}^{-1}$ :

$$(u, v) \mapsto (u \otimes v) \circ \ell_{\mathcal{V}}^{-1}.$$

Note that this is indeed well-defined, since  $u : * \rightarrow a$  and  $v : * \rightarrow b$  means that we obtain the composition

$$* \xrightarrow{\ell_{\mathcal{V}}^{-1}} * \otimes * \xrightarrow{u \otimes v} a \otimes b.$$

Now we show  $\lambda$  is natural. That is, for any pair  $(f, g)$  with  $f : a \rightarrow c$  and  $g : b \rightarrow d$  in  $\mathcal{V}$ , the diagram

$$\begin{array}{ccc} \mathcal{V}(*, a) \times \mathcal{V}(*, b) & \xrightarrow{f_* \times g_*} & \mathcal{V}(*, c) \times \mathcal{V}(*, d) \\ \lambda_{a,b} \downarrow & & \downarrow \lambda_{c,d} \\ \mathcal{V}(*, a \otimes b) & \xrightarrow{(f \otimes g)_*} & \mathcal{V}(*, c \otimes d) \end{array}$$

should commute. Indeed, the diagram expresses the equality

$$(f \otimes g) \circ (u \otimes v) \circ \ell_{\mathcal{V}}^{-1} = ((f \circ u) \otimes (g \circ v)) \circ \ell_{\mathcal{V}}^{-1},$$

which follows simply by functoriality of the monoidal product. Next, we need to show there is an arrow  $1 \rightarrow \mathcal{V}(*, *)$ , where  $1$  denotes the singleton. To choose this canonically, let  $\varphi = \text{id}_*$ . Thus, a canonical arrow exists, and therefore,  $(-)_* = \mathcal{V}(*, -)$  is a 2-functor  $\mathcal{V}\text{-Cat} \rightarrow \text{Cat}$ .  $\blacksquare$

We now modify our notation by dropping the subscript 0. For a  $\mathcal{V}$ -category  $\underline{\mathcal{C}}$ , we denote by  $\mathcal{C}$  the underlying category, and other underlying structures may be inferred from context. We also obtain from this proof an explicit description of the notion of the underlying natural transformation. Let  $\underline{\alpha} : F \Rightarrow G$  be  $\mathcal{V}$ -natural, and define the underlying natural transformation  $\alpha$  componentwise via

$$\alpha_x = \mathcal{V}(*, \underline{\alpha}_x) \circ \text{id}_* = (\alpha_x)_*.$$

**Lemma 3.4.** *Let  $\mathcal{V}$  be closed, and let  $\underline{\mathcal{C}}$  be a  $\mathcal{V}$ -category. The following are equivalent:*

1.  $x, y \in \underline{\mathcal{C}}$  are isomorphic as objects in  $\mathcal{C}$ ;
2. The functors  $\mathcal{C}(x, -), \mathcal{C}(y, -) : \mathcal{C} \rightarrow \text{Set}$  are naturally isomorphic;
3. The unenriched functors  $\underline{\mathcal{C}}(x, -), \underline{\mathcal{C}}(y, -) : \mathcal{C} \rightarrow \mathcal{V}$  are naturally isomorphic;
4. The  $\mathcal{V}$ -functors  $\underline{\mathcal{C}}(x, -), \underline{\mathcal{C}}(y, -) : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{V}}$  are  $\mathcal{V}$ -isomorphic.

**Proof.** By proposition 3.1, underlying is a functor, and hence (4) implies (3). Recall by definition and a previous exercise that  $\mathcal{C}(x, -) = \mathcal{V}(*, -) \circ \underline{\mathcal{C}}(x, -)$ , and thus (3) implies (2) via horizontal composition of the natural isomorphism with the identity for the underlying functor. By the Yoneda lemma, (2) implies (1). Finally, (1) implies (4) via the following direct construction.

Let  $x, y \in \underline{\mathcal{C}}$  be isomorphic in  $\mathcal{C}$ . Recall a  $\mathcal{V}$ -natural transformation  $\alpha : \underline{\mathcal{C}}(x, -) \Rightarrow \underline{\mathcal{C}}(y, -)$  is a collection of arrows in  $\mathcal{V}$  such that for all  $a, b \in \underline{\mathcal{C}}$ , the diagram

$$\begin{array}{ccc} \underline{\mathcal{C}}(a, b) & \xrightarrow{\underline{\mathcal{C}}(x, -)} & \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(x, b)) \\ \underline{\mathcal{C}}(y, -) \downarrow & & \downarrow (\alpha_b)_* \\ \underline{\mathcal{V}}(\underline{\mathcal{C}}(y, a), \underline{\mathcal{C}}(y, b)) & \xrightarrow{(\alpha_a)_*} & \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, b)) \end{array}$$

commutes. Let  $f : y \rightarrow x$  be the isomorphism in question, and equivalently  $f : * \rightarrow \underline{\mathcal{C}}(y, x)$  in  $\mathcal{V}$ . Since precomposition  $f^*$  is an arrow  $\underline{\mathcal{C}}(x, -) \rightarrow \underline{\mathcal{C}}(y, -)$ , define  $f_a^*$  as precomposition  $\underline{\mathcal{C}}(x, a) \rightarrow \underline{\mathcal{C}}(y, a)$ . Hence we obtain since  $\mathcal{V}$  is closed the adjunction

$$f_a^* \in \mathcal{V}(* \otimes \underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, a)) \cong \mathcal{V}(*, \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, a))).$$

Here we abuse notation and write  $f_*^a$  for  $f_*^a \circ \lambda_{\underline{\mathcal{C}}(x,a)}$ , for  $\lambda$  the left unitor. This remark will be important later. Define  $\alpha_a$  as the adjunct of  $f_a^*$  above. Let  $g = f^{-1}$ , and define  $\beta_a$  as the adjunct of  $g_a^*$ , forming another transformation  $\beta : \underline{\mathcal{C}}(y, -) \Rightarrow \underline{\mathcal{C}}(x, -)$ . The claim is that the horizontal composites  $\beta \circ \alpha = 1$  and  $\alpha \circ \beta = 1$ .

Recall the horizontal composite is defined pointwise as

$$(\beta \circ \alpha)_a : * \xrightarrow{\lambda_*} * \otimes * \xrightarrow{\beta_a \otimes \alpha_a} \underline{\mathcal{V}}(\underline{\mathcal{C}}(y, a), \underline{\mathcal{C}}(x, a)) \otimes \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, a)) \xrightarrow{\circ} \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(x, a)).$$

An annoying expansion of the precompositions shows that this composite is  $\text{id}_{\underline{\mathcal{C}}(x,a)}$ . It remains to be shown that these definitions are indeed natural, which is fairly trivial. ■

Adjunctions and equivalences can be defined in any 2-category. We use this fact to define them in the 2-categories  $\mathcal{V}\text{-Cat}$ , and indeed these lead to the correct notions.

**Definition 3.7** ( *$\mathcal{V}$ -equivalence*). A  $\mathcal{V}$ -equivalence of categories is a  $\mathcal{V}$ -functor  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$  that is

- *essentially surjective*: every  $d \in \underline{\mathcal{D}}$  is isomorphic in  $\mathcal{D}$  to some object  $F(c)$ ;
- $\mathcal{V}$ -fully faithful: for each  $c, c' \in \underline{\mathcal{C}}$ , the arrow  $F_{c,c'}$  is an isomorphism in  $\mathcal{V}$ .

The usual proof shows that this is precisely the same as the other definition; that is a  $\mathcal{V}$ -equivalence of categories  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$  is a  $\mathcal{V}$ -functor such that there exists a  $\mathcal{V}$ -functor  $G : \underline{\mathcal{D}} \rightarrow \underline{\mathcal{C}}$  which is an inverse up to isomorphism:

$$F \circ G = 1_{\underline{\mathcal{D}}}, \quad G \circ F = 1_{\underline{\mathcal{C}}}.$$

♥

**Definition 3.8** ( *$\mathcal{V}$ -adjunction*). A  $\mathcal{V}$ -adjunction consists of  $\mathcal{V}$ -functors

$$\underline{\mathcal{C}} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow{G} \end{array} \underline{\mathcal{D}}$$

together with either of the following equivalent criteria:

- $\mathcal{V}$ -natural isomorphisms  $\underline{\mathcal{D}}(F(c), d) \cong \underline{\mathcal{C}}(c, G(d))$  in  $\mathcal{V}$ ;
- $\mathcal{V}$ -natural transformations  $\eta : 1 \Rightarrow GF$  and  $\varepsilon : FG \Rightarrow 1$  satisfying the triangle identities

$$G\varepsilon \circ \eta G = 1_G, \quad \varepsilon F \circ F\eta = 1_F.$$

♥

One example is as follows: we will see later that the localization  $\text{sSet} \rightarrow \text{Ho}(\text{sSet}) =: \mathcal{H}$  is lax monoidal, and hence there is a functor  $\text{sSet-Cat} \rightarrow \mathcal{H}\text{-Cat}$ . An equivalence of simplicially enriched categories (called a DK-equivalence for Dwyer and Kan) is a simplicial functor where the corresponding  $\mathcal{H}$ -functor is a  $\mathcal{H}$ -equivalence.

### 3.6 Simplicial Categories

We explore simplicial categories as an example of these notions with important applications to homotopy theory. A simplicial category  $\underline{\mathcal{C}}$  induces a simplicial object in  $\text{Cat}$ , denoted  $\mathcal{C}_\bullet : \Delta^{\text{op}} \rightarrow \text{Cat}$  and defined as follows. The category  $\mathcal{C}_n$  takes as objects the same as  $\underline{\mathcal{C}}$ , and as arrows is defined as

$$\mathcal{C}_n(x, y) := \underline{\mathcal{C}}(x, y)_n.$$

Hence, arrows  $x \rightarrow y$  in  $\mathcal{C}_n$  are  $n$ -simplices in the hom-space  $\underline{\mathcal{C}}(x, y)$ . The induced actions of the morphisms in  $\mathbf{\Delta}$  on the hom-spaces define functors which are identities on objects. Also note that the category  $\mathcal{C}_0$  is the underlying category of  $\underline{\mathcal{C}}$ .

For example, recall that  $\mathbf{sSet}$  is closed cartesian via the definition  $\mathbf{sSet}(X, Y)_n := \mathbf{sSet}(\Delta^n \times Y, X)$ . Hence  $\underline{\mathbf{sSet}}$  is a simplicial object in (large) categories. Objects of each category are simplicial sets, and morphisms  $X \rightarrow Y$  in the  $n$ th category are morphisms  $\underline{\mathbf{sSet}}(X, Y)_n$  of simplicial sets with our previous definition. The face and degeneracy operators act by precomposition on the representable part of the domain.

As a fairly obvious partial converse, any simplicial object  $\mathcal{C}_\bullet : \mathbf{\Delta}^{\text{op}} \rightarrow \mathbf{Cat}$  with functors  $d_i : \mathcal{C}_n \rightarrow \mathcal{C}_{n-1}$ ,  $s_i : \mathcal{C}_n \rightarrow \mathcal{C}_{n+1}$  identities on objects corresponds to a simplicial category, with  $\underline{\mathcal{C}}(x, y)_n := \mathcal{C}_n(x, y)$ . The face and degeneracy functors specify the simplicial actions.

Simplicial functors  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$  are collections of functors  $F_n : \mathcal{C}_n \rightarrow \mathcal{D}_n$  that *commute with the simplicial operator functors*, hence having the same underlying object functions. For example,  $F : \underline{\mathcal{C}} \rightarrow \underline{\mathbf{sSet}}$  gives a simplicial set  $F(x)$  for each  $x \in \underline{\mathcal{C}}$ , and a morphism  $F_{x,y} : \underline{\mathcal{C}}(x, y) \rightarrow \underline{\mathbf{sSet}}(x, y)$  for each  $x, y$ ; equivalently a *function* mapping each  $n$ -simplex in  $\underline{\mathcal{C}}(x, y)$  to a map  $\Delta^n \times F(x) \rightarrow F(y)$  natural in  $n$ , meaning that face and degeneracies of the map correspond to the faces and degeneracies of the  $n$ -simplex.

A simplicial natural transformation between simplicial functors  $F, G : \underline{\mathcal{C}} \rightarrow \underline{\mathcal{D}}$  is a collection of arrows in the underlying category  $\mathcal{D}$  for each object of  $\underline{\mathcal{C}}$ , satisfying the enriched naturality condition. Here, that enriched naturality condition says that the underlying natural transformation is natural between the underlying functors, but also that the degenerate images form natural transformations. For example, the collection of images  $s_0(\alpha_x) \in \mathcal{D}_1(F(x), G(x))$  forms a natural transformation  $F_1 \Rightarrow G_1$ ; and in general the unique degeneracy operator  $[n] \rightarrow [0]$  forms a natural transformation  $F_n \Rightarrow G_n$ .

### 3.7 Tensors and Cotensors

We will now prove a statement that was stated earlier. The proof was delegated to this section for pedagogical purposes. Given  $(\mathcal{V}, \times, *)$  closed monoidal<sup>2</sup>, we need to show that  $\underline{\mathcal{V}}(u \times v, w) \cong \underline{\mathcal{V}}(u, \underline{\mathcal{V}}(v, w))$ . Note for any  $x \in \mathcal{V}$  that there is a sequence of natural isomorphisms

$$\mathcal{V}(x, \underline{\mathcal{V}}(u \times v, w)) \cong \mathcal{V}(x \times u \times v, w) \cong \mathcal{V}(x \times u, \underline{\mathcal{V}}(v, w)) \cong \mathcal{V}(x, \underline{\mathcal{V}}(u, \underline{\mathcal{V}}(v, w)))$$

where we have used various results discussed previously. Hence, these two elements represent the same functor, and by the Yoneda lemma they are isomorphic. We also have from our previous discussion that for any  $v \in \mathcal{V}$ , the represented functor  $\underline{\mathcal{V}}(v, -) : \mathcal{V} \rightarrow \mathcal{V}$  is a  $\mathcal{V}$ -functor (as viewed as a functor between the enriched structures). It follows that the left adjoint  $- \times v : \mathcal{V} \rightarrow \mathcal{V}$  also inherits  $\mathcal{V}$ -functor structure. Given  $\eta^v$  the unit of the adjunction at  $v$ , we very simply have

$$\underline{\mathcal{V}}(u, w) \xrightarrow{(\eta_w^v)^*} \underline{\mathcal{V}}(u, \underline{\mathcal{V}}(v, w \times v)) \cong \underline{\mathcal{V}}(u \times v, w \times v).$$

Define the required action on hom-objects as this composition.

This generalizes. Let  $\underline{\mathcal{M}}$  and  $\underline{\mathcal{N}}$  be  $\mathcal{V}$ -categories with  $F \dashv G$  an adjunction between the underlying categories

$$\mathcal{N}(F(m), n) \cong \mathcal{M}(m, G(n)).$$

<sup>2</sup>Here I adhere to the book's convention of using  $\times$  instead of  $\otimes$  for the monoidal product. This means  $\times$  does not denote the product, but a monoidal product.

We can try to enrich this adjunction to an isomorphism in  $\mathcal{V}$ . If we want to use the same idea as previously, meaning we want  $\underline{\mathcal{N}}(F(m), n)$  and  $\underline{\mathcal{M}}(m, G(n))$  to represent the same functor, leads us to consider  $\mathcal{V}(v, \underline{\mathcal{M}}(m, G(n)))$ . However there's not a clear way to proceed from here, and we require some special adjunction conditions. We can prove this if each  $\underline{\mathcal{M}}(m, -)$  and  $\underline{\mathcal{N}}(n, -)$  has a left adjoint preserved by  $F$ , or dually if the contravariant hom-functors have right adjoints preserved by  $G$ . We will return to this situation shortly after working through the definitions this motivates.

**Definition 3.9 (Tensorred).** A  $\mathcal{V}$ -category  $\underline{\mathcal{M}}$  is *tensorred* if for each  $v \in \mathcal{V}$  and  $m \in \underline{\mathcal{M}}$ , there is an object  $v \otimes m \in \underline{\mathcal{M}}$  and isomorphisms

$$\underline{\mathcal{M}}(v \otimes m, n) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n)).$$

♡

By Yoneda, there is a unique way to make  $\otimes : \mathcal{V} \times \underline{\mathcal{M}} \rightarrow \underline{\mathcal{M}}$  a bifunctor natural in all three variables. Hence each  $\underline{\mathcal{M}}(m, -)$  admits a left  $\mathcal{V}$ -adjoint  $- \otimes m$ .

**Definition 3.10 (Cotensorred).** A  $\mathcal{V}$ -category  $\underline{\mathcal{M}}$  is *cotensorred* if for each  $v \in \mathcal{V}$  and  $m \in \underline{\mathcal{M}}$ , there is an object  $n^v$  and isomorphisms

$$\underline{\mathcal{M}}(m, n^v) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n)).$$

♡

The dual statements are true: there is a unique bifunctor  $(-)^- : \mathcal{V}^{\text{op}} \times \underline{\mathcal{M}} \rightarrow \underline{\mathcal{M}}$  with isomorphisms natural in all variables, and each  $\underline{\mathcal{M}}(-, n)$  has a right  $\mathcal{V}$ -adjoint  $n^-$ .

For example, a closed  $\mathcal{V}$ -category is a monoidal category which is tensorred, with  $\otimes = \times$  simply the monoidal product, and which is cotensorred with  $\underline{\mathcal{V}}(u, v) = v^u$ . Hence any locally small category with products and coproducts is enriched, tensorred, and cotensorred over  $\text{Set}$  with the usual cartesian monoidal structure: the cotensore is the copower, and the tensor is the power.

**Lemma 3.5.** *Let  $(\mathcal{V}, \times, *)$  be closed symmetric monoidal, and let  $\underline{\mathcal{M}}$  be a tensorred  $\mathcal{V}$ -category. Then tensor product is unital and associative.*

**Proof.** Recall that we have natural isomorphisms

$$\mathcal{V}(w, \underline{\mathcal{V}}(*, v)) \cong \mathcal{V}(w \times *, v) \cong \mathcal{V}(w, v).$$

Thus  $\underline{\mathcal{V}}(*, v)$  and  $v$  represent the same functor, and are thus isomorphic by the Yoneda lemma. Thus we obtain

$$\underline{\mathcal{M}}(* \otimes m, n) \cong \underline{\mathcal{V}}(*, \underline{\mathcal{M}}(m, n)) \cong \underline{\mathcal{M}}(m, n).$$

Therefore,  $* \otimes m$  and  $m$  represent the same  $\mathcal{V}$ -functor, so by lemma 3.4 we have  $* \otimes m \cong m$ . Since these isomorphisms are natural, we have that there is a natural isomorphism  $* \otimes - \cong 1$ , hence  $\otimes$  is unital.

Similarly we obtain

$$\underline{\mathcal{M}}((v \times w) \otimes m, n) \cong \underline{\mathcal{V}}(v \times w, \underline{\mathcal{M}}(m, n)) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{V}}(w, \underline{\mathcal{M}}(m, n))) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(w \otimes m, n)) \cong \underline{\mathcal{M}}(v \otimes (w \otimes m), n).$$

This proves the associativity property

$$(v \times w) \otimes m \cong v \otimes (w \otimes m).$$

**Proposition 3.2.** *Let  $\underline{\mathcal{M}}, \underline{\mathcal{N}}$  be tensored and cotensored  $\mathcal{V}$ -categories and  $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$  an adjunction between the underlying categories. Then any of the following determines the others:*

- A  $\mathcal{V}$ -adjunction  $\underline{\mathcal{N}}(F(m), n) \cong \underline{\mathcal{M}}(m, G(n))$ ;
- A  $\mathcal{V}$ -functor  $F$  and natural isomorphisms  $F(v \otimes m) \cong v \otimes F(m)$ ;
- A  $\mathcal{V}$ -functor  $G$  and natural isomorphisms  $G(n^v) \cong G(n)^v$ .

**Proof.** Left as an exercise. It suffices to show they are equivalent criteria. First we show 1 implies 2. Let  $F \dashv G$  induce a  $\mathcal{V}$ -adjunction. We want to show  $F(v \otimes m) \cong v \otimes F(m)$ . By the  $\mathcal{V}$ -adjunction, we have

$$\underline{\mathcal{N}}(F(v \otimes m), n) \cong \underline{\mathcal{M}}(v \otimes m, G(n)).$$

By definition of a tensored category, we have

$$\underline{\mathcal{M}}(v \otimes m, G(n)) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, G(n))).$$

By lemma 3.4, since  $\underline{\mathcal{M}}(m, G(n)) \cong \underline{\mathcal{N}}(F(m), n)$  are isomorphic, they represent the same  $\mathcal{V}$ -functor, hence

$$\underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, G(n))) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{N}}(F(m), n)) \cong \underline{\mathcal{V}}(v \otimes F(m), n),$$

where the last isomorphism once again follows by definition of the tensor. Thus,  $F(v \otimes m)$  and  $v \otimes F(m)$  represent the same  $\mathcal{V}$ -functor, so by lemma 3.4 they are isomorphic. Since all the isomorphisms here were natural, so is that isomorphism.

Now we show 2 implies 3. Since  $\mathcal{M}$  is tensored and cotensored,

$$\underline{\mathcal{M}}(v \otimes m, n) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n)) \cong \underline{\mathcal{M}}(m, n^v).$$

The same is true for  $\mathcal{N}$ . The following sequence of isomorphisms makes heavy use of this fact, and makes use of 2-functoriality of the underlying object functor to lower these  $\mathcal{V}$ -adjunctions to regular adjunctions. We have

$$\begin{aligned} \mathcal{N}(F(v \otimes m), n) &\cong \mathcal{N}(v \otimes F(m), n) \\ &\cong \mathcal{V}(v, \underline{\mathcal{N}}(F(m), n)) \\ &\cong \mathcal{V}(v, \underline{\mathcal{M}}(m, G(n))) \\ &\cong \mathcal{M}(m, G(n)^v) \\ \mathcal{V}(v, \underline{\mathcal{N}}(F(m), n)) &\cong \mathcal{N}(F(m), n^v) \\ &\cong \mathcal{M}(m, G(n^v)). \end{aligned}$$

By the same logic as before, we have the isomorphism, as required.

Now we show 3 implies 1. Since  $G(n^v) \cong G(n)^v$ , we have

$$\underline{\mathcal{M}}(m, G(n^v)) \cong \underline{\mathcal{M}}(m, G(n)^v) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, G(n))).$$



We also have since  $F \dashv G$

$$\mathcal{M}(m, G(n^v)) \cong \mathcal{N}(F(m), n^v) \cong \mathcal{V}(v, \underline{\mathcal{N}}(F(m), n)).$$

By the same logic as previously, we are done. ■

**Theorem 3.1.** *Let  $F : \mathcal{V} \rightleftarrows \mathcal{U} : G$  be an adjunction between closed symmetric monoidal categories such that  $F$  is strong monoidal. Then any tensored, cotensored, and enriched  $\mathcal{U}$ -category can be canonically enriched, tensored, and cotensored over  $\mathcal{V}$ .*

**Proof.** Some paper from 1974 shows that  $F$  strong monoidal implies  $G$  is lax monoidal, and hence by the change of base theorem, or more generally proposition 3.1, we can change the base of enrichment from  $\mathcal{U}$  to  $\mathcal{V}$ . Given that  $\mathcal{M}$  is a tensored and cotensored  $\mathcal{U}$ -category

$$\underline{\mathcal{M}}_{\mathcal{U}}(u \otimes m, n) \cong \underline{\mathcal{U}}(u, \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \cong \underline{\mathcal{M}}_{\mathcal{U}}(m, n^u),$$

we define the tensor  $- \star - : \mathcal{V} \times \mathcal{M} \rightarrow \mathcal{M}$ , cotensor  $- \pitchfork - : \mathcal{V}^{\text{op}} \times \mathcal{M} \rightarrow \mathcal{M}$ , and internal hom  $\underline{\mathcal{M}}_{\mathcal{V}}(-, -) : \mathcal{M}^{\text{op}} \times \mathcal{M} \rightarrow \mathcal{V}$  by

$$v \star m := F(v) \otimes m, \quad v \pitchfork m := m^{F(v)}, \quad \underline{\mathcal{M}}_{\mathcal{V}}(m, n) := G\underline{\mathcal{M}}_{\mathcal{U}}(m, n).$$

We will show the required natural isomorphism

$$\underline{\mathcal{M}}_{\mathcal{V}}(v \star m, n) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n)) \cong \underline{\mathcal{M}}(m, v \pitchfork n)$$

in the same way as before, by showing they represent the same functor  $\mathcal{V}^{\text{op}} \rightarrow \text{Set}$ . The first isomorphism is given as

$$\begin{aligned} \mathcal{V}(x, \underline{\mathcal{M}}_{\mathcal{V}}(v \star m, n)) &= \mathcal{V}(x, G\underline{\mathcal{M}}_{\mathcal{U}}(F(v) \otimes m, n)) \\ &\cong \mathcal{U}(F(x), \underline{\mathcal{M}}_{\mathcal{U}}(F(v) \otimes m, n)) \\ &\cong \mathcal{M}(F(x) \otimes (F(v) \otimes m), n) \\ &\cong \mathcal{M}((F(x) \times F(v)) \otimes m, n) \\ &\cong \mathcal{U}(F(x) \times F(v), \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\ &\cong \mathcal{U}(F(x \times v), \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\ &\cong \mathcal{V}(x \times v, G\underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\ &= \mathcal{V}(x \times v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n)) \\ &\cong \mathcal{V}(x, \underline{\mathcal{V}}(v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n))). \end{aligned}$$

The second isomorphism is given by going the reverse direction. Note that dualizing associativity of the tensor gives the property

$$n^{u \times v} \cong (n^u)^v.$$

Hence we obtain

$$\begin{aligned}
\mathcal{V}(x, \underline{\mathcal{V}}(v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n))) &\cong \mathcal{V}(x \times v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n)) \\
&= \mathcal{V}(x \times v, G\underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\
&\cong \mathcal{U}(F(x \times v), \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\
&\cong \mathcal{U}(F(x) \times F(v), \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\
&\cong \mathcal{M}(m, n^{F(x) \times F(v)}) \\
&\cong \mathcal{M}(m, n^{F(v) \times F(x)}) \\
&\cong \mathcal{M}(m, (n^{F(v)})^{F(x)}) \\
&= \mathcal{M}(m, (v \multimap n)^{F(x)}) \\
&\cong \mathcal{U}(F(x), \underline{\mathcal{M}}_{\mathcal{U}}(m, v \multimap n)) \\
&\cong \mathcal{V}(x, G\underline{\mathcal{M}}_{\mathcal{U}}(m, v \multimap n)) \\
&= \mathcal{V}(x, \underline{\mathcal{M}}_{\mathcal{V}}(m, v \multimap n)).
\end{aligned}$$

**Corollary 3.1.** *The adjunction  $F \dashv G$  of the previous ??, hereafter called a strong monoidal adjunction, is a  $\mathcal{V}$ -adjunction with respect to the  $\mathcal{V}$ -structure induced on  $\mathcal{U}$ .*

**Proof.** By the yoneda lemma and the definitions above,

$$\begin{aligned}
\mathcal{V}(w, \underline{\mathcal{U}}_{\mathcal{V}}(F(v), u)) &\cong \mathcal{V}(w, G\underline{\mathcal{U}}_{\mathcal{U}}(F(v), u)) \cong \mathcal{U}(F(w), \underline{\mathcal{U}}_{\mathcal{U}}(F(v), u)) \cong \mathcal{U}(F(w) \times F(v), u) \\
&\cong \mathcal{U}(F(w \times v), u) \cong \mathcal{V}(w \times v, G(u)) \cong \mathcal{V}(w, \underline{\mathcal{V}}(v, G(u))).
\end{aligned}$$

Results following from this show that the categories of based simplicial sets and based topological spaces are enriched, tensored, and cotensored. For example, the cotensor  $Y^K$  in  $\underline{\mathbf{sSet}}_*$  is  $\underline{\mathbf{sSet}}_*(K_+, Y)$ .

More interestingly, we will show later that the geometric realization is a strong monoidal left adjoint  $|-| \dashv S$ . Because the convenient category of topological spaces  $\mathbf{Top}$  is closed monoidal, it is thus tensored, cotensored, and enriched over simplicial sets with definitions as  $K \otimes X := |K| \times X$  and  $X^K := \mathbf{Top}(|K|, X)$ . The simplicial enrichment has as hom-objects  $S(\underline{\mathbf{Top}}(X, Y))$ . Hence this is a simplicial adjunction.

**Exercise.** Let  $\mathcal{M}$  be cocomplete. Show that  $\mathcal{M}^{\Delta^{\text{op}}}$ , the category of simplicial objects in  $\mathcal{M}$ , is simplicially enriched and tensored, with  $(K \otimes X)_n := K_n \cdot X_n$  the copower.

**Proof.** The proof is messy and I don't want to copy it down. The internal hom is defined the same way as the closed structure on  $\mathbf{sSet}$ , as one would expect:

$$\underline{\mathcal{M}}^{\Delta^{\text{op}}}(X, Y)_n := \mathcal{M}^{\Delta^{\text{op}}}(\Delta^n \otimes X, Y).$$

Simplicial maps are induced in the expected way via precomposition. Since a closed structure is the monoidal product also being a tensor, this definition agrees with the definition for the closed monoidal structure on  $\mathbf{sSet}$ .

For example, an unenriched adjunction  $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$  promotes to an adjunction given by

levelwise postcomposition

$$F_* : \mathcal{M}^{\Delta^{\text{op}}} \rightleftarrows \mathcal{N}^{\Delta^{\text{op}}} : G_*.$$

Since  $F_*$  is left adjoint, it preserves coproducts, and thus tensors. By proposition 3.2,  $F_* \dashv G_*$  is thus a simplicial adjunction.

Another interesting result is the Dold-Kan correspondence: there is an equivalence of categories  $\mathcal{A}b^{\Delta^{\text{op}}} \rightleftarrows \mathcal{C}h_{\geq 0}(\mathbb{Z})$ .

### 3.8 Simplicial homotopy and simplicial model categories

If  $\underline{\mathcal{M}}$  is a simplicial category, we obtain a canonical notion of homotopy in the underlying category  $\mathcal{M}$ . Recall that  $\Delta^0$  is the unit for the monoidal product in  $\mathbf{sSet}$ . A *homotopy* between morphisms  $f, g : m \rightarrow n$  is given as the commutative diagram

$$\begin{array}{ccc} \Delta^0 & & \\ d^1 \downarrow & \searrow f & \\ \Delta^1 & \xrightarrow{H} & \underline{\mathcal{M}}(m, n) \\ d^0 \uparrow & \nearrow g & \\ \Delta^0 & & \end{array}$$

If  $\underline{\mathcal{M}}$  is tensored, then we can realize the homotopy as a homotopy in the underlying category by recalling that we have an isomorphism

$$\mathbf{sSet}(-, \underline{\mathcal{M}}(m, n)) \cong \mathcal{M}(- \otimes m, n).$$

Thus the diagram transposes as

$$\begin{array}{ccc} \Delta^0 \otimes m \cong m & & \\ d^1 \otimes 1 \downarrow & \searrow f & \\ \Delta^1 \otimes m & \xrightarrow{H} & n \\ d^0 \otimes 1 \uparrow & \nearrow g & \\ \Delta^0 \otimes m \cong m & & \end{array}$$

Similarly, if  $\mathcal{M}$  is cotensored, then we obtain a homotopy in  $\mathcal{M}$  as

$$\begin{array}{ccc} & n \cong n^{\Delta^0} & \\ & \uparrow n^{d^1} & \\ m & \xrightarrow{H} n^{\Delta^1} & \\ & \downarrow n^{d^0} & \\ & n \cong n^{\Delta^0} & \end{array}$$

$\begin{array}{ccc} f \nearrow & & \nwarrow g \\ & m & \end{array}$

Thus we obtain a notion of a simplicial homotopy equivalence if  $\mathcal{M}$  is either tensored or cotensored over simplicial sets.

**Remark.** Some explanation of why this definition of homotopy makes sense. The diagram says that the composition  $H \circ d^1 = f$  and  $H \circ d^0 = g$ . Recall that  $d_i$  is the face operator which skips  $i$ . Thus  $d^i$  here is postcomposition by the relevant face operator, and thus we can think of  $f$  as being inserted into  $\Delta^1$  at 0, and  $g$  inserted at 1. Thus a homotopy is a morphism which agrees with  $f$  when restricted to 0 and agrees with  $g$  when restricted to 1, similar to the topological notion. Tensoring/cotensoring then simply recovers the definition in the underlying category.

**Definition 3.11 (Geometric Realization).** If  $\mathcal{M}$  is simplicially enriched, tensored, and cocomplete, the *geometric realization* of a simplicial object  $X_\bullet$  in  $\mathcal{M}$  is

$$|X_\bullet| := \int^{n \in \Delta} \Delta^n \otimes X_n.$$

These coends define a functor  $|-| : \mathcal{M}^{\Delta^{\text{op}}} \rightarrow \mathcal{M}$ . ♥

If  $\mathcal{M}$  is simplicially tensored, there are two reasonable definitions for simplicial tensors on  $\mathcal{M}^{\Delta^{\text{op}}}$ . One can define tensors pointwise for any diagram category  $\mathcal{M}^{\mathcal{D}}$ : given  $K \in \mathbf{sSet}$  and  $X : \mathcal{D} \rightarrow \mathcal{M}$ , define  $K \otimes X$  as the functor  $d \mapsto K \otimes X(d)$ . The category of simplicial objects is then a special case thereof. However, we prefer to define tensors as done earlier, via  $(K \otimes X)_n := K_n \cdot X_n$ .

**Lemma 3.6.** *Let  $\mathcal{M}$  be simplicially enriched, tensored, cotensored, and admits geometric realizations. Then geometric realization preserves tensors.*

Will be proven later. The key is that tensor admits pointwise right adjoints, and thus preserves colimits in both variables. We present this result now due to the following corollary.

**Corollary 3.2.** *If  $\mathcal{M}$  is simplicially enriched, tensored, and cotensored, then geometric realization preserves simplicial homotopy equivalences.*

**Proof.** Given a simplicial homotopy equivalence  $fg, gf \sim 1$  between simplicial objects  $X_\bullet, Y_\bullet$  in  $\mathcal{M}^{\Delta^{\text{op}}}$ , since geometric realization preserves tensors, geometric realization produces simplicial homotopy equivalences in  $\mathcal{M}$  between the geometric realizations of the simplicial objects. ■

The chapter is ended with the motivation we sought at the end of the previous chapter. The following lemma defines some properties of simplicial model categories, which will be properly introduced later. This result suffices as a working understanding for now.

**Lemma 3.7.** *A simplicial model category*

- *is complete and cocomplete;*
- *is tensored, cotensored, and enriched over simplicial sets;*
- *has subcategories of cofibrant and fibrant objects preserved by tensoring and cotensoring, respectively, with any simplicial set;*
- *admits a left deformation  $Q$  into the cofibrant objects and a right deformation  $R$  into the fibrant objects;*
- *is a saturated homotopical category;*
- *has simplicial homotopy equivalences among the weak equivalences;*
- *has the property that the internal hom preserves weak equivalences between cofibrant objects in its first variable provided the second variable is fibrant, and preserves weak equivalences between fibrant objects in its second variable provided the first is cofibrant.*

Examples include  $\mathbf{sSet}$ ,  $\mathbf{sSet}_*$ ,  $\mathcal{T}\mathbf{op}$ ,  $\mathcal{T}\mathbf{op}_*$ . There are analogous definitions of  $\mathcal{V}$ -model categories, which are tensored, cotensored, and enriched over  $\mathcal{V}$  instead of  $\mathbf{sSet}$ . A homotopical version of theorem 3.1 says that if the strong monoidal adjunction is Quillen, then any  $\mathcal{U}$ -model category is a  $\mathcal{V}$ -model category. The adjunctions

$$(-)_+ : \mathbf{sSet} \rightleftarrows \mathbf{sSet}_* : U, \quad |-| : \mathbf{sSet} \rightleftarrows \mathcal{T}\mathbf{op} : S$$

are strong monoidal and Quillen, provided  $\mathcal{T}\mathbf{op}$  is once again a convenient category. Hence, any based simplicial or topological model category is a simplicial model category. The converse is not always true however. We determine that simplicial model categories are an appropriate place to define homotopy limits and colimits.

## Chapter 4

### The unreasonably effective (co)bar construction

The bar originated from Eilenberg and Mac Lane in homological algebra as a tool for constructing resolutions. We present a version of it at the right level of generality for this text, although it generalizes further. We will choose our notation to reflect the fact that the bar construction generalizes to enriched contexts, but for this section we will focus on simplicial categories. Thus,  $\mathcal{V}$  should be interpreted in general as  $\mathbf{sSet}$ , and  $\mathcal{M}$  should represent a simplicially enriched, tensored, and cotensored category. The enrichment is not as important as is the notion of a simplicial homotopy.

#### 4.1 Functor tensor products

**Definition 4.1** (Functor Tensor Product). Let  $- \otimes - : \mathcal{V} \times \mathcal{M} \rightarrow \mathcal{M}$  be any bifunctor, and let  $F : \mathcal{D} \rightarrow \mathcal{M}$  and  $G : \mathcal{D}^{\text{op}} \rightarrow \mathcal{M}$  be functors. The functor tensor product  $G \otimes_{\mathcal{D}} F$  is defined as the coend

$$G \otimes_{\mathcal{D}} F := \int^{\mathcal{D}} G \otimes F.$$

♥

The codomain is allowed to be a category other than  $\mathcal{M}$ , but we don't need this level of generalization. Usually,  $\otimes$  will be the tensor in some tensored  $\mathcal{V}$ -category.

There are many examples. For example, given a right  $R$ -module  $A : R^{\text{op}} \rightarrow \underline{\mathcal{A}b}$  and a left  $R$ -module  $B : R \rightarrow \underline{\mathcal{A}b}$ , the usual tensor product of  $R$ -modules is the functor tensor product

$$A \otimes_R B = \int^R A \otimes_{\mathbb{Z}} B.$$

Another example is that if  $*$  :  $\mathcal{D}^{\text{op}} \rightarrow \mathbf{sSet}$  is the constant functor at the terminal object, then for any  $F : \mathcal{D} \rightarrow \mathbf{sSet}$ ,

$$* \otimes_{\mathcal{D}} F = \int^{\mathcal{D}} * \times F \cong \int^{\mathcal{D}} F \cong \varinjlim F.$$

Recall that pointwise Kan extensions are computed as

$$\text{Lan}_K F(d) = \int^{c \in \mathcal{C}} \mathcal{D}(K(c), d) \cdot F(c) = \mathcal{D}(K(-), d) \otimes_{\mathcal{C}} F.$$

In particular, taking  $\Delta^\bullet : \Delta \rightarrow \mathcal{T}op$  as the functor mapping to the standard  $n$ -simplex, the yoneda lemma gives the geometric realization of a simplicial set as

$$|X| = \int^{n \in \Delta} \mathbf{sSet}(\Delta^n, X) \cdot X_n \cong \mathbf{sSet}(\Delta^\bullet, X) \otimes_{\Delta} \Delta^\bullet \cong X_\bullet \otimes_{\Delta} \Delta^\bullet.$$

For the geometric realization of simplicial objects in a category tensored over simplicial sets, as given in definition 3.11, we have

$$|X| := \Delta^\bullet \otimes_{\Delta^{\text{op}}} X_\bullet.$$

Here  $\Delta^\bullet$  is the Yoneda embedding  $\Delta \rightarrow \mathbf{sSet}$  for simplicial categories. We always use this except when  $X$  is a simplicial set, in which case convention dictates that the tensor is the copower, which ends up returning the simplicial set  $X$  itself. We instead define  $|X|$  as a topological space, as above.

## 4.2 The bar construction

Let  $\mathcal{M}$  be simplicially enriched, tensored, and cotensored, and let  $G : \mathcal{D}^{\text{op}} \rightarrow \mathbf{sSet}$  and  $F : \mathcal{D} \rightarrow \mathcal{M}$ .

**Definition 4.2 (Bar).** The two-sided simplicial bar construction is a simplicial object  $B_\bullet(G, \mathcal{D}, F)$  in  $\mathcal{M}$  whose  $n$ -simplices are defined by the coproduct

$$B_n(G, \mathcal{D}, F) := \coprod_{\vec{d}: [n] \rightarrow \mathcal{D}} G(d_n) \otimes F(d_0),$$

where  $\vec{d}$  is a shorthand for a sequence  $d_0 \rightarrow d_1 \rightarrow \dots \rightarrow d_n$  of composable arrows in  $\mathcal{D}$ . ♥

**Remark.** This is a very important remark. Note that we are taking the coproduct along all possible chains of length  $n$  from an object  $d_0$  to an object  $d_n$ . Fix some list of objects  $(d_0, \dots, d_n)$ . We then obtain that the number of times  $G(d_n) \otimes F(d_0)$  factors into the coproduct as a result of this chain is equivalent to the number of possible paths along this fixed path; equivalently the product

$$\prod_{i=0}^{n-1} \mathcal{D}(d_i, d_{i+1}).$$

Hence we represent the above coproduct equivalently as

$$B_n(G, \mathcal{D}, F) = \coprod_{(d_0, \dots, d_n)} \left( \prod_{i=0}^{n-1} \mathcal{D}(d_i, d_{i+1}) \right) \cdot (G(d_n) \otimes F(d_0)),$$

where the coproduct now only runs along distinct paths of *objects*, instead of distinct paths of arrows. View the hom-sets as discrete simplicial sets (see explanation/definition in below exercise). Then (also with explanation as below), the tensor can be viewed as a product, which is isomorphic to the copower. By uniqueness of adjunction, we then obtain the representation via tensors

$$B_n(G, \mathcal{D}, F) = \coprod_{(d_0, \dots, d_n)} \left( \bigotimes_{i=0}^n \mathcal{D}(d_i, d_{i+1}) \right) \otimes G(d_n) \otimes F(d_0).$$

These are all other common and useful expressions of the simplicial two-sided bar construction.

The diagram  $\vec{d}$  is simply an  $n$ -simplex in the nerve  $N(\mathcal{D})$ . Thus, we define the degeneracy and face maps as reindexing the coproduct according to the degeneracy and face maps in  $N(\mathcal{D})$ . The  $n$ th face map is defined by applying  $G$  to  $d_{n-1} \rightarrow d_n$  to obtain a map  $G(d_n) \otimes F(d_0) \rightarrow G(d_{n-1}) \otimes F(d_0)$  and includes this at the restriction of  $\vec{d}$  along  $d^n$ . The 0th face map is defined similarly. We will see later that the two-sided bar is a weighted colimit.

**Definition 4.3** (Bar). The *bar construction* is the geometric realization of the simplicial bar construction

$$B(G, \mathcal{D}, F) = |B_\bullet(G, \mathcal{D}, F)| = \Delta^\bullet \otimes_{\Delta^{\text{op}}} B_\bullet(G, \mathcal{D}, F) = \int^{n \in \Delta^{\text{op}}} \Delta^n \otimes B_n(G, \mathcal{D}, F).$$

♡

Note here that since  $B_\bullet(G, \mathcal{D}, F) : \Delta^{\text{op}} \rightarrow \mathcal{M}$ , and since  $\Delta^\bullet$  is the standard simplicial set given by Yoneda, the tensor product is simply the tensor in  $\mathcal{M}$ .

Some examples. If  $F, G$  are the constant functors mapping to the terminal object, then  $B_\bullet(*, \mathcal{D}, *) = N(\mathcal{D})$ . We are particularly interested in the case when  $G$  is a representable functor  $G = \mathcal{D}(-, d)$ . This follows since

$$\mathcal{D}(-, d) \otimes_{\mathcal{D}} F \cong F(d), \quad G \otimes_{\mathcal{D}} \mathcal{D}(d, -) \cong G(d)$$

which is true for general functor tensor products, by the coYoneda lemma. By the Yoneda lemma, the assignment

$$d \mapsto B(\mathcal{D}(-, d), \mathcal{D}, F)$$

is functorial, and we write  $B(\mathcal{D}, \mathcal{D}, F) : \mathcal{D} \rightarrow \mathcal{M}$  for this functor. Allowing  $F$  to vary over this also gives a functor  $B(\mathcal{D}, \mathcal{D}, -) : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}^{\mathcal{D}}$ .

**Exercise.** Let  $\mathcal{M} = \text{Set}$ . Show that  $B_\bullet(\mathcal{D}(-, d), \mathcal{D}, *) \cong N(\mathcal{D}/d)$ , so that  $B_\bullet(\mathcal{D}, \mathcal{D}, *)$  is naturally isomorphic to  $N(\mathcal{D}/-) : \mathcal{D} \rightarrow \text{sSet}$ .

**Proof.** Recall the slice category  $\mathcal{D}/d = (\mathcal{D} \downarrow d)$ . Note also that we view  $\text{Set}$  as enriched over  $\text{sSet}$  by taking hom-sets to be *discrete simplicial sets*, meaning face and degeneracies are all 1, and  $X_n = X_0$  for all  $n$ . By this same logic, we can view  $\mathcal{D}(-, d)$  as a functor to  $\text{sSet}$  instead of  $\text{Set}$ . Under this definition, a tensor product  $- \otimes - : \text{sSet} \times \text{Set} \rightarrow \text{Set}$  must be left-adjoint to the internal hom of  $\text{Set}$ , so by lemma 3.4 we need only show tensor is left adjoint to regular hom, since the underlying functor of the discrete simplicial set representing hom is hom. Indeed, this is simply the standard cartesian monoidal structure on  $\text{Set}$ , so we obtain that tensor can be defined as

$$K \otimes X := K_0 \times X.$$

Thus, we obtain

$$B_n = \coprod_{\vec{d}: [n] \rightarrow \mathcal{D}} \mathcal{D}(d_n, d) \times 1 \cong \coprod_{\vec{d}: [n] \rightarrow \mathcal{D}} \mathcal{D}(d_n, d) = \coprod_{(d_0, \dots, d_n)} \left( \prod_{i=0}^{n-1} \mathcal{D}(d_i, d_{i+1}) \right) \times \mathcal{D}(d_n, d).$$

$n$ -simplices of the two-sided simplicial bar are thus paths  $d_0 \rightarrow d_n$ , appended with a morphism  $d_n \rightarrow d$ . This determines an  $n$ -simplex in the nerve  $N(\mathcal{D} \downarrow d)$  as follows: let  $f_i : d_i \rightarrow d_{i+1}$  and  $g : d_n \rightarrow d$  together constitute an  $n$ -simplex in  $B_\bullet$ . For each  $0 \leq i \leq n$ , define an object  $\alpha_i : d_i \rightarrow d$  in the slice category as the composition

$$\alpha_i : d_i \xrightarrow{f_i} d_{i+1} \xrightarrow{f_{i+1}} \dots \xrightarrow{f_{n-1}} d_n \xrightarrow{g} d.$$



We then obtain that each  $f_i$  is a morphism  $\alpha_i \rightarrow \alpha_{i+1}$  in the slice category by definition, since we can rewrite the above definition as an inductive definition given by

$$\alpha_i = \alpha_{i+1} \circ f_i, \quad \alpha_n = g.$$

Hence,  $f_\bullet$  is a chain of morphisms in  $(\mathcal{D} \downarrow d)$ , and thus an  $n$ -simplex in the nerve. Hence we obtain an obvious map by dropping  $g$ . It is easiest from here to note that the map is manifestly surjective, and it is injective obviously along  $i < n$ , but also for  $\alpha_n$  since different choices of  $g$  give a different object here. Hence we have the set isomorphism, as needed.

Raising the isomorphism to that of simplicial sets requires showing the isomorphism commutes with the face and degeneracy operators. We must here recall the definition of the face and degeneracy operators for the nerve and the bar. Recall that in the nerve, face operators  $d^i : N(\mathcal{D} \downarrow d)_n \rightarrow N(\mathcal{D} \downarrow d)_{n-1}$  are given by removing the  $i$ th object and composing the adjacent morphisms. That is, given an  $n$ -simplex  $\sigma = (f_0, \dots, f_{n-1})$ , we define

$$d^i(\sigma)_j = \begin{cases} \sigma_j, & 0 \leq j < i-1 \\ \sigma_i \circ \sigma_{i-1}, & j = i-1 \\ \sigma_{j+1}, & j \geq i \end{cases}.$$

This of course does not cover the edge cases  $i = 0, n$ ; those are solved more simply by simply removing the trailing object and the trailing arrow. Face maps  $s^i : N(\mathcal{D} \downarrow d)_n \rightarrow N(\mathcal{D} \downarrow d)_{n+1}$  are defined more simply by replacing the  $i$ th object with two adjacent copies of it, with the arrow between them the identity morphism. The  $n$ -simplices of the two sided simplicial bar can be seen as indexed over the  $n$ -simplices of the nerve, so it becomes reindexed in accordance to applying the face and degeneracy operators of the nerve. This makes the fact that they commute with the given isomorphism fairly obvious. ■

Dually,  $B_\bullet(*, \mathcal{D}, \mathcal{D}(d, -)) \cong N(d \downarrow \mathcal{D})$  and hence  $B_\bullet(*, \mathcal{D}, \mathcal{D}) : \mathcal{D}^{\text{op}} \rightarrow \mathbf{sSet}$ .

### 4.3 The cobar construction

We now spend some time explaining in depth the dual notion to the two sided simplicial bar construction. Let  $F, G : \mathcal{D} \rightarrow \mathcal{M}$ . Recall from previously that we have the natural isomorphism

$$\text{Nat}(G, F) \cong \int_{d \in \mathcal{D}} \mathcal{M}(G(d), F(d)).$$

In the presence of a bifunctor  $\{-, -\} : \mathcal{V}^{\text{op}} \times \mathcal{M} \rightarrow \mathcal{M}$ , we can define the *functor cotensor product* (sometimes *functor hom* for some reason) in the expected way for  $G : \mathcal{D} \rightarrow \mathcal{V}$  and  $F : \mathcal{D} \rightarrow \mathcal{M}$ :

$$\{G, F\}^{\mathcal{D}} := \int_{d \in \mathcal{D}} \{G(d), F(d)\}.$$

Often,  $\{-, -\}$  will be a cotensor, copower, or oftentimes if  $\mathcal{V} = \mathcal{M}$  we just use  $\text{hom}$ .

If  $\mathcal{M}$  is censored over  $\mathbf{sSet}$ , dual to the notion of a geometric realization of a simplicial object  $X_\bullet \otimes_{\Delta^{\text{op}}} \Delta^\bullet$  is the *totalization* of a cosimplicial object  $X^\bullet : \Delta \rightarrow \mathcal{M}$ , given as

$$\text{Tot } X^\bullet := \{\Delta^\bullet, X^\bullet\}^{\Delta}.$$

Here,  $\Delta^\bullet$  denotes the covariant Yoneda embedding  $\Delta \rightarrow \mathbf{sSet}$ .

**Definition 4.4** (Cosimplicial cobar). The *cosimplicial cobar construction*  $C^\bullet(G, \mathcal{D}, F)$  is a cosimplicial object in  $\mathcal{M}$  given by

$$C^n(G, \mathcal{D}, F) := \prod_{\vec{d}: [n] \rightarrow \mathcal{D}} F(d_n)^{G(d_0)}.$$

Here, exponential notation denotes the cotensor in  $\mathcal{M}$ . ♥

Cosimplicial degeneracies and (inner) faces are once again indexed by the corresponding maps in the nerve  $N(\mathcal{D})$ . The 0th face map uses  $G(d_0) \rightarrow G(d_1)$  in  $\mathcal{V}$  to define  $F(d_n)^{G(d_1)} \rightarrow F(d_n)^{G(d_0)}$  by contravariance in the first slot of cotensor. One first projects along the components of the product indexed by the face map (drops the one the face map drops), and then postcomposes by this map. The  $n$ th face map is defined similarly. We then have

**Definition 4.5** (Cobar). The *cobar construction*  $C(G, \mathcal{D}, F)$  is the totalization of the cosimplicial cobar construction

$$C(G, \mathcal{D}, F) := \text{Tot } C^\bullet(G, \mathcal{D}, F) = \{\Delta^\bullet, C^\bullet(G, \mathcal{D}, F)\}^\Delta = \int_{n \in \Delta} \{\Delta^n, C^n(G, \mathcal{D}, F)\}.$$

♥

#### 4.4 Simplicial replacements and colimits

An explanation of the applications of these constructions to homotopy theory is in order. Let  $F : \mathcal{D} \rightarrow \mathcal{M}$  be a diagram. We know already how to compute its colimit as a coequalizer of arrows between coproducts, but we also know that  $\varinjlim_{\mathcal{D}}$  need not be homotopical. We will see in the next chapter that a “fattened up colimit” will preserve pointwise weak equivalences<sup>1</sup> between diagrams.

Let  $\mathcal{D} := \Delta^{\text{op}}$  and  $\mathcal{M}$  simplicial and tensored. Then one way to achieve this is via its geometric realization<sup>2</sup>, as defined previously. One might then try to replace diagrams  $F : \mathcal{D} \rightarrow \mathcal{M}$  with simplicial objects which have the same colimit. To each small  $\mathcal{D}$ , we have the simplicial set  $N(\mathcal{D})$ . The data of a simplicial set  $X$  can be encoded into the *category of simplicies*  $\mathbf{e}lX$  taking simplices  $\sigma \in X_n$  as objects. Morphisms  $\alpha : \sigma \rightarrow \sigma'$  are simplicial operators  $\alpha : [n] \rightarrow [m]$  such that  $X(\alpha)(\sigma') = \sigma$ . Note the similarity to the category of elements for a general functor<sup>3</sup>, and note the forgetful functor  $\mathbf{e}lX \rightarrow \Delta$  given by mapping  $n$ -simplices to  $[n]$ , and the similar expected map on faces and degeneracies.

Putting these ideas together suggests we consider  $\mathbf{e}lN(\mathcal{D})$ . This is given by the category of finite composable strings in  $\mathcal{D}$ . Note that since morphisms are somewhat “reversed” in this category, faces can be seen as *factoring* composites or *appending* strings to the end, and similarly degeneracies can be seen as *deleting* identities. This is somewhat the opposite of their action in the nerve. We denote the forgetful functor  $\Sigma : \mathbf{e}lN(\mathcal{D}) \rightarrow \Delta$ .

Note also there is a functor  $S : \mathbf{e}lN(\mathcal{D}) \rightarrow \mathcal{D}^{\text{op}}$  given by projecting on the source of the string of arrows onto the source, and with the relevant induced action by the morphisms. Taking opposite categories, we might want to consider precomposing our original diagram  $F : \mathcal{D} \rightarrow \mathcal{M}$  by  $S$ ,

<sup>1</sup>A pointwise weak equivalence is a natural way to induce a homotopical structure on a functor category, by taking as weak equivalences all natural weak equivalences.

<sup>2</sup>The geometric realization of a simplicial object is not in general its homotopy colimit

<sup>3</sup>Indeed this can be expressed as the comma category  $(\Delta^\bullet \downarrow X)$  for  $\Delta$  the Yoneda embedding

and then taking the left Kan extension

$$\begin{array}{ccc}
 & \Delta^{\text{op}} & \\
 \Sigma \nearrow & \uparrow & \nwarrow \text{Lan}_{\Sigma} S \\
 (\mathbf{e}N(\mathcal{D}))^{\text{op}} & \xrightarrow{F \circ S} & \mathcal{M}
 \end{array}$$

This gives us a simplicial object in  $\mathcal{M}$  associated to  $F$ . Note that precomposition  $S^* : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}^{(\mathbf{e}N(\mathcal{D}))^{\text{op}}}$  is fully faithful, and thus the counit of the adjunction  $\text{Lan}_S \dashv S^*$  is an isomorphism, hence so is the natural transformation above. Next, letting  $\Delta(-)$  denote the constant functor at  $-$ , we have by universality of colimit and Kan extension

$$\mathcal{M}(\varinjlim_{\mathcal{E}} \text{Lan}_K F, m) \cong \mathcal{M}^{\mathcal{E}}(\text{Lan}_K F, \Delta(m)) \cong \mathcal{M}^{\mathcal{D}}(F, \Delta(m)) \cong \mathcal{M}(\varinjlim_{\mathcal{D}} F, m).$$

Hence, we obtain the colimit of  $F$  as

$$\varinjlim_{\mathcal{D}} F \cong \varinjlim_{\mathcal{D}} \text{Lan}_S(F \circ S) \cong \varinjlim_{\mathbf{e}N(\mathcal{D})^{\text{op}}} (F \circ S) \cong \varinjlim_{\Delta^{\text{op}}} \text{Lan}_{\Sigma}(F \circ S).$$

We can construct this via our earlier pointwise construction via

$$\text{Lan}_{\Sigma}(F \circ S)_n = \varinjlim \left( (\Sigma \downarrow [n]) \xrightarrow{U} (\mathbf{e}N(\mathcal{D}))^{\text{op}} \xrightarrow{S} \mathcal{D} \xrightarrow{F} \mathcal{M} \right).$$

More explicitly, objects in the slice category  $(\Sigma \downarrow [n])$  are sequences  $\vec{d} : [m] \rightarrow \mathcal{D}$  together with morphisms  $[n] \rightarrow [m] = \Sigma(\vec{d})$  in  $\Delta$ . One obtains a terminal object in the comma category via an identity map, and the category can be shown to simply be given as a disjoint union of categories fully determined by this terminal object. The colimit of a functor over a category with a terminal object is simply evaluating at that terminal object. Computing further, the forgetful functor simply returns the composable string  $\vec{d}$ , and composing with  $S$  returns the starting point  $d_0$ , so we obtain that the colimit is computed as a coproduct

$$(\text{Lan}_{\Sigma}(F \circ S))_n = \coprod_{\vec{d} : [n] \rightarrow \mathcal{D}} F(d_0) = B_n(*, \mathcal{D}, F).$$

**Lemma 4.1.** *To each diagram  $F : \mathcal{D} \rightarrow \mathcal{M}$ , there is a simplicial object  $B_{\bullet}(*, \mathcal{D}, F)$  and a cosimplicial object  $C^{\bullet}(*, \mathcal{D}, F)$  such that*

$$\varinjlim_{\mathcal{D}} F \cong \varinjlim_{\Delta^{\text{op}}} B_{\bullet}(*, \mathcal{D}, F), \quad \varprojlim_{\mathcal{D}} F \cong \varprojlim_{\Delta} C^{\bullet}(*, \mathcal{D}, F).$$

**Remark.** The proof that the colimit of a Kan extension is isomorphic to the colimit of the original diagram extends to weighted colimits, in particular functor tensor products, in particular geometric realizations. Let  $\mathcal{M}$  be cocomplete and tensored over simplicial sets. A simplicial object  $X_{\bullet} : \Delta^{\text{op}} \rightarrow \mathcal{M}$  is *n-skeletal* if it is isomorphic to the left Kan extension of its *n*-truncation  $X_{\leq n}$

$$\mathcal{M}^{\Delta^{\text{op}}} \xleftarrow[\text{res}_{\leq n}]{\text{Lan}_{\perp}} \mathcal{M}^{\Delta^{\text{op}}_{\leq n}}.$$

This has the following application for computing geometric realizations.

**Lemma 4.2.** *If  $X$  is  $n$ -skeletal, then*

$$|X| \cong \Delta^\bullet \otimes_{\Delta^{\text{op}}} X_\bullet \cong \Delta^\bullet \otimes_{\Delta^{\text{op}}} \text{Lan } X_{\leq n} \cong \Delta_{\leq n}^\bullet \otimes_{\Delta_{\leq n}^{\text{op}}} X_{\leq n},$$

where the Kan extension is computed along the inclusion functor. In other words, it is isomorphic to its  $n$ -truncation tensored with the restricted Yoneda embedding.

#### 4.5 Augmented simplicial objects and extra degeneracies

Now we consider the computation of something called “the homotopy type” of the geometric realization of a simplicial object in a tensored, cotensored, and simplicially enriched category  $\mathcal{M}$ . Recall an augmented simplicial object includes also the empty ordinal  $0 = [-1]$ , meaning it is a simplicial object  $X$  and an object  $X_{-1}$ , together with a map  $d_0 : X_0 \rightarrow X_{-1}$  such that the two maps  $X_1 \rightrightarrows X_0 \rightarrow X_{-1}$  agree.

In a complete category  $\mathcal{M}$ , any simplicial object obtains an augmentation via the unique map to the terminal object. More explicitly, we define  $X_{-1} = *$ , and the two maps manifestly agree by universality. We can also take  $*$  as the constant simplicial set, and ask when the map  $X \rightarrow *$  is a simplicial homotopy equivalence.

For our expository discussion, we take  $\mathcal{M} = \text{Set}$  for simplicity. Exploring this requires us to consider the notion of a contracting homotopy  $X \rightarrow *$ , which roughly is a method to deform each simplex of  $X$  down to a point. As an example, we could express this as

- a vertex  $* \in X_0$ ;
- for each  $x \in X_0$ , a 1-simplex from  $*$  to  $x_0$ ;
- for each  $f \in X_1$ , a 2-simplex whose initial vertex is  $*$ , whose 0th face is  $f$ , and whose other faces are the previous 1-simplices;
- for each  $\sigma \in X_n$ , an  $n + 1$ -simplex whose initial vertex is  $*$ , whose 0th face is  $\sigma$ , and whose other faces are the previously specified  $n$ -simplices.

Note here that the way we define a 1-simplex *between* 0-simplices is via the face maps. In other words, a *contracting homotopy* is a set of maps  $s_{-1} : X_n \rightarrow X_{n+1}$  for each  $n > -1$  which are left-inverses (hereafter sections) of the 0th face maps  $d_0 : X_{n+1} \rightarrow X_n$  and satisfy the relevant simplicial identities.

This extends to augmented simplicial objects by taking  $s_{-1} : X_{-1} \rightarrow X_0$  to be a section of  $d_0 : X_0 \rightarrow X_{-1}$ . Sometimes these are called “backwards” contracting homotopies, and the forwards variant is given by taking the opposite simplicial object given by reversing the ordering of each set  $[n]$ . The maps of a forwards contracting homotopy are usually called  $s_{n+1} : X_n \rightarrow X_{n+1}$ . Contracting homotopies of all types are often called *extra degeneracies*. We now reconcile this notion with that of homotopy.

**Lemma 4.3.** *An augmentation and extra degeneracies for a simplicial object  $X$  in a cocomplete category  $\mathcal{M}$  is equivalent to specifying an object  $X_{-1} \in \mathcal{M}$  and a retract diagram*

$$X_{-1} \xrightarrow{s_{-1}} X_0 \xrightarrow{d_0} X_{-1}$$

whose maps define a simplicial homotopy equivalence between  $X$  and the constant simplicial object  $X_{-1}$ .

The proof uses the previous exercise which shows that for  $\mathcal{M}$  cocomplete,  $\mathcal{M}^{\Delta^{\text{op}}}$  is simplicially enriched and tensored, with tensor given by levelwise copower. Here note that we say there is a simplicial homotopy equivalence between these objects means there are morphisms  $X \rightrightarrows X_{-1}$  which are a simplicial homotopy equivalence (in this case, they are given above). If  $\mathcal{M}$  is a simplicial model category, lemma 3.7 implies this map is a weak equivalence.

**Corollary 4.1.** *If an augmented simplicial object  $X \rightarrow X_{-1}$  in a simplicial model category admits either a forwards or backwards contracting homotopy, then the natural map  $|X| \rightarrow X_{-1}$  is a weak equivalence.*

**Remark.** It is not in general true that the homotopy colimit and the geometric realization of a simplicial object coincide. However, given a simplicial object which admits an augmentation and contracting homotopy, the homotopy colimit is always weakly equivalent to its augmentation. This is proven in a few chapters.

For example, let  $G$  be a group, and write  $EG = B(G, G, *)$ . Here the notation indicates that  $*$  identifies the terminal set, the functor  $G$  identifies the constant simplicial set at  $G$ , and the tensor product is given as before via  $K \otimes X = K_0 \times X$ . We then obtain

$$B_n(G, G, *) = \coprod_{\vec{d}: [n] \rightarrow G} G(d_n) \otimes * \cong \coprod_{\vec{d}: [n] \rightarrow G} G \cong \coprod_{*} \prod_{i=0}^{n-1} G(*, *) = G^n,$$

where we view  $G$  as a group in the classical way at the end. This simplicial set is augmented by the singleton  $*$  and admits a backwards contracting homotopy that inserts identities into the left-most group in each product. Hence  $EG = |B_{\bullet}(G, G, *)|$  is a contractible  $G$ -space. Recall from topology contractible spaces are those for which the identity is null-homotopic. By corollary 4.1, we have that  $EG$  is weakly equivalent in a suitable simplicial model structure to the singleton, which is the categorical translation of “null homotopic”. Hence the term “contractible  $G$ -space”.

A more important example is as follows. Recall  $B(\mathcal{D}, \mathcal{D}, F)$  is shorthand for the functor  $d \mapsto B(\mathcal{D}(-, d), \mathcal{D}, F)$ . The above example can be viewed this way since  $* \mapsto G(*, *)$  is the only hom-set, so we provide the following generalization. The simplicial object  $B_{\bullet}(\mathcal{D}(-, d), \mathcal{D}, F)$  admits an augmentation and an extra degeneracy. The augmentation

$$\coprod_{d' \in \mathcal{D}} \mathcal{D}(d', d) \otimes F(d') \rightarrow F(d)$$

is by composition. More explicitly, tensor here is given by the copower, so we obtain a copower of  $F(d')$  indexed by maps  $f \in \mathcal{D}(d', d)$ . Applying  $F$  to each of these maps gives a collection of arrows  $F(f) : F(d') \rightarrow F(d)$ , so universality of coproduct gives a unique map as seen above. As a consequence, this process is natural in  $d$ . The extra degeneracies, which insert the identity at  $1_d$ , are *not* natural in  $d$ . However, lemma 4.3 still gives that the natural map  $\varepsilon : B(\mathcal{D}, \mathcal{D}, F) \Rightarrow F$  via the augmentation admits pointwise homotopy inverses; aka each  $\varepsilon_d$  is a homotopy equivalence.

## Chapter 5

### Homotopy limits and colimits: the theory

Recall that since  $\varprojlim$  and  $\varinjlim$  are not necessarily homotopic  $\mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$ , we want to compute their derived functors. Since colimits are left adjoints, we try to compute  $\mathbb{L}\varinjlim$ ; respectively,  $\mathbb{R}\varprojlim$ . The choice of derived functor does not matter if  $\mathcal{M}$  is saturated (the localization inverts precisely those maps which are weak equivalences), which it always is in practice (for example, in all model categories).

We will see later in the text that our current method is not the only answer; there are approaches via finding a Reedy categorical structure on  $\mathcal{D}$ , or by taking  $\mathcal{M}$  to be a cofibrantly generated or combinatorial model category. The former are fairly specialized, and the latter are computationally difficult, so we specialize instead in simplicial model categories  $\mathcal{M}$  by producing deformations from the bar and cobar constructions. We use the properties axiomatized by lemma 3.7 to do this. Next chapter explores examples of interesting simplicial model categories, and later we explore “local” universality conditions of the homotopy (co)limit.

#### 5.1 The homotopy limit and colimit functors

We write  $\mathrm{hocolim} : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$  for the left derived functor of  $\varinjlim$ , and  $\mathrm{holim}$  for the right derived functor of  $\varprojlim$ . We start by fixing at least one construction satisfying this property on the point-set level. This chapter will be dedicated to showing the following theorems, which define a specific construction of homotopy (co)limits. We first prove the relevant corollary, since the theorem is rather long.

**Theorem 5.1.** *Let  $\mathcal{M}$  be a simplicial model category with cofibrant replacement  $Q$  and fibrant replacement  $R^1$ . The pair*

$$B(\mathcal{D}, \mathcal{D}, Q(-)) : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}^{\mathcal{D}} \quad B(\mathcal{D}, \mathcal{D}, Q(-)) \xrightarrow{\varepsilon_Q} Q \xrightarrow{q} 1$$

*is a left deformation for  $\varinjlim : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$ , where we recall our notation gives the functor*

$$F \mapsto B(\mathcal{D}, \mathcal{D}, Q(F)).$$

*Dually,*

$$C(\mathcal{D}, \mathcal{D}, R(-)) : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}^{\mathcal{D}} \quad 1 \xrightarrow{r} R \xrightarrow{\eta_R} C(\mathcal{D}, \mathcal{D}, R(-))$$

*is a right deformation for  $\varprojlim : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$ .*

<sup>1</sup>Every arrow admits a factorization as a weak equivalence followed by a fibration; in particular, if this fibration is to the terminal object, we find that the object is weakly equivalent to a fibrant object. Since model categories in particular have this factorization be functorial, we obtain a fibrant replacement functor  $R : \mathcal{M} \rightarrow \mathcal{M}$  replacing objects with their fibrant replacement. Dually, we obtain a cofibrant replacement. Source: Hovey, *nLab*

**Corollary 5.1.** *If  $\mathcal{M}$  is a simplicial model category and  $\mathcal{D}$  is any small category, then the functors  $\varinjlim, \varprojlim : \mathcal{M}^{\mathcal{D}} \rightarrow \mathcal{M}$  admit left and derived functors, called hocolim and holim, respectively, defined by*

$$\mathrm{hocolim}_{\mathcal{D}} := \mathbb{L} \varinjlim_{\mathcal{D}} \cong B(*, \mathcal{D}, Q(-)) \quad \text{and} \quad \mathrm{holim}_{\mathcal{D}} := \mathbb{R} \varprojlim_{\mathcal{D}} \cong C(*, \mathcal{D}, R(-)).$$

**Proof.** Note here that  $*$  denotes the constant functor at the terminal object in  $\mathbf{sSet}$ , so recall that

$$* \otimes_{\mathcal{D}} - := \int^{d \in \mathcal{D}} * \otimes - \cong \int^{d \in \mathcal{D}} - \cong \varinjlim -$$

since  $\Delta^0$  is also the unit for the monoidal (cartesian) product in  $\mathbf{sSet}$ . By theorem 5.1, the left derived functor of colim exists, and by theorem 2.1 it is given as precomposing by the left deformation

$$\mathbb{L} \varinjlim_{\mathcal{D}}(-) \cong \varinjlim_{\mathcal{D}}(-) \circ B(\mathcal{D}, \mathcal{D}, Q(-)) = \varinjlim_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(-)) \cong * \otimes_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(-)).$$

Fubini's theorem for ends<sup>a</sup> gives us that functor tensor products commute past either variable in the two-sided bar construction. More precisely,

$$\begin{aligned} * \otimes_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(-)) &= \int^{d \in \mathcal{D}} * \otimes B(\mathcal{D}(-, d), \mathcal{D}, Q(-)) \\ &= \int^{d \in \mathcal{D}} * \otimes \int^{[n] \in \Delta} \Delta^n \otimes B_n(\mathcal{D}, (-, d), \mathcal{D}, Q(-)) \\ &\cong \int^{d \in \mathcal{D}} \int^{[n] \in \Delta} * \otimes \Delta^n \otimes B_n(\mathcal{D}(-, d), \mathcal{D}, Q(-)) \\ &\cong \int^{[n] \in \Delta} \int^{d \in \mathcal{D}} * \otimes \Delta^n \otimes B_n(\mathcal{D}(-, d), \mathcal{D}, Q(-)) \\ &\cong \int^{[n] \in \Delta} \int^{d \in \mathcal{D}} (* \times \Delta^n) \otimes B_n(\mathcal{D}(-, d), \mathcal{D}, Q(-)) \\ &\cong \int^{[n] \in \Delta} \int^{d \in \mathcal{D}} (\Delta^n \times *) \otimes B_n(\mathcal{D}(-, d), \mathcal{D}, Q(-)) \\ &\cong \int^{[n] \in \Delta} \int^{d \in \mathcal{D}} \Delta^n \otimes * \otimes \coprod_{\vec{d}: [m] \rightarrow \mathcal{D}} \mathcal{D}(d_n, d) \otimes Q(-)(d_0) \\ &\cong \int^{[n] \in \Delta} \int^{d \in \mathcal{D}} \Delta^n \otimes \coprod_{\vec{d}: [m] \rightarrow \mathcal{D}} * \otimes \mathcal{D}(d_n, d) \otimes Q(-)(d_0) \\ &\cong \int^{[n] \in \Delta} \int^{d \in \mathcal{D}} \Delta^n \otimes B(* \otimes \mathcal{D}(-, d), \mathcal{D}, Q(-)) \\ &\cong \int^{[n] \in \Delta} \Delta^n \otimes B_n(* \otimes_{\mathcal{D}} \mathcal{D}, \mathcal{D}, Q(-)) \\ &\cong B(* \otimes_{\mathcal{D}} \mathcal{D}, \mathcal{D}, Q(-)). \end{aligned}$$

The steps involve the fact that left adjoint functors preserve colimits, lemma 3.5, and definitions 3.9 and 3.11. One could state that this suffices as well as a proof of lemma 3.6. All one needs to understand the proof is to make precise the domains and codomains of each functor, then simply work through the tensors, viewing sets as enriched over simplicial sets as discussed previously whenever needed. Hence

$$* \otimes_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(-)) \cong B(* \otimes_{\mathcal{D}} \mathcal{D}, \mathcal{D}, Q(-)) \cong B(*, \mathcal{D}, Q(-)).$$

This last isomorphism follows by the coYoneda lemma

$$F(c) \cong \int^{d \in \mathcal{D}} \mathcal{D}(d, c) \cdot F(d).$$

We can view this as tensoring down into  $\mathbf{Set}$ , and being isomorphic to the cartesian product. Hence we have  $* \otimes_{\mathcal{D}} \mathcal{D}$ , evaluated at  $d$ , gives the constant functor  $d \mapsto *$  by evaluation.

The dual statement follows by the dual string of isomorphisms:

$$\mathbb{R} \varprojlim_{\mathcal{D}} (-) \cong \varprojlim_{\mathcal{D}} C(\mathcal{D}, \mathcal{D}, R(-)) \cong \{*, C(\mathcal{D}, \mathcal{D}, R(-))\}^{\mathcal{D}}.$$

The cotensor is an end, so by the same logic it commutes through. However since cobar is contravariant in its first variable, limits become colimits, or more precisely the cotensor becomes a tensor:

$$\{*, C(\mathcal{D}, \mathcal{D}, R(-))\}^{\mathcal{D}} \cong C(* \otimes_{\mathcal{D}} \mathcal{D}, \mathcal{D}, R(-)) \cong C(*, \mathcal{D}, R(-)).$$

■

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<sup>a</sup>this is imo the best name for a theorem in basic category theory, shoutout my man mac lane

With this fact established, theorem 5.1 remains to be shown. We begin with a series of lemmas.

**Lemma 5.1.** *The natural weak equivalence  $\varepsilon : B(\mathcal{D}, \mathcal{D}, F) \Rightarrow F$  makes  $B(\mathcal{D}, \mathcal{D}, -)$  a left deformation for  $\mathcal{M}^{\mathcal{D}}$ .*

**Proof.** Recall from the last part of chapter 4 that  $B_{\bullet}(\mathcal{D}(-, d), \mathcal{D}, F)$  admits an augmentation and extra degeneracy, which induces a natural map  $\varepsilon : B(\mathcal{D}, \mathcal{D}, F) \Rightarrow F$  which is a pointwise simplicial homotopy equivalence. This is a nontrivial fact and I will gloss over it for now. and Since  $\mathcal{M}$  is a simplicial model category, lemma 3.7 makes this a pointwise weak equivalence. Hence the two-sided bar construction  $B(\mathcal{D}, \mathcal{D}, -)$  is a deformation for  $\mathcal{M}^{\mathcal{D}}$ , since we recall that is simply an endofunctor and a natural weak equivalence to 1. ■

We then obtain the deformation in theorem 5.1 by composing  $\varepsilon$  with the left deformation  $q : Q \Rightarrow 1$  to the cofibrant objects of the simplicial model category, which exists by lemma 3.7. This is certainly a deformation, so it suffices to show it's indeed a deformation for the colimit functor. This means we must show that colimit preserves pointwise weak equivalences between diagrams in a full subcategory of  $\mathcal{M}^{\mathcal{D}}$  containing the image of  $B(\mathcal{D}, \mathcal{D}, Q(-))$ .

**Lemma 5.2.** *Let  $(Q, q)$  be a left deformation on  $\mathcal{M}$  and  $F : \mathcal{M} \rightarrow \mathcal{N}$ . If*

- $FQ$  is homotopical;
- $FqQ : FQ^2 \Rightarrow FQ$  is a natural weak equivalence;



then  $(Q, q)$  is a left deformation for  $F$ , meaning  $F$  is homotopical on a full subcategory of cofibrant objects if  $F$  preserves weak equivalences in the image of  $Q$  and also preserves the weak equivalences  $qQ$ .

**Proof.** Let  $x, y \in \mathcal{M}$ . The obvious thing to do is to show weak equivalences in  $\mathcal{M}(Q(x), Q(y))$  are preserved by  $F$ . For any  $f : Q(x) \rightarrow Q(y)$ , we have that  $FQ(f)$  is a weak equivalence, since  $FQ$  is homotopical. Naturality of  $FqQ$  gives

$$\begin{array}{ccc} FQ^2(x) & \xrightarrow{FQ(w)} & FQ^2(y) \\ F(q_{Q(x)}) \downarrow & & \downarrow F(q_{Q(y)}) \\ FQ(x) & \xrightarrow{F(w)} & FQ(y) \end{array}$$

This reads as the equality

$$F(w) \circ F(q_{Q(x)}) = F(q_{Q(y)}) \circ FQ(w).$$

Since  $FqQ$  is a natural weak equivalence, we have that  $F(w)$  is a weak equivalence by the 2-of-3 property: the right side is a weak equivalence since weak equivalences are closed under composition, and  $F(q_{Q(x)})$  is a weak equivalence. ■

By lemma 5.2, it suffices to show that  $B(*, \mathcal{D}, Q(-)) \cong \varinjlim_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(-))$  is homotopical, and that the composite

$$\varinjlim_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(B(\mathcal{D}, \mathcal{D}, Q(F)))) \xrightarrow{\varinjlim_{\mathcal{D}} \cdot \varepsilon} \varinjlim_{\mathcal{D}} Q(B(\mathcal{D}, \mathcal{D}, Q(F))) \xrightarrow{\varinjlim_{\mathcal{D}} \cdot q} \varinjlim_{\mathcal{D}} B(\mathcal{D}, \mathcal{D}, Q(F))$$

is a natural weak equivalence. We show this in the next section.

## 5.2 Homotopical aspects of the bar construction

To finish the proof, we must explore the homotopical properties of the bar construction, which follow from the fact that the two-sided simplicial bar is *Reedy cofibrant* when  $G$  and  $F$  are pointwise cofibrant. A Reedy cofibrant simplicial object is one in which the inclusion of the degenerate  $n$ -simplices into all  $n$ -simplices is a cofibrant. A degenerate  $n$ -simplex is supposed to be a simplex in the image of a degeneracy map, but I have no idea what this means in non-concrete categories. Anyways the inclusion is called the latching map and the degenerate  $n$ -simplices are called the  $n$ th latching object.

**Lemma 5.3.** *Let  $\mathcal{D}$  be small and  $\mathcal{M}$  be a simplicial model category. If  $F : \mathcal{D} \rightarrow \mathcal{M}$  is pointwise cofibrant, then  $B_{\bullet}(*, \mathcal{D}, F)$  is Reedy cofibrant.*

**Proof.** Recall  $B_n(*, \mathcal{D}, F)$  is the coproduct over  $N(\mathcal{D})_n$  of the image under  $F$  of the first object in the composable sequence of arrows. Hence, the  $n$ th latching object sits inside  $B_n(*, \mathcal{D}, F)$  as the coproduct indexed by degenerate simplices in  $N(\mathcal{D})_n$ . By a lemma proved in 6 chapters, the cofibrations, and hence the cofibrant objects, are closed under coproducts. By the same lemma, cofibrations are closed under pushout, and hence coproduct inclusions of cofibrant objects are also cofibrations. By hypothesis, every object in one of these coproducts is cofibrant, so the above

observations imply that the  $n$ th latching map is a cofibration. ■

I have no idea what this means I copied it directly from the book. I might review it when I learn what these words mean. For now, note this explains the role of the functor  $Q$  in the bar construction. Postcomposition by  $Q$  yields a pointwise cofibrant replacement of the diagram  $F$ , so the theorem will apply.

Recall that tensoring with any simplicial set preserves cofibrant objects. Thus,  $B_\bullet(G, \mathcal{D}, F)$  is Reedy cofibrant by the same argument for any pointwise cofibrant  $F$  with codomain in a simplicial model category, and  $G : \mathcal{D}^{\text{op}} \rightarrow \text{sSet}$ . We now introduce a standard result that will this time be proven in 9 chapters, which we need to prove the current result.

**Theorem 5.2.** *If  $\mathcal{M}$  is a simplicial model category, then*

$$|-| : \mathcal{M}^{\Delta^{\text{op}}} \rightarrow \mathcal{M}$$

*is left Quillen with respect to the Reedy model structure. In particular,  $|-|$  sends Reedy cofibrant simplicial objects to cofibrant objects and preserves pointwise weak equivalences between them.*

**Remark.** This is the strongest result we can give. It is not true that geometric realization preserves all pointwise homotopy equivalences, only those between cofibrant objects. This is why it is not a homotopy colimit always, since it is not in general homotopical.

**Corollary 5.2.** *The functor  $B(G, \mathcal{D}, -)$  preserves weak equivalences between pointwise cofibrant objects. In particular,  $B(*, \mathcal{D}, Q(-))$  is homotopical. Dually, when  $F$  is pointwise cofibrant,  $B(-, \mathcal{D}, F)$  preserves weak equivalences.*

**Proof.** Since  $B_\bullet(G, \mathcal{D}, F)$  is Reedy cofibrant for  $F$  pointwise cofibrant, by theorem 5.2 so is  $B(G, \mathcal{D}, F)$ . In particular, pointwise weak equivalences between these objects are preserved, so  $B(G, \mathcal{D}, -)$  preserves the relevant weak equivalences between pointwise cofibrant objects. Since  $Q$  is a left deformation into cofibrant objects, we obtain that  $B(*, \mathcal{D}, Q(-))$  is homotopical. ■

This shows the first objective of the proof. Now we must show the relevant natural map is a weak equivalence. Naturality gives

$$\begin{array}{ccc} B(\mathcal{D}, \mathcal{D}, Q(B(\mathcal{D}, \mathcal{D}, Q(F)))) & \xrightarrow{\varepsilon_{QB}} & Q(B(\mathcal{D}, \mathcal{D}, Q(F))) \\ \downarrow B(\mathcal{D}, \mathcal{D}, q) & & \downarrow q \\ B(\mathcal{D}, \mathcal{D}, B(\mathcal{D}, \mathcal{D}, Q(F))) & \xrightarrow{\varepsilon_B} & B(\mathcal{D}, \mathcal{D}, Q(F)) \end{array}$$

By the discussion following lemma 5.3,  $B_\bullet(\mathcal{D}, \mathcal{D}, Q(F))$  is pointwise Reedy cofibrant, since  $Q$  is a deformation into cofibrant objects. By theorem 5.2,  $B(\mathcal{D}, \mathcal{D}, Q(F))$  is pointwise cofibrant. Taking the colimit of the left side of the diagram gives

$$B(*, \mathcal{D}, q) : B(*, \mathcal{D}, Q(B(\mathcal{D}, \mathcal{D}, Q(F)))) \rightarrow B(*, \mathcal{D}, B(\mathcal{D}, \mathcal{D}, F)).$$

corollary 5.2 says that  $B(*, \mathcal{D}, -)$  preserves weak equivalences between pointwise cofibrant diagrams, hence this map is a weak equivalence. It remains only to be shown that the image of the bottom map under colim is a weak equivalence, by the 2-of-3 property.

**Lemma 5.4.** *When  $F$  is pointwise cofibrant, the functor  $\varinjlim_{\mathcal{D}}$  preserves the weak equivalence*

$$\varepsilon_{B(\mathcal{D}, \mathcal{D}, F)} : B(\mathcal{D}, \mathcal{D}, B(\mathcal{D}, \mathcal{D}, F)) \rightarrow B(\mathcal{D}, \mathcal{D}, F).$$

**Proof.** Recall that  $\varepsilon_F : B(\mathcal{D}, \mathcal{D}, F) \rightarrow F$  is the geometric realization of a map between simplicial objects determined by the maps

$$\mathcal{D}(d_n, d) \times \left( \prod_{i=0}^{n-1} \mathcal{D}(d_i, d_{i+1}) \right) \cdot F(d_0) \rightarrow F(d).$$

These maps compose the chains of arrows in  $\mathcal{D}$  to obtain a map  $d_0 \rightarrow d$ , then evaluate this at  $F$  to obtain  $F(d_0) \rightarrow F(d)$ . Since functor tensor products commute with both variables of the two-sided bar construction, and since  $D(-, d) \otimes_{\mathcal{D}} F \cong F(d)$ , we have

$$B(\mathcal{D}, \mathcal{D}, F) \cong B(\mathcal{D}, \mathcal{D}, \mathcal{D}, \otimes_{\mathcal{D}} F) \cong B(\mathcal{D}, \mathcal{D}, \mathcal{D}) \otimes_{\mathcal{D}} F \xrightarrow{\varepsilon \otimes_{\mathcal{D}} 1} \mathcal{D} \otimes_{\mathcal{D}} F \cong F.$$

Here,  $\varepsilon$  is the geometric realization of the map of simplicial objects

$$\varepsilon : B_{\bullet}(\mathcal{D}(-, d), \mathcal{D}, \mathcal{D}(d', -)) \rightarrow \mathcal{D}, (d', d)$$

defined by composing the arrows. This augmented simplicial object admits both forward and backwards contracting homotopies, which shows that  $G \otimes_{\mathcal{D}} \varepsilon$  is a natural weak equivalence. The argument is then computed by a commutative diagram that I don't want to copy down. ■

## Chapter 6

# Homotopy limits and colimits: the practice

## 6.1 Convenient categories of spaces

We now actually define various convenient categories of spaces that we have been referencing, such that we might obtain a complete cocomplete closed symmetric monoidal category, which is sometimes referred to as a “cosmos” since it is the best place to do enriched category theory. We go back to writing  $\mathcal{Top}$  to denote the standard category of topological spaces. The main issue with this category is it’s failiure to be closed under the cartesian monoidal structure:

**Theorem 6.1.** *The product, unit, and internal hom for any symmetric monoidal closed structure on  $\mathcal{Top}$  necessarily have the cartesian product, singleton, and set of continuous functions as underlying sets. However, there is no such structure in which the monoidal product is the cartesian product, hence  $\mathcal{Top}$  is not cartesian closed.*

The details of the proof are that one cannot topologize the set  $\mathcal{Top}(X, Y)$  such that evaluation  $\mathcal{Top}(X, Y) \times X \rightarrow Y$  is continuous and satisfies the relevant universal property:  $\mathcal{Top}(X, Y)$  should be terminal in the category of spaces  $W$  with a continuous map  $W \times X \rightarrow Y$ . The situation is even worse for pointed spaces.

**Lemma 6.1.** *The smash product in  $\mathcal{Top}_*$  is not associative. In particular, if  $\mathbb{Q}$  and  $\mathbb{N}$  inherit the subspace topology of  $\mathbb{R}$  with basepoint 0, then  $(\mathbb{Q} \wedge \mathbb{Q}) \wedge \mathbb{N}$  is not homeomorphic to  $\mathbb{Q} \wedge (\mathbb{Q} \wedge \mathbb{N})$ .*

We thus need to suggest alternatives to  $\mathcal{Top}$ , which contain as many of the important topological spaces as possible, in particular CW complexes, metric spaces, manifolds, and preferably even Hausdorff spaces.

**Definition 6.1** (Compactly Closed and  $k$ -Spaces). A subspace  $A \subseteq X$  is *compactly closed* if for any continuous function  $f : K \rightarrow X$  with  $K$  compact Hausdorff, the preimage  $f^{-1}(A)$  is closed in  $K$ . A space  $X$  is called a  $k$ -space if every compactly closed subspace of  $X$  is closed. ♥

We write  $k\mathcal{Top}$  for the full subcategory of  $k$ -spaces. The inclusion  $k\mathcal{Top} \hookrightarrow \mathcal{Top}$  admits a right adjoint  $k$ , called  $k$ -ification, which retopologizes the space  $X$  such that closed subsets of  $kX$  are precisely the compactly closed subsets of  $X$ . The identity  $kX \rightarrow X$  is continuous and is universal in the sense that  $kX$  is the finest topology such that  $K \rightarrow X$  factors through  $kX \rightarrow X$ .

**Theorem 6.2.** *The category  $k\mathcal{Top}$  is complete and cocomplete. Colimits are formed as in  $\mathcal{Top}$ , and limits are formed via applying  $k$  to the corresponding limit in  $\mathcal{Top}$ . The category  $k\mathcal{Top}$  is cartesian closed.*

The proof of cartesian closure is in an article I cannot access. The assertion about limits

follows somehow idk. The main point is that the product of two  $k$ -spaces is the  $k$ -ification of the usual product topology. If one of these spaces is locally compact the topology is the same, but in general the  $k$ -ification is a finer topology. This modification means that geometric realization  $|-| : \mathbf{sSet} \rightarrow \mathbf{kTop}$  preserves products, which it didn't when the codomain was  $\mathbf{Top}$ .

**Lemma 6.2.** *If  $X$  and  $Y$  are simplicial sets, then the geometric realization of their product is the product of their geometric realization.*

A function is  $k$ -continuous if and only if its composite with a continuous function  $K \rightarrow X$  is continuous, for  $K$  compact Hausdorff. Clearly a  $k$ -continuous function with domain as a  $k$ -space is continuous. The cartesian closed structure on  $\mathbf{kTop}$  is the  $k$ -ification of the compact-open topology, which we recall was defined as generated by

$$\{\{f : X \rightarrow Y \mid f(A) \subseteq U\} \mid A \in \mathcal{T}_X^c, \quad U \in \mathcal{T}_Y\}.$$

Here,  $\mathcal{T}_X^c$  is the set of all compact subsets. The  $k$ -ification of this topology only differs if  $X$  has non-Hausdorff compact subspaces.

$k$ -continuity is nonlocal unless compact subsets of  $X$  are locally compact, which is true if  $X$  is Hausdorff. We have another assertion that fixes this issue, giving a better notion of convergence for sequences of continuous functions.

**Definition 6.2** (Weak Hausdorff/Compactly Generated). A space  $X$  is *weak Hausdorff* if the image of any continuous map with compact Hausdorff domain is closed. A weakly Hausdorff  $k$ -space is called *compactly generated*. ♥

Of course, all Hausdorff spaces are weak Hausdorff; we call Hausdorff  $T_2$  and weak Hausdorff  $T_1$ . Examples of compactly generated spaces include metric spaces, CW complexes, and topological manifolds. Hence we obtain many of the important examples in topology. The idea is that the compact Hausdorff subspaces of a compactly generated space determine the topology. In particular, maps  $X \rightarrow Y$  between compactly generated spaces are continuous if and only if the restriction to any compact Hausdorff subspaces of  $X$  are continuous.

We write  $\mathbf{CGTop}$  for the category of compactly generated spaces. The inclusion  $\mathbf{CGTop} \hookrightarrow \mathbf{kTop}$  has a left adjoint, which maps  $K$ -spaces to their maximal weak Hausdorff quotient. Important to note here is that this adjunction is monadic, so we obtain that  $\mathbf{CGTop} \rightarrow \mathbf{kTop}$  creates limits. Similarly, the left adjoint carries colimits down to  $\mathbf{CGTop}$ . Hence, limits formed in  $\mathbf{CGTop}$  are the same as in  $\mathbf{kTop}$ , which is again applying  $k$  to the ordinary limit. In particular, geometric realization is a strong monoidal functor  $|-| : \mathbf{sSet} \rightarrow \mathbf{CGTop}$ . Many colimits are weak Hausdorff already, but the ones that aren't are poorly behaved under the left adjoint  $w$  of the function. Hence we only work in  $\mathbf{CGTop}$  when the relevant colimits do not need weak Hausdorffication.

**Theorem 6.3.** *The categories  $\mathbf{kTop}$  and  $\mathbf{CGTop}$  are cartesian closed. The categories  $\mathbf{kTop}_*$  and  $\mathbf{CGTop}_*$  are closed symmetric monoidal with respect to the smash product.*

## 6.2 Simplicial model categories of spaces

We now turn our attention to exploring simplicial homotopy structures and homotopy limits therein. The main example is that of  $\mathbf{sSet}$ . All objects are cofibrant, and fibrant objects are Kan complexes. These are simplicial sets in which *all* horns admit fillers. We define the  $i$ th horn of the  $n$ -simplex  $\Lambda_i^n$  as follows: the  $j$ th face map  $d^j : \Delta^{n-1} \rightarrow \Delta^n$  as a morphism of simplicial sets is defined as

postcomposing with the coface map  $\delta^j$ . The image  $\text{im}(d^j)$  is a subsimplicial set of  $\Delta^n$  defined levelwise as

$$\text{im}(d^j)_m = \text{im}(d_m^j).$$

Here we note that morphisms of simplicial sets have components for each  $m$ . The simplicial operators are simply the restrictions of those for  $\Delta^n$  to the image. We then define the horn as the union

$$\Lambda_i^n := \bigcup_{j \in [n], j \neq i} \text{im}(d^j : \Delta^{n-1} \rightarrow \Delta^n).$$

The union is once again defined levelwise, and since each simplicial set is a subsimplicial set of  $\Delta^n$ , we still have that the restriction of the simplicial operators is valid. The geometric intuition here is that the  $j$ th coface map inserts the standard  $(n-1)$ -simplex across from the  $j$ th vertex, hence we obtain the geometric notion of a horn via this construction. A Kan complex is then a simplicial set  $X_\bullet$  in which all horns admit fillers, meaning for any diagram of solid arrows

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & X_\bullet \\ \downarrow & \nearrow \text{dotted} & \\ \Delta^n & & \end{array}$$

there exists a dotted arrow as indicated, rendering the diagram commutative. Note that all Kan complexes are weak Kan complexes, which require only that inner horns admit fillers (so  $i \neq 0, n$ ), and hence are quasi-categories. Regular Kan complexes in fact define  $\infty$ -groupoids. I don't know why.

The usual geometric realization is strong monoidal as seen earlier in lemma 6.2 when the codomain is restricted to a convenient category of spaces. Quillen showed that it is left Quillen, so we have that  $\mathbf{kTop}$  and  $\mathcal{CGTop}$  are *simplicial model categories*. We have proven part of this fact, see theorem 3.1. All objects are fibrant in these categories.

There is a strong monoidal Quillen adjunction with leftdjoint  $(-)_+ : \mathbf{sSet} \rightarrow \mathbf{sSet}_*$ , as seen earlier. This makes based simplicial sets into a simplicial model category with all objects cofibrant, and fibrant objects as based Kan complexes. We also obtain

**Proposition 6.1.** *Geometric realization extends to a strong monoidal left Quillen functor  $|-| : \mathbf{sSet}_* \rightarrow \mathcal{CGTop}_*$ .*

In particular, the monoidal categorical structures of  $\mathcal{CGTop}_*$  and  $\mathbf{kTop}$  under the smash product are simplicial model structures. We can show that it is strong monoidal, however. The smash product is a pushout, and since geometric realization is a left adjoint, it preserves the pushout. It also commutes with the coproduct and product per above. Again, all objects are fibrant, and cofibrant objects are non-degenerately based cell complexes.

We have now lost all use for the category  $\mathcal{Top}$ , and for the purposes of this book (or at least this part), all limits, especially those used in the bar construction, are sufficiently well-behaved as to make the distinction between  $\mathbf{kTop}$  and  $\mathcal{CGTop}$  immaterial. Hence all results proven are valid for either of them, so we write  $\mathcal{Top}$  to denote either one of these categories.

### 6.3 Warnings and simplifications

**Remark.** Given  $\mathcal{M}$  homotopical, we always define the weak equivalences in  $\mathcal{M}^{\mathcal{D}}$  to be defined pointwise; hence weak equivalences are pointwise natural weak equivalences. I swap notation briefly

to  $\text{Fun}$  instead of exponential to make the following diagram more clear. Universality of localization gives a map

$$\begin{array}{ccc} \text{Fun}(\mathcal{D}, \mathcal{M}) & \xrightarrow{\text{Fun}(\mathcal{D}, \gamma)} & \text{Fun}(\mathcal{D}, \text{Ho } \mathcal{M}) \\ \gamma \downarrow & \nearrow & \\ \text{Ho}(\text{Fun}(\mathcal{D}, \mathcal{M})) & & \end{array}$$

but this is only an equivalence of categories in very special cases, for example when  $\mathcal{D}$  is discrete.

The diagonal functor  $\Delta : \mathcal{M} \rightarrow \mathcal{M}^{\mathcal{D}}$  is homotopical and is thus its own derived functor.<sup>1</sup> Since  $\varinjlim \dashv \Delta \dashv \varprojlim$ , in particular the colimit, as seen before the adjunction promotes as

$$\mathbf{L} \varinjlim \dashv \mathbf{L} \Delta \cong \Delta.$$

Interestingly, this adjoint version of the diagonal functor may not be the diagonal functor  $\text{Ho } \mathcal{M} \rightarrow \text{Ho}(\mathcal{M})^{\mathcal{D}}$ , unless the comparison is an *equivalence*. Hence, in absense of an equivalence, homotopy colimits need not be colimits in the homotopy category.

Fix a field  $k$ . A chain complex  $A_{\bullet} \in \text{Ch}(k)$  is quasi-isomorphic to its homology, when differentials between the homologies are given as 0 morphisms. Among chain complexes with differentials given by 0, quasi-isomorphisms are just pointwise isomorphisms by definition of homology. Hence inverting quasi-isomorphisms in  $\text{Ho}(\text{Ch}(k))$  is equivalent to the category of graded vector spaces, since a complex with 0 differentials is precisely a graded vector space. The latter is complete and cocomplete, but these (co)limits are not homotopy (co)limits. “ $\text{Ch}(k)$  is a stable  $(\infty, 1)$ -category so homotopy pullbacks are also homotopy pushouts, which mutatis mutandis is not true for graded vector spaces.” I don’t know what this means, all the way from the remark about stable  $\infty$ -categories to “mutatis mutandis”. Additionally, homotopy categories usually have very few limits and colimits. For example, not all coequalizers exist in  $\text{Ho } \mathcal{J}\text{op}$ .

We provide the following simpliciation in most cases. By corollary 5.1, any  $F : \mathcal{D} \rightarrow \mathcal{M}$  with codomain a simplicial model category has  $B(*, \mathcal{D}, Q(F))$  has the right homotopy type for  $\text{hocolim } F$ . However, if the diagram is already cofibrant, we have that  $B(*, \mathcal{D}, F)$  and  $B(*, \mathcal{D}, Q(F))$  are weakly equivalent by corollary 5.2, and the correction via  $Q$  is not needed. The book rephrases this as:  $Q$  simply corrects the homotopy type, but if it is already correct, we prefer a simpler construction.

**Remark.** In  $\mathcal{J}\text{op}$ , viewed as a simplicial model cateogry, not all objects are cofibrant. They are all fibrant, so we can always compute the homotopy limit without the correction, but “a folklore result” shows that we can also compute homotopy colimits without correction as well. We will produce this proof later, after considering Reedy category theory more rigorously.

## 6.4 Sasample homotopy colimits

<sup>1</sup>omg i finally figured out derived functors: its just the best approximation to make it homotopical! the total derived functor is the kan extension which is automatically homotopical but its between homotopy categories so like who cares, but the derived functor is a new functor that goes between the same categories as the original, but is homotopical, and agrees with the kan extension when postcomposed with delta and the kan extension is precomposed with gamma.